

# High-dimensional Covariance Estimation From Quadratic Measurements via Nonconvex Learning

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**Abstract**—The covariance matrix is a fundamental second-order statistic in signal and information processing that quantifies the linear relationships among multiple variables. When data evolves rapidly, or the acquisition devices have limited processing power and storage—particularly given the large dimensionality of modern datasets—covariance estimation becomes challenging. To address these challenges, it is desirable to estimate the covariance matrix from a single pass over the data with compressed measurements. In this paper, we study covariance matrix estimation in high dimensions based on quadratic measurements, assuming the covariance matrix exhibits a sparse structure. We formulate the problem as a least-squares estimation with nonconvex sparsity-inducing penalties. To efficiently compute this estimator, we develop a multistage convex relaxation algorithm based on the majorization-minimization (MM) algorithmic framework. We comprehensively characterize the computational and statistical properties of the iterates from the algorithm by leveraging restricted strong convexity and Hessian smoothness. Specifically, we show that the proposed algorithm achieves (local) quadratic convergence to approximate solutions for each convex relaxation subproblem and eventually achieves a sparse approximate local optimum after only a limited of convex relaxation subproblems. Furthermore, we show that the solution obtained possesses oracle statistical properties in the Frobenius norm under mild assumptions. Numerical simulations support the theoretical findings and validate the performance of the proposed estimator.

**Index Terms**—Compressive covariance sensing, compressive covariance sketching, quadratic measurements model, rank-one measurements model, majorization-minimization, positive definiteness, sparsity, nonconvex statistical optimization.

## I. INTRODUCTION

The rapid advancements in communication technologies have led to an increasing demand for processing ultra-wideband signals, introducing new challenges in signal sampling and processing [1]–[7]. Compressed sensing (CS) has emerged as a promising framework for signal reconstruction, leveraging the inherent sparsity of signals in appropriate bases to achieve remarkable compression efficiency and low computational complexity [8]–[13]. This sparsity allows high dimensional signals to be efficiently sub-sampled using random linear projections and accurately reconstructed from low dimensional measurements [14]–[19]. In many practical applications, however, random processes are involved. In such settings, the reconstruction of original signals itself is not meaningful [20]–[23]. Instead, the focus shifts towards extracting critical quantities from signals, such as second-order or even higher-order statistics, which contain valuable features

and provide reliable information about the random processes [24]–[26].

The covariance matrix, as a fundamental second-order statistic, is pivotal to numerous tasks in signal processing [27]–[29] and machine learning [30]. Specific applications include spectral estimation [31], adaptive filtering [32], beamformer design [33], principal component analysis [34], linear and quadratic discriminant analysis [35], among others. The covariance matrix is not directly observable and must, hence, be estimated from empirical data. Consider a zero-mean random vector  $\mathbf{x} \in \mathbb{R}^d$  with covariance matrix  $\Sigma^*$ . A widely used estimator  $\Sigma^*$  is the sample covariance matrix (SCM)  $\mathbf{S} = \frac{1}{n} \sum_{t=1}^n \mathbf{x}_t \mathbf{x}_t^\top$ . The computation of SCM relies on the availability of complete data samples [36], [37]. However, obtaining a sufficient number of complete data samples for a reliable estimator is computationally expensive and resource-intensive [38], [39], especially when data evolves rapidly or sampling devices have limited processing power and storage. The problem becomes more pronounced when we are faced with modern high-dimensional datasets.

In lieu of conducting complete measurements, estimating the covariance matrix based on compressed measurements has emerged as a viable alternative. Consider a compressed measurement of  $\mathbf{x}$  in the form  $\mathbf{a}^\top \mathbf{x}$ , where  $\mathbf{a}$  is a sensing vector. The variance of  $\mathbf{a}^\top \mathbf{x}$ ,  $\mathbf{a}^\top \Sigma^* \mathbf{a}$ , is referred to as quadratic measurements (or rank-one measurements) [40]. It serves as a compressed sketch of  $\Sigma^*$ , and can be effectively stored and processed. XXX, our target is to efficiently reconstruct the covariance matrix  $\Sigma^*$  from quadratic measurements [41]–[43]. A central question in this framework is whether it is possible to design the sensing vectors such that the resulting compressive sketches exhibit statistically desirable properties, enabling accurate reconstruction of  $\Sigma^*$  from a minimal number of measurements. While  $\Sigma$  may generally not be identifiable under arbitrary sensing designs, reliable recovery is possible with cleverly designed  $\mathbf{a}$  when  $\Sigma$  possesses certain latent structures, such as sparsity, in alignment with the principles of CS [9]–[11], [15], [16], [44]–[46]. In high-dimensional covariance matrix estimation problems, where the sample size is small relative to the ambient dimension of the underlying parameter to be estimated [27], [37], [39], [47]–[49]. The sparsity assumption is actually well-justified and widely adopted by many practical applications [40], [50], [51]. In the literature, covariance estimation from quadratic measurements has been investigated [40]–[42]. In [52], [53], the authors explored the recovery of second-order statistics of a cyclostationary signal from random linear measurements, assuming that the covariance matrix is approximately sparse. They focused on

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the sampling process and proposed an  $\ell_1$ -minimization based recovery method without performance guarantees. The authors in [40] presented a unified framework for compressing and recovering sparse covariance matrices using the  $\ell_1$ -norm heuristic. They showed that  $\mathcal{O}(s \log(d^2/s))$  measurements are sufficient for compressing a  $d$ -dimensional sparse covariance matrix with  $s$  non-zero entries. Concurrently, *Dasarathy et al.* [42] sketched the covariance matrix  $\Sigma$  as  $A\Sigma B^\top$ , assuming  $\Sigma$  exhibits distributed sparsity, with sketching matrices  $A$  and  $B$  constructed from expander graphs. It is proved that the sample complexity for compressing distributed sparse covariance matrices is  $\mathcal{O}(\sqrt{d} \log d)$ . [WenbinWang: Consider breaking them down into the following parts: compressed seeing, covariance estimation, and the quadratic measurement model]

directly sensing the true covariance, which is not reasonable. All existing XXX, It is well-known that methods based on the  $\ell_1$ -norm, e.g., Lasso [54], introduces a non-negligible bias into the resulting estimator [55], which compromises the estimation accuracy. To alleviate this bias effect, nonconvex penalties such as smoothly clipped absolute deviation (SCAD) penalty [55], minimax concave penalty (MCP) [56], and capped  $\ell_1$ -regularization [57] have been proposed as alternatives. It has been demonstrated in [55]–[57] that the nonconvex penalized regression can effectively eliminate the estimation bias in Lasso and achieve improved statistical convergence rates.

However, the theory of nonconvex penalty in the context of covariance estimation from compressed measurements remains underexplored. In this paper, we investigate the quadratic measurement model for covariance matrix estimation. We show that for a more general class of sub-Gaussian sensing vectors than those considered in prior works [40], it is possible to derive the covariance matrix exactly and establish theoretical performance guarantees. We formulate the problem as a least-squares estimation with nonconvex sparsity-inducing penalties. However, the analysis of the theoretical performance guarantee is complicated due to the nonconvexity of the objective function. To address this, we introduce an efficient multi-stage convex relaxation algorithm based on the majorization-minimization (MM) framework, solving the original problem via a sequence of convex subproblems. XXX Based on the MM framework, we prove that the proposed estimator achieves faster statistical convergence rates compared to the conventional estimator using the  $\ell_1$ -norm penalty. Furthermore, under mild assumptions, we rigorously prove that the proposed estimator exhibits oracle statistical properties. Numerical simulations are provided to support the theoretical findings and validate the superior performance of the proposed estimator.

## II. PRELIMINARIES

In this section, we introduce a comprehensive framework for CCS, followed by the proposed estimator with a non-convex penalty.

### A. The Quadratic Measurement Model

Consider  $n$  independent and identically distributed (i.i.d.) observations  $\{\mathbf{x}_i\}_{i=1}^n$ , each drawn from a zero-mean random

vector  $\mathbf{x}$ . The quadratic measurement model is given by

$$y_i = \frac{1}{n} \sum_{t=1}^n |\mathbf{a}_i^\top \mathbf{x}_t|^2 + \eta_i = \mathbf{a}_i^\top \mathbf{S} \mathbf{a}_i + \eta_i, \quad i = 1, \dots, m, \quad (1)$$

where  $\{y_i\}_{i=1}^m$  are a sequence of measurements,  $\{\mathbf{a}_i\}_{i=1}^m$  are the sensing vectors, and  $\{\eta_i\}_{i=1}^m$  are additive noise components assumed to be drawn from a sub-exponential distribution with mean 0 and variance proxy  $\sigma^2$ .  $\mathbf{S} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^\top$  is the SCM, which can be decomposed as  $\mathbf{S} = \Sigma^* + \mathbf{E}$ , where  $\Sigma^*$  is the true covariance matrix, and  $\mathbf{E}$  is the bias term.

For convenience, we define  $\mathbf{A}_i := \mathbf{a}_i \mathbf{a}_i^\top$  as the corresponding sensing matrix, so that  $y_i = \langle \mathbf{A}_i, \mathbf{S} \rangle + \eta_i$ . Denote the measurement vector as  $\mathbf{y} := [y_1, \dots, y_m]^\top$  and the noise vector as  $\boldsymbol{\eta} := [\eta_1, \dots, \eta_m]^\top$ . Defining the linear operator  $\mathcal{A}(\mathbf{S}) : \mathbb{R}^{d \times d} \mapsto \mathbb{R}^m$ , which maps a matrix  $\mathbf{S} \in \mathbb{R}^{d \times d}$  to the set  $\{\langle \mathbf{A}_i, \mathbf{S} \rangle\}_{i=1}^m$ , the  $m$  quadratic measurements can be expressed as

$$\mathbf{y} = \mathcal{A}(\mathbf{S}) + \boldsymbol{\eta}.$$

It is noteworthy that the measurement model in (1) has been previously explored in [40], [58], [59]. The model presented in (1) is significantly simpler to implement and entails lower computational complexity when compared to full-rank measurement matrices with independent and identically distributed (i.i.d.) entries. [WenbinWang: Just talk about the model, the rest of the unnecessary content later.]

### B. Practical Examples

The quadratic measurement model in (1) has numerous practical applications across various domains. We briefly review two examples: Spectrum estimation of stochastic processes from energy measurements data sciences and direction-of-arrival (DOA) estimation. For further examples and detailed discussions, readers are referred to the works [40], [42], [59].

**Spectrum Estimation** In the context of spectrum estimation for stochastic processes based on energy measurements, one typically employs a sensing vector  $\mathbf{a}_i$ , which can be implemented using random demodulators [60]. The energy measurements are averaged over  $N$  instances  $\{\mathbf{x}_t\}_{1 \leq t \leq N}$ , leading to observations in the form of quadratic forms.

**DOA** In the context of DOA, the objective is to determine the angles of arrival for a finite number of signal sources. This problem can be framed as a sparse power spectrum estimation task, which naturally leads to a quadratic measurement model. A wide range of techniques have been developed to address this challenge [61]–[64].

### C. Proposed Estimator

Based on X, we estimate the unknown covariance matrix  $\Sigma^*$  through solving the following optimization problem

$$\min_{\Sigma \succ 0} \left\{ \frac{1}{2m} \|\mathbf{y} - \mathcal{A}(\Sigma)\|_2^2 - \tau \log \det \Sigma + \sum_{i,j} p_\lambda(|\Sigma_{ij}|) \right\}, \quad (2)$$

where the first term aims to minimize the empirical errors; the second term incorporates a logarithmic barrier function to

enforce positive definiteness of  $\Sigma$ , with  $\tau > 0$  [65]<sup>1</sup>; the third term **XXX**, where  $p_\lambda(\cdot)$  represents a penalty function with  $\lambda > 0$ . We consider a class of non-convex penalty functions  $p_\lambda(\cdot)$  satisfying the following assumptions.

**Assumption 1.** *The function  $p_\lambda(t) : \mathbb{R} \rightarrow \mathbb{R}$  satisfies:*

- $p_\lambda(t)$  is **symmetric around zero** with  $p_\lambda(0) = 0$ , nondecreasing on the nonnegative, differentiable almost everywhere on  $(0, +\infty)$ , and subdifferentiable at  $t = 0$ ;
- $0 \leq p'_\lambda(t_1) \leq p'_\lambda(t_2) \leq \lambda$  for all  $t_1 \geq t_2 \geq 0$  and  $\lim_{t \rightarrow 0^+} p'_\lambda(t) = \lambda$ ;
- There exists an  $\alpha > 0$  such that  $p'_\lambda(t) = 0$  for  $t \geq \alpha\lambda$ ;

It is suggested that a well-designed penalty function should yield an estimator with three properties: unbiasedness, sparsity, continuity [55], which coincides with the three conditions in Assumption 1. Two representative examples are:

- Smooth clipped absolute deviation penalty [55]: This penalty is given by

$$p'_\lambda(t) = \begin{cases} \lambda, & \text{for } 0 < t \leq \lambda, \\ \frac{b\lambda - t}{b-1}, & \text{for } \lambda \leq t \leq b\lambda, \\ 0, & \text{for } t \geq b\lambda, \end{cases}$$

where  $b > 2$  is an additional tuning parameter.

- Minimax concave penalty [57]: This penalty is defined as follows:

$$p_\lambda(t) = \text{sign}(t) \lambda \cdot \int_0^{|t|} \left(1 - \frac{z}{\lambda b}\right)_+ dz$$

for some  $b > 0$ .

### III. OPTIMIZATION ALGORITHM

In this section, we develop a numerical algorithm to solve problem (2). We first revisit the MM framework, which plays a crucial role in the forthcoming theoretical analysis. We then introduce an MM-based multistage convex relaxation algorithm. Furthermore, we describe the solution method employed at each stage of the MM algorithm: a proximal Newton homotopy algorithm.

#### A. The MM Framework

**[plot]** The MM algorithm framework is an iterative process. Suppose we want to minimize a real-valued function  $F(\mathbf{x})$ . Instead of directly minimizing  $F(\mathbf{x})$ , the algorithm focuses on solving a series of simple optimization problems sequentially. The process begins with a feasible initial point  $\mathbf{x}^0$  and continues iteratively through  $\mathbf{x}^1, \mathbf{x}^2, \dots$  until a specified convergence criterion is satisfied. Specifically, the basic procedure of the MM algorithm framework encompasses two main steps:

- **Majorization Step:** In this step, the goal is to select or construct a surrogate function  $\bar{F}(\mathbf{x} | \mathbf{x}^{k-1})$  which equals the value of the original objective function  $F(\mathbf{x})$  at the

<sup>1</sup>While covariance matrices are typically semi-positive definite, practical applications often require that estimators be strictly positive definite, even for finite sample sizes. This property is particularly important in supervised learning contexts, such as in quadratic discriminant analysis and covariance regularized regression [66].

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**Algorithm 1:** The MM-Based Multistage Convex Relaxation Algorithm for Solving (2).

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**Input:**  $\{y_i, \mathbf{a}_i\}_{i=1}^m, \tau, \lambda$ ;  
**1 Initialize**  $\Sigma^0 = \mathbf{I}$   
**2 for**  $k = 1, 2, \dots, K$  **do**  
**3**    $\Lambda_{ij}^{k-1} = p'_\lambda(|\Sigma_{ij}^{k-1}|)$ ;  
**4**    $\Sigma^k = \arg \min_{\Sigma} F(\Sigma)$ ;  
**5**    $k = k + 1$ ;  
**6 end**  
**Output:**  $\Sigma^k$

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current iteration point  $\mathbf{x}^{k-1}$ , but is an upper bound for  $F(\mathbf{x})$  at all other points. Specifically,  $\bar{F}(\mathbf{x} | \mathbf{x}^{k-1}) \geq F(\mathbf{x})$  for all  $\mathbf{x}$  and  $\bar{F}(\mathbf{x}^{k-1} | \mathbf{x}^{k-1}) = F(\mathbf{x}^{k-1})$ .

- **Minimization Step:** After the surrogate function is determined, the next step is to find the point  $\mathbf{x}^k$  that minimizes  $\bar{F}(\mathbf{x} | \mathbf{x}^{k-1})$ , such that  $\mathbf{x}^k \in \arg \min \bar{F}(\mathbf{x} | \mathbf{x}^{k-1})$ .

These two steps are executed in an alternating fashion, with each iteration aimed at reducing or at least not increasing the value of the objective function  $F(\mathbf{x})$ , ensuring  $F(\mathbf{x}^k) \leq F(\mathbf{x}^{k-1})$ .

#### B. A Multistage Convex Relaxation Algorithm

We now introduce a MM-based multistage convex relaxation algorithm to solve the problem (2).

In each iteration of the algorithm, a weighted  $\ell_1$ -norm serves as the surrogate function for  $\sum_{i,j} p_\lambda(\Sigma_{ij})$ . Specifically, for each  $1 \leq k \leq K$ , we minimize the **following convex relaxation subproblem**:

$$F_k(\Sigma) = f(\Sigma) + \sum_{i,j} p'_\lambda(|\Sigma_{ij}^{k-1}|) |\Sigma_{ij}|, \quad (3)$$

where

$$f(\Sigma) = \frac{1}{2m} \|\mathbf{y} - \mathcal{A}(\Sigma)\|_2^2 - \tau \log \det \Sigma.$$

and  $\Sigma^k$  represents the solution to the  $k$ -th subproblem.

According to the Karush-Kuhn-Tucker (KKT) conditions, the optimal solution  $\hat{\Sigma}$  for each subproblem satisfies the first-order optimal condition:

$$\mathbf{0} \in \nabla f(\hat{\Sigma}) + \Lambda^k \odot \hat{\Xi},$$

where  $\nabla f(\Sigma) = -\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma)) - \tau \Sigma^{-1}$ , with  $\mathcal{A}^*$  the conjugate operator of  $\mathcal{A}$ ;  $\Lambda^k$  is the regularized parameter matrix with entries  $\Lambda_{ij} = p'_\lambda(|\hat{\Sigma}_{ij}|) \in [0, \lambda]$  and  $\hat{\Xi} \in \partial \|\hat{\Sigma}\|_1$ .

Given that the problem (3) lacks an analytical solution, we seek a sub-optimal solution, as defined in Definition 2, and terminate the iterations when the approximate KKT conditions are satisfied.

**Definition 2.** Let  $\varepsilon$  denote the tolerance level. The solution  $\Sigma^k$  is considered  $\varepsilon$ -optimal for the  $k$ -th subproblem (??) if the condition  $\omega(\Sigma^k) \leq \varepsilon$  is satisfied, where

$$\omega(\Sigma^k) = \min_{\Xi \in \partial \|\Sigma^k\|_1} \left\| \nabla f(\Sigma^k) + \Lambda^{k-1} \odot \Xi \right\|_{\max}.$$

We present the following assumption on the pre-specified tolerance level  $\varepsilon$ , which is essential for the convergence analysis of multistage convex relaxation.

**Assumption 3.** For each iteration  $k \geq 1$  of the convex relaxation process (??), we assume the accuracy parameter  $\varepsilon$  as follows:

$$\varepsilon = \frac{c_6}{\sqrt{mn}} \leq \frac{\lambda}{8},$$

where  $c_6$  is a predetermined small constant.

### C. Proximal Newton Algorithm

At each stage, the subproblem is obviously convex, enabling the application of various solution techniques. In this study, we adopt the proximal Newton method to enhance computational efficiency. The proximal Newton algorithm solves a proximal subproblem in each iteration, which involves updating the current estimate using both gradient and Hessian information. Specifically, the algorithm computes a Newton direction to serve as the descent direction, followed by determining an appropriate step size that ensures a sufficient decrease in the objective function.

We denote the  $t$ -th iteration of proximal Newton method within the  $k$ -th stage as  $\Sigma_t^k$ . We construct a quadratic approximation by considering the second-order Taylor expansion of  $f(\Sigma)$ :

$$f_t^k(\Sigma) = f(\Sigma_t) + \langle \nabla f(\Sigma_t), \Sigma - \Sigma_t \rangle + \frac{1}{2} \|\Sigma - \Sigma_t\|_{\nabla^2 f(\Sigma_t)}^2. \quad (4)$$

We then solve

$$\Sigma_{t+\frac{1}{2}} = \underset{\Sigma \succ 0}{\operatorname{argmin}} \bar{F}(\Sigma \mid \Sigma_t, \Lambda), \quad (5)$$

where  $\bar{F}(\Sigma \mid \Sigma_t, \Lambda) = \bar{f}(\Sigma \mid \Sigma_t) + \|\Lambda \odot \Sigma\|_1$ . Denote  $\Delta \Sigma_t = \Sigma_{t+\frac{1}{2}} - \Sigma_t$  as the Newton direction for the function  $F(\Sigma)$ . Particularly, we adopt coordinate descent algorithms combined with an active set strategy to get the Newton direction, which can improve the time complexity. The detailed computation will be introduced in the Appendix B.

An additional backtracking line searching procedure is included to guarantee the descent of the objective value. We try to find a step size  $\beta \in (0, 1]$  such that the Armijo condition [67] holds. Specially, we start with a fixed constant  $\mu \in (0.5, 1)$  and update  $\beta = \mu^q$  from  $q = 0$  with a constant decrease rate until we find the smallest  $q \in \mathbb{N}$  such that

$$F(\Sigma_t + \beta \Delta \Sigma_t) \leq F(\Sigma_t) + \alpha \beta \delta_t,$$

where  $\alpha \in (0, 0.5)$  and

$$\delta_t^k = \langle \nabla f(\Sigma_t), \Delta \Sigma_t \rangle - \|\Lambda \odot \Sigma_t\|_1 + \|\Lambda \odot (\Sigma_t + \Delta \Sigma_t)\|_1. \quad (6)$$

Then  $\Sigma_{t+1}$  is set as  $\Sigma_{t+1} = \Sigma_t + \beta \Delta \Sigma_t$ . The whole proximal Newton algorithm is summarized in Algorithm 2.

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### Algorithm 2: Proximal Newton Algorithm for (??).

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**Input:**  $\Sigma^{k-1}, \Lambda^{k-1}, \varepsilon$ ;  
**1 Initialize**  $t = 0, \Sigma_t^k = \Sigma^{k-1}, \mu = 0.8, \alpha = 0.3$   
**2 repeat**  
**3**     $\Sigma_{t+\frac{1}{2}}^k = \underset{\Sigma \succ 0}{\operatorname{argmin}} \bar{F}(\Sigma \mid \Sigma_t^k, \Lambda^{k-1})$   
**4**     $\Delta \Sigma_t^k = \Sigma_{t+\frac{1}{2}}^k - \Sigma_t^k$   
**5**     $\delta_t = \langle \nabla f(\Sigma_t^k), \Delta \Sigma_t^k \rangle - \|\Lambda^{k-1} \odot \Sigma_t^k\|_1$   
**6**        $+ \|\Lambda^{k-1} \odot (\Sigma_t^k + \Delta \Sigma_t^k)\|_1$   
**7**     $\beta = 1, q = 0$   
**8**    **repeat**  
**9**        $\beta = \mu^q$   
**10**        $q = q + 1$   
**11**       **if**  $\Sigma_t^k + \beta \Delta \Sigma_t^k \preceq 0$  **then**  
**12**          **continue**;  
**13**       **end**  
**14**    **until**  $F(\Sigma_t^k + \beta \Delta \Sigma_t^k) \leq F(\Sigma_t^k) + \alpha \beta \delta_t$ ;  
**15**     $\Sigma_{t+1}^k = \Sigma_t^k + \beta \Delta \Sigma_t^k$   
**16**     $t = t + 1$   
**17 until**  $\omega_{\Lambda^{k-1}}(\Sigma_t^k) \leq \varepsilon$ ;  
**Output:**  $\Sigma^k = \Sigma_t^k$

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## IV. THEORETICAL RESULTS

In this section, we first introduce some technical assumptions and important definitions. Following these, we establish the statistical convergence rate of our proposed covariance estimator and the iteration complexity of the proposed algorithm. The proofs are provided in the supplementary material.

We present several fundamental assumptions on the sensing vectors  $\mathbf{a}_i$ .

**Assumption 4.** We assume that the sensing vectors  $\mathbf{a}_i$ 's ( $1 \leq i \leq m$ ) are i.i.d. sub-Gaussian random variables. Each vector  $\mathbf{a}_i$  contains  $d$  elements, denoted as  $a_{i,j}$  ( $1 \leq j \leq d$ ), which are characterized by the following conditions:

$$\mathbb{E}[(\mathbf{a}_i)_j] = 0, \quad \mathbb{E}[(\mathbf{a}_i)_j^2] = 1. \quad (7)$$

Assumption 4 asserts that each row of the design matrix  $\mathbf{A}$  is composed of i.i.d. sub-exponential random variables.

### A. Restricted Eigenvalue and Sparse Eigenvalue

In this section, we investigate the sparse eigenvalue condition, which is crucial for establishing the following theorem on the universal recovery of CCS.

It is well-established in CS that perfect recovery from a minimal number of samples is achievable if the dimensionality reduction projection preserves the signal strength within the class of matrices under consideration [9], [11]. Several incoherence measures, such as the Uniform Uncertainty Principle (UUP) and the Restricted Isometry Property (RIP), have been proposed to characterize this [68]. These concepts similarly



facilitate the adaptive recovery of CCS. In [40], the mixed-norm  $\text{RIP-}\ell_2/\ell_1$  was introduced, which combines the Frobenius norm and the  $\ell_1$  norm to assess input and output strengths, respectively. An analogous criterion,  $\text{RIP-}\ell_1/\ell_1$ , has also been applied to CCS recovery [58]. **XXX**

Bickel et al. [69] introduced the Restricted Eigenvalue (RE) condition, demonstrating that it is one of the weakest and most general conditions in sparse reconstruction problems. This condition provides a relaxation of the UUP [56], [70]. Here, we present one version of the RE condition, as described by Bickel et al. [69].

**Definition 5** (Restricted Eigenvalue). For some integer  $0 < s_0 < d$  and a positive constant  $c_0$ , the  $\text{RE}(s_0, c_0, \mathbf{X})$  for the matrix  $\mathbf{X} \in \mathbb{R}^{d \times d}$  necessitates that the following inequality holds:

$$\forall \mathbf{u} \neq 0, \min_{\substack{J \subset \{1, \dots, d\}, \\ |J| \leq s_0}} \min_{\|\mathbf{u}_{-J}\|_1 \leq c_0 \|\mathbf{u}_J\|_1} \frac{\|\mathbf{X}\mathbf{u}\|_2}{\|\mathbf{u}_J\|_2} > 0, \quad (8)$$

where  $\mathbf{v}_J$  denotes the subvector of  $\mathbf{v} \in \mathbb{R}^d$  restricted to the subset  $J$  of  $\{1, \dots, d\}$ .

In the following, we introduce the Sparse Eigenvalues (SE) condition as delineated by Fan et al. [71], which exhibits a significant correlation with the RE properties. For the detailed relationships between RE and SE, please refer to Appendix A.

**Definition 6** (Sparse Eigenvalue). For a positive integer  $0 < s_0 < d$ , the localized sparse eigenvalues are defined as

$$\begin{aligned} \rho_{s_0, r}^+ &= \sup \left\{ \frac{\mathbf{u}^\top \nabla^2 f(\Sigma) \mathbf{u}}{\mathbf{u}^\top \mathbf{u}} \mid \|\mathbf{u}\|_0 \leq s_0, \|\Sigma - \Sigma^*\|_F \leq r \right\}; \\ \rho_{s_0, r}^- &= \inf \left\{ \frac{\mathbf{u}^\top \nabla^2 f(\Sigma) \mathbf{u}}{\mathbf{u}^\top \mathbf{u}} \mid \|\mathbf{u}\|_0 \leq s_0, \|\Sigma - \Sigma^*\|_F \leq r \right\}. \end{aligned}$$

The SE condition has been widely studied within the domain of high-dimensional sparse recovery and compressed sensing [69], [72]–[75]. The SE condition with parameter  $s_0$  holds for i.i.d. sub-exponential random matrices, provided that  $m = \mathcal{O}(s_0 \log^2 d / s_0)$  [76]. This condition is satisfied with probability at least  $1 - c_1 \exp(-c_2 \sqrt{m})$ , where  $c_1, c_2 > 0$  are universal numerical constants.

Hence, we present the following assumption:

**Assumption 7.** *There exists a universal constant  $c_3$  such that for  $\tilde{s} \geq c_3 s^*$ , the SE property is guaranteed with parameters  $\rho_{2s^*+2\tilde{s}, r}^-$  and  $\rho_{2s^*+2\tilde{s}, r}^+$  satisfying*

$$0 < \rho_{2s^*+2\tilde{s}, r}^- < \rho_{2s^*+2\tilde{s}, r}^+ < +\infty$$

*with probability at least  $1 - c_4 \exp(-c_5 \sqrt{m})$  for  $c_4, c_5 > 0$ .*

Assumption 7 asserts that the sparse eigenvalues of the Hessian matrix  $\nabla^2 f(\Sigma)$  are lower and upper bounded when  $\Sigma$  is adequate sparsity and in close proximity to  $\Sigma^*$  with high probability (w.h.p). In other words,  $\nabla^2 f(\Sigma)$  possesses both bounded maximum and non-zero minimum sparse eigenvalues over a local cone.

## B. Assumptions

We present several assumptions about the true covariance matrix  $\Sigma^*$ . Define the support set of  $\Sigma^*$  as  $\mathcal{S}^* = \{(i, j) \mid \Sigma_{ij}^* \neq 0\}$ , with  $s^*$  representing its size, i.e.,  $s^* = |\mathcal{S}^*|$ . In the following, we impose some mild conditions on the  $\Sigma^*$ .

**Assumption 8.** *For the ground truth covariance matrix  $\Sigma^*$ , there exist finite upper and positive lower bounds on its eigenvalues. Specifically, there exists a constant  $\kappa \geq 1$  that satisfies*

$$0 < \frac{1}{\kappa} \leq \lambda_{\min}(\Sigma^*) \leq \lambda_{\max}(\Sigma^*) \leq \kappa < \infty.$$

Assumption 8 is a common premise in the literature concerning the estimation of sparse covariance matrices [37], [48], [65], [77]. This assumption offers several advantages, and readers are encouraged to consult [65] for further details.

**Assumption 9.** *Given the true covariance matrix  $\Sigma^*$ , there exist universal constants  $\alpha, c_7$  such that*

$$\|\Sigma_{\mathcal{S}^*}^*\|_{\min} = \min_{(i,j) \in \mathcal{S}^*} |\Sigma_{ij}^*| \geq (\alpha + c_7) \lambda,$$

*where  $\alpha$  is a constant introduced in Assumption 1, and  $c_7 \in (0, \alpha)$  satisfies  $p'_\lambda(c_7 \lambda) \geq \frac{\lambda}{2}$ .*

Assumption 9 is referred to as the minimum signal strength condition, commonly applied in the analysis of non-convex penalized regression problems [55], [57], [71]. Additionally, it is essential to select  $\lambda$  appropriately to ensure that the regularization is sufficiently large to eliminate irrelevant coordinates, thereby yielding that the solution is adequately sparse.

**Assumption 10.** *There exists a generic constant  $c_8$  such that*

$$\lambda = c_8 \sqrt{\frac{\log d}{m}} \geq 4 (\|\nabla f(\Sigma^*)\|_{\max} + \varepsilon).$$

Assumption 10 sets the tuning parameter  $\lambda$  in the order of  $\sqrt{\frac{\log d}{mn}}$ . It prevents  $\lambda$  from becoming overly large as the number of measurements  $m$  or the number of data  $n$  increases, thereby ensuring that the estimators maintain a close correspondence with the true model parameters.

**Definition 11.** Define a local region around  $\Sigma^*$  as

$$\mathcal{B}(\Sigma^*, r) = \{\Sigma \succ 0 \mid \|\Sigma - \Sigma^*\|_F \leq r\}.$$

In our analysis, we set the radius  $r$  as  $\frac{\rho_{2s^*+2\tilde{s}}^-}{4\tau\kappa}$ . Under Assumption 10, we prove that the solution produced by our algorithm will fall within the local region  $\mathcal{B}\left(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}^-}{4\tau\kappa}\right)$  after a finite number of iterations.

## C. Statistical Guarantees and Consequences

The following presents the main result, which demonstrates the contraction property of the solution path  $\{\tilde{\Sigma}^k\}_{k \geq 1}$ .

**Theorem 12** (Contraction Property). *Consider the estimator in (2) and suppose the Assumptions 4, 7, 8, and 9 hold. Then with*

probability exceeding  $1 - c_4 \exp(-c_5 \sqrt{m})$  for some  $c_4, c_5 > 0$ , the  $\varepsilon$ -optimal solution  $\tilde{\Sigma}^k$  is bounded by:

$$\begin{aligned} \|\tilde{\Sigma}^k - \Sigma^*\|_F &\leq \frac{1}{\rho_{2s^*+2\tilde{s}}} \left( \underbrace{\|(\nabla f(\Sigma^*))_{S^*}\|_F}_{\text{oracle rate}} + \underbrace{\varepsilon \sqrt{s^*}}_{\text{optimization error}} \right) \\ &\quad + \underbrace{\delta \|\tilde{\Sigma}^{k-1} - \Sigma^*\|_F}_{\text{contraction}}, \end{aligned}$$

for  $1 \leq k \leq K$ , where  $\delta \in (0, 1)$  is the contraction factor, provided that  $m = \mathcal{O}((s^* + \tilde{s}) \log^2(d/(s^* + \tilde{s})))$ .

**Remark 13.** The oracle estimator  $\hat{\Sigma}^O$  is defined with prior knowledge of the true support set  $S^*$ , and is given by

$$\hat{\Sigma}^O = \arg \min_{\Sigma} f(\Sigma) \quad \text{s.t. } \Sigma_{\bar{S}^*} = \mathbf{0}.$$

The statistical convergence rate of this oracle estimator, commonly referred to as the oracle rate, provides a benchmark for evaluating the theoretical upper limit of estimator performance when the true support set is known. Specifically, the distance between  $\hat{\Sigma}^O$  and  $\Sigma^*$  is characterized by  $\|\hat{\Sigma}^O - \Sigma^*\|_F \lesssim \|(\nabla f(\Sigma^*))_{S^*}\|_F$ . This inequality indicates that the oracle estimator can theoretically approximate the true parameter matrix  $\Sigma^*$  very closely when the true support sets are known.

Theorem 12 elaborates the estimation error between the  $\varepsilon$ -optimal solution  $\tilde{\Sigma}^k$  and the ground truth  $\Sigma^*$  is constrained by three primary factors: the oracle rate, the optimization error, and a contraction term. The oracle rate represents the error bound achievable under ideal conditions using the optimal strategy; the optimization error reflects the performance gap between the actual optimization algorithm and the optimal strategy; the contraction term indicates the rate at which the error converges to its minimum value as the number of iterations increases. The ensuing results specify the exact statistical convergence rate.

**Corollary 14.** Let  $\mathbf{x}$  be a sub-Gaussian random vector with zero mean and covariance  $\Sigma^*$  and  $\{\mathbf{x}_i\}_{i=1}^n$  be a collection of i.i.d. samples from  $\mathbf{x}$ . Suppose that Assumptions 4, 7, 3, 8, and 9 hold, that is,  $\lambda \asymp \sqrt{\frac{\log d}{mn}}$  and  $\varepsilon \lesssim \sqrt{\frac{1}{mn}}$ . If  $\tau \lesssim \sqrt{\frac{1}{mn}} \|(\Sigma^*)^{-1}\|_{\max}^{-1}$ , then the  $\varepsilon$ -optimal solution  $\tilde{\Sigma}^1$  satisfies

$$\|\tilde{\Sigma}^1 - \Sigma^*\|_F \lesssim \sqrt{\frac{s^* \log d}{mn}}$$

with high probability (w.h.p.).

**Corollary 15.** Let  $\mathbf{x}$  be a sub-Gaussian random vector with zero mean and covariance  $\Sigma^*$  and  $\{\mathbf{x}_i\}_{i=1}^n$  be a collection of i.i.d. samples from  $\mathbf{x}$ . Suppose that Assumptions 4, 7, 3, 8, and 9 hold, that is,  $\lambda \asymp \sqrt{\frac{\log d}{mn}}$  and  $\varepsilon \lesssim \sqrt{\frac{1}{mn}}$ . If  $\tau \lesssim \sqrt{\frac{1}{mn}} \|(\Sigma^*)^{-1}\|_{\max}^{-1}$ , and  $K \gtrsim \log(\lambda \sqrt{mn}) \gtrsim \log \log d$ , then the  $\varepsilon$ -optimal solution  $\tilde{\Sigma}^k$  satisfies

$$\|\tilde{\Sigma}^k - \Sigma^*\|_F = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right),$$

w.h.p.

Corollary 14 and Corollary 15 are direct consequences of Theorem 12. The former addresses the scenario where  $k = 1$  and describes a contraction property resulting from the MM-based multistage convex relaxation algorithm. It demonstrates that to achieve the oracle rate, the optimization error  $\varepsilon$  must be chosen such that  $\varepsilon \leq \frac{\|(\nabla f(\Sigma^*))_{S^*}\|_F}{\sqrt{s^*}}$ , and the parameter  $K$  must be sufficiently large. The latter implies that, with minimal assumptions, solving no more than approximately  $\log \log d$  convex problems is enough to achieve the oracle rate  $\sqrt{\frac{s^*}{mn}}$ .

#### D. Computational Theory

**Lemma 16.** Under Assumptions 7 and 8,  $f(\Sigma)$  is  $(\rho_{2s^*+2\tilde{s}}^- + \frac{16\tau^3}{\kappa^2(4\tau + \rho_{2s^*+2\tilde{s}}^-)^2})$ -strongly convex and  $(\rho_{2s^*+2\tilde{s}}^+ + \frac{16\tau^3\kappa^2}{(4\tau - \rho_{2s^*+2\tilde{s}}^-)^2})$ -smooth in the region  $\mathcal{B}(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}^-}{4\tau\kappa})$ , where  $\kappa$  is defined in Assumption 8,  $\rho_{2s^*+2\tilde{s}}^-$ ,  $\rho_{2s^*+2\tilde{s}}^+$  is defined in Assumption 7 and  $\tau$  is the positive tuning hyper-parameter.

Lemma 16 ensures that within the specified region, the function exhibits strong convexity and smoothness properties, which are essential for stability and convergence analysis. Next, we present the explicit iteration complexity of the proposed algorithm. We begin by characterizing the convergence for the first stage.

**Theorem 17.** For  $k = 1$ , suppose Assumption 4, 7, 3, 8, and 9 hold. After a sufficient number of iterations  $T < \infty$ , for all  $t \geq T$ , the following conditions are satisfied:  $\|\Sigma_t^1 - \Sigma^*\|_F \leq r$ ,  $\|(\Sigma_t^1)_{\bar{S}^*}\|_0 \leq \tilde{s}$  and

$$\|\Sigma_{t+1}^1 - \hat{\Sigma}^1\|_F \leq \frac{\tau\kappa^3}{\rho_{2s^*+2\tilde{s}}^-} \|\Sigma_t^1 - \hat{\Sigma}^1\|_F^2.$$

Moreover, we need at most  $T + \log \log(\frac{3\rho_{2s^*+2\tilde{s}}^+}{\varepsilon})$  iterations to terminate the proximal Newton algorithm.

**Theorem 18.** For  $k \geq 2$ , suppose 4, 7, 3, 8, and 9 hold. For all iterations  $t$ , we have  $\|\Sigma_t^k - \Sigma^*\|_F \leq r$ ,  $\|(\Sigma_t^k)_{\bar{S}^*}\|_0 \leq \tilde{s}$ , which guarantees  $\beta = 1$ , and

$$\|\Sigma_{t+1}^k - \hat{\Sigma}^k\|_F \leq \frac{\tau\kappa^3}{\rho_{2s^*+2\tilde{s}}^-} \|\Sigma_t^k - \hat{\Sigma}^k\|_F^2.$$

Moreover, the proximal Newton algorithm requires at most  $\log \log(\frac{3\rho_{2s^*+2\tilde{s}}^+}{\varepsilon})$  iterations to terminate.

Theorem 17 demonstrates that the solution enters the ball  $\mathcal{B}(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}^-}{4\tau\kappa})$  following the initial stage. To satisfy the approximate KKT conditions, the number of iterations is capped at  $\mathcal{O}(T + \log \log(\frac{3\rho_{2s^*+2\tilde{s}}^+}{\varepsilon}))$ . The resulting solution  $\tilde{\Sigma}^1$  exhibits notable qualities, laying the foundation for improved computational efficiency in the subsequent stages of our proximal Newton algorithm. Theorem 18 states that the sparsity of the solution is preserved throughout the iterations, and the solution remains within  $\mathcal{B}(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}^-}{4\tau\kappa})$ . Due to the sparsity of the solution, the algorithm exhibits quadratic convergence. Moreover, the algorithm achieves a logarithmic number of

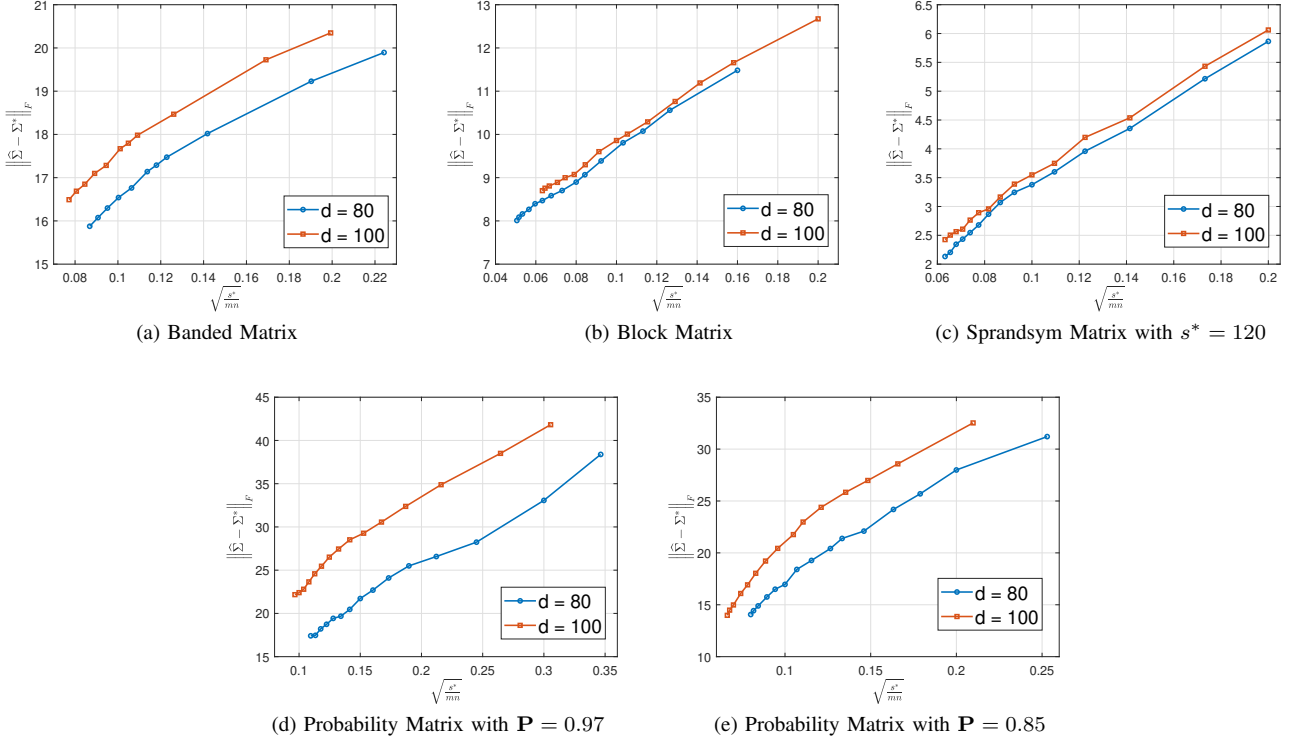


Fig. 1. The oracle rate for different dimensions of different covariance models<sup>2</sup>.

iterations,  $\mathcal{O}(\log \log(\frac{3\rho_{2s^*+2\tilde{s}}^+}{\epsilon}))$ , to satisfy the approximate KKT condition.

## V. NUMERICAL EXPERIMENTS

In this section, we examine the practical recovery performance of the proposed estimator and provide numerical results for the proposed algorithm for sparse CCS. Section V-A describes the generation of synthetic data and the measurement criteria used in the paper. In Section V-B and Section V-C, the statistical theoretical results of Section IV-C and the computational theoretical results of Section IV-D are demonstrated, respectively. Section V-D compares the relative estimation error of our proposed estimator and existing  $\ell_1$  estimator [40]. The non-convex penalty function chosen for these experiments is the MCP, with a constant setting of  $b = 2$  across all trials. The selection of the tuning parameters  $\lambda$  and  $\tau$  is determined by applying five-fold cross-validation. All methods are implemented in MATLAB and run on an Intel i7 – 10700 2.90 GHz  $\times 8$  with 16 GB of RAM. All results are averaged on 100 Monte Carlo realizations.

### A. Synthetic Datasets and Measurement Criteria

For different fixed  $d$ , we generate  $n$  independent data points by sampling i.i.d. from a multivariate normal distribution  $\mathcal{N}(0, \Sigma^*)$ . In this paper, we consider three typical covariance models and two additional models with random sparsity. The details of each model are as follows:

- Banded matrix:

$$\Sigma_{ij}^* = \begin{cases} 1 - \frac{|i-j|}{10}, & |i-j| \leq 10, \\ 0, & \text{otherwise.} \end{cases}$$

- Block matrix: The indices  $1, 2, \dots, d$  are partitioned into 10 ordered groups of equal size with

$$\Sigma_{ij}^* = \begin{cases} 1, & i = j, \\ 0.5, & i \text{ and } j (i \neq j) \text{ belong to the same group,} \\ 0, & \text{otherwise.} \end{cases}$$

- Toeplitz matrix:  $\Sigma_{ij}^* = 0.75^{|i-j|}$ .
- Spandsym matrix:  $\Sigma^*$  is generated using the “spandsym” built-in function in MATLAB with a sparsity controlled by the parameter  $s^*$ .
- Probability matrix:  $\Sigma^*$  is generated with off-diagonal entries drawn i.i.d. from a uniform  $(-1, 1)$  distribution. These entries are set to zero with a fixed probability  $\mathbf{P}$ . Finally, a multiple of the identity was added to the resulting matrix.

The first two models are sparse, while the third one is approximately sparse. The fourth model is positive, symmetric, and sparse when  $s^*$  is small. The fifth model is positive definite, well-conditioned, and can be sparse when  $\mathbf{P}$  is appropriate (in this case, either  $\mathbf{P} = 0.97$  or  $\mathbf{P} = 0.85$  to simulate a very sparse and a somewhat sparse model).

The estimation performance is measured by the absolute error and relative error under both the Frobenius norm and the spectral norm, respectively. Specifically, the Frobenius

<sup>2</sup>The Toeplitz model is not reported due to the fact that it is not sparse but approximately sparse.

absolute error (FAE) is defined as  $\|\hat{\Sigma} - \Sigma^*\|_F$ , where  $\hat{\Sigma}^3$  is the estimated covariance matrix and  $\Sigma^*$  is the true value. Similarly, the Frobenius relative error (FRE) is defined as  $\frac{\|\hat{\Sigma} - \Sigma^*\|_F}{\|\Sigma^*\|_F}$ . The selection performance is examined by the false positive rate (FPR) and the true positive rate (TPR), defined as, respectively

$$\text{FPR} = \frac{\#\{(i, j) : \hat{\Sigma}_{ij} \neq 0 \text{ \& } \Sigma_{ij}^* = 0\}}{\#\{(i, j) : \Sigma_{ij}^* = 0\}},$$

$$\text{TPR} = \frac{\#\{(i, j) : \hat{\Sigma}_{ij} \neq 0 \text{ \& } \Sigma_{ij}^* \neq 0\}}{\#\{(i, j) : \Sigma_{ij}^* \neq 0\}}.$$

### B. Demonstration of Oracle Rate

The oracle rate derived for the proposed estimator and algorithm in Section IV-C is shown to be dependent on the sparsity level  $s^*$ , the number of measurements  $m$ , and the number of samples  $n$ . To demonstrate this theoretical result, we apply our proposed algorithm to synthetic datasets, as described in Section V-A. Figure 1 illustrates the estimation performance across various matrix configurations differentiated by dimensions  $d$ , sparsity level  $s^*$ , measurement count  $m$ , and number of samples  $n$ . The  $x$ -axis is  $\sqrt{\frac{s^*}{mn}}$ , and the  $y$ -axis represents FAE. Two distinct lines depict the FAE for dimensions  $d = \{80, 100\}$ . We observe that an increase in measurement size  $m$  and sample count  $n$  corresponds to a reduction in FAE, visually demonstrating how the estimation quality of the covariance matrix varies with the oracle rate. According to Corollary 15, the Frobenius norm difference  $\|\hat{\Sigma} - \Sigma^*\|_F$  converges to the statistical errors as the stage index  $k$  increases, following the order of  $\mathcal{O}\left(\sqrt{\frac{s^*}{mn}}\right)$ . We can see that the estimation errors grow approximately linearly with the theoretical rate, which validates our theoretical guarantee.

### C. Demonstration of Computational Rate

Figure 2 illustrates a detailed analysis of the convergence rates observed at each stage of the computational process. In the initial phase, the algorithm exhibits sub-linear convergence, attributed to the absence of strong convexity in the optimization problem. As the computation progresses and the solution approaches the sparse region, the convergence dynamics undergo a significant transformation. Within this region, feasible solutions become increasingly sparse, and the objective function effectively reduces to a low-dimensional form. This reduced form is characterized by strong convexity and smoothness properties that are crucial for accelerated convergence. Consequently, the algorithm achieves a quadratic convergence rate for  $k \geq 2$ . Notably, even for  $k = 1$ , where the convergence rate is initially sub-linear, the algorithm transitions into the contraction region and subsequently attains a quadratic convergence rate.

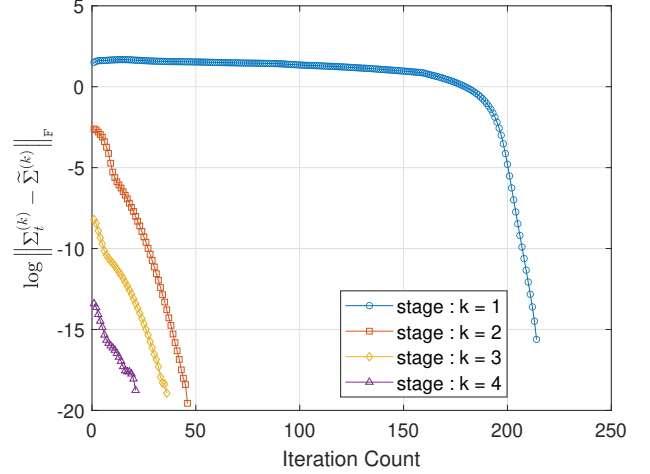


Fig. 2. Computational rate of convergence in each stage for simulation experiment. The  $x$ -axis is the iteration count  $t$  within the  $k$ -th subproblem. A distinct phase transition in the algorithmic convergence rate from sub-linear to super-linear is observable as the iterations progress into the sparse region.

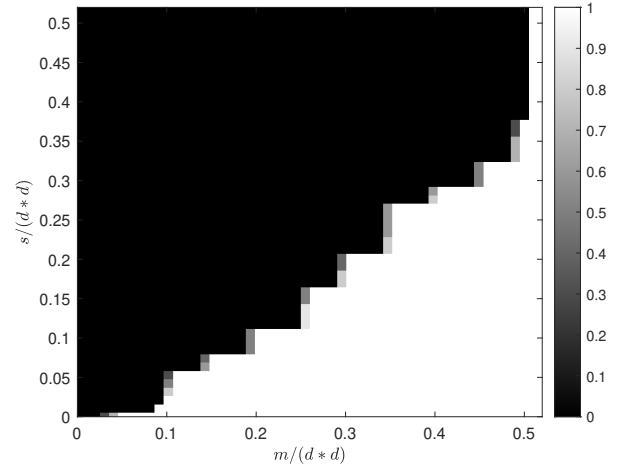


Fig. 3. The rate of successful covariance reconstruction when  $d = 100$ .

### D. Simulation Experiment Comparisons

We conduct a series of Monte Carlo trials to evaluate the performance of our proposed CCS algorithm using “Sprandsym matrix model” with  $d = 100$  and  $n = 50$ . The efficacy of the algorithm is analyzed under varying sparsity levels and noise conditions, and comparisons are made against existing estimators.

To evaluate the empirical probability of successful recovery, we use a color-coded matrix representation for each cell in Figure 3. To mitigate the influence of sample size  $n$ , we directly sense the true covariance matrix  $\Sigma^*$ . A recovery is considered successful if the solution  $\hat{\Sigma}$  satisfies  $\frac{\|\hat{\Sigma} - \Sigma^*\|_F}{\|\Sigma^*\|_F} \leq 10^{-3}$ .

Next, we examine the FRE of the estimated covariance matrices across different sparsity levels

<sup>3</sup>In the whole Section V, we use  $\hat{\Sigma}$  to denote a generic covariance estimator, which can be the estimation results obtained by various algorithms and estimators.



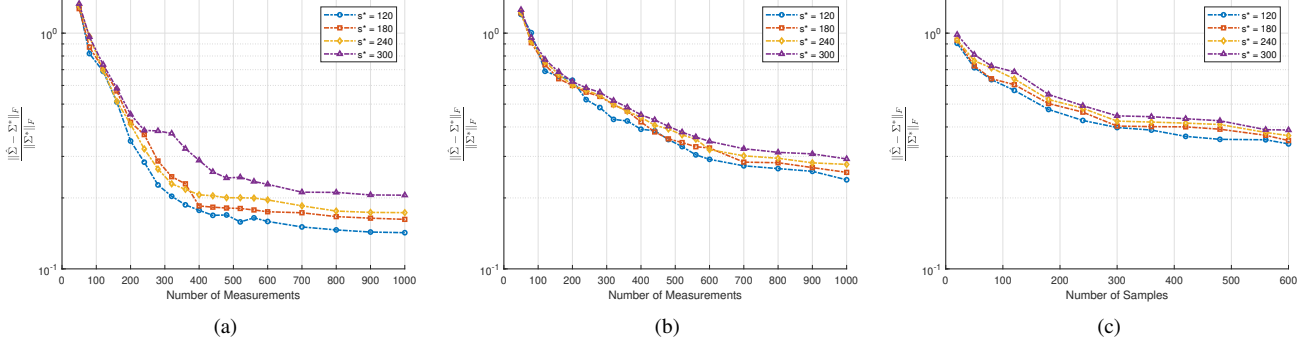


Fig. 4. The FRE of the estimated covariance matrices is examined in two contexts: (a) the true covariance under varying levels of sparsity in the absence of noise; (b) the sample covariance under different sparsity levels when subjected to a noise parameter of  $\gamma = 0.1$  and  $n = 50$ ; (c) the sample covariance under different sparsity levels when subjected to a noise parameter of  $\gamma = 0.1$  and  $m = 300$ ;

( $s^* \in \{120, 180, 240, 300\}$ ). Results are depicted in Figure 4a, obtained in the absence of noise and while sensing the true covariance  $\Sigma^*$  with a fixed sample count  $n = 50$ . These results illustrate an inverse relationship between FRE and the number of measurements  $m$ , where an increase in  $m$  correlates with enhanced estimation accuracy. Additionally, for a fixed  $m$ , a lower  $s^*$  value, which corresponds to higher sparsity, results in a reduced FRE, highlighting the influence of sparsity on estimation precision.

To simulate more realistic conditions, we introduce additive noise into each measurement. The noise  $\eta_i$  is modeled as a sub-exponential random variable scaled by  $\gamma$ , specifically  $\gamma \cdot \mathcal{M}(0, 1)$ . The influence of noise, particularly at  $\gamma = 10^{-1}$ , on the average relative error is illustrated in Figure 4b. Despite the presence of noise, the proposed estimator demonstrates resilience, maintaining consistent trends in FRE across varying measurements and sparsity levels.

Further, we examine the variations in FRE with respect to the sample size, as shown in Figure 4c, with the number of measurements fixed at  $m = 300$ . To investigate the impact of the noise intensity, we analyze the performance of the estimator for  $s^* = 300$  under different noise levels, as depicted in Figure 5. The results reveal that at lower  $m$ , noise intensity marginally affects recovery accuracy. However, as  $m$  increases, the impact of noise diminishes, leading to improved recovery precision. This underscores the importance of sufficient data in mitigating noise effects and enhancing estimation reliability.

Finally, we perform a comparative analysis between our proposed estimator and an existing  $\ell_1$ -norm-based estimator [40] under a consistent noise level ( $\gamma = 10^{-1}$ ). The comparative results, shown in Figure 6, demonstrate that the proposed estimator, leveraging a non-convex penalty, consistently outperforms the  $\ell_1$ -norm-based estimator in terms of Frobenius norm error. These findings validate our theoretical claim that employing a non-convex penalty can significantly enhance accuracy in CCS.

## VI. DISCUSSION

### A. Bilinear Rank-one Measurement Model

Our research provides compelling evidence that covariance matrices can be effectively reconstructed from a limited

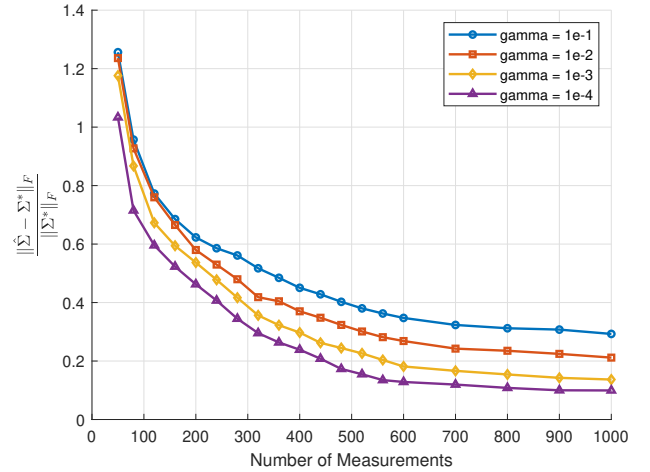


Fig. 5. The FRE of the estimated covariance matrices for different noise levels when  $s^* = 300$ .

number of quadratic measurements, provided the underlying structure exhibits sparsity. Furthermore, we extend our analysis to the bilinear rank-one measurement model, where measurements are expressed as  $y_i = \mathbf{a}_i^\top \Sigma \mathbf{b}_i$ , with the sensing vectors  $\mathbf{a}_i$  and  $\mathbf{b}_i$  being independently generated. This extension validates our findings in the context of an asymmetric sensing framework, which holds significant potential for applications in diverse fields, including communication [78], imaging [79], and machine learning [80].

The bilinear model introduces additional complexity due to the interaction between two independent sensing vectors,  $\mathbf{a}_i$  and  $\mathbf{b}_i$ . Nonetheless, it is straightforward to demonstrate that this configuration satisfies Assumption 8, a critical criterion for ensuring a successful recovery. The derivation closely follows that of the quadratic model, with appropriate adaptations to address the bilinear structure. Notably, we establish the Hessian matrix associated with the bilinear model retains symmetry, thereby preserving the computational efficiency and efficacy of our proposed estimator. Consequently, the bilinear measurement model inherits the oracle property, further underscoring the robustness and broad applicability of our approach.

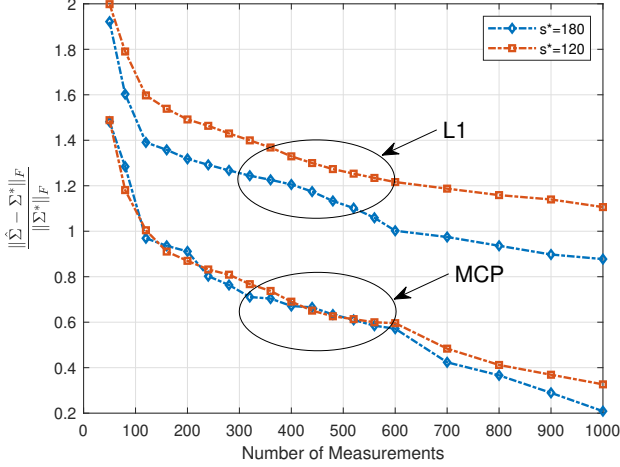


Fig. 6. The FRE of the estimated covariance matrices for different sparsity levels with noise ( $\ell_1$  v.s. MCP).

### B. CCS for variable with non-zero mean

In the preceding discussion, we consider the scenario in which the mean signal vector is assumed to be zero. However, this assumption frequently does not hold true in numerous practical applications. When the mean vector is unknown, conventional methods often involve first estimating the mean vector using robust techniques, followed by a subsequent estimation of the covariance matrix based on this mean estimate. This two-step procedure introduces additional tuning parameters and increases both statistical variability and computational complexity.

To address these challenges, we propose an end-to-end methodology that leverages a pairwise difference approach, thereby eliminating the need for direct mean estimation. Specifically, we define  $N := n(n-1)/2$  and consider the pairwise differences

$$\{\tilde{\mathbf{x}}_1, \tilde{\mathbf{x}}_2, \dots, \tilde{\mathbf{x}}_N\} = \{\mathbf{x}_1 - \mathbf{x}_2, \mathbf{x}_2 - \mathbf{x}_3, \dots, \mathbf{x}_{n-1} - \mathbf{x}_n\},$$

which are identically distributed with zero mean. The covariance matrix of these differences is given by  $\text{cov}(\tilde{\mathbf{x}}) = 2\Sigma$ , which is a U-statistic. While the original sample covariance is calculated as  $\Sigma_{\text{sam}} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$ , where  $\bar{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i$ , the pairwise difference approach allows us to equivalently express it as

$$\Sigma_{\text{sam}} = \frac{1}{N} \sum_{i=1}^N \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^\top / 2.$$

This reformulation simplifies the estimation process by obviating the direct calculation of the mean, thereby reducing computational complexity and mitigating statistical errors associated with separate mean and covariance estimations.

## VII. CONCLUSION

In this paper, we conduct a detailed investigation into high-dimensional sparse covariance matrix estimation within the framework of the quadratic measurements model. This model

is relevant for scenarios with stringent processing and memory constraints, such as real-time data acquisition systems. Our results establish that sparse covariance matrices can be reconstructed with high accuracy using a minimal number of quadratic measurements, requiring significantly reduced storage requirements. We provide rigorous theoretical analysis and empirical validations to support these conclusions. Notably, our proposed estimators exhibit superior statistical convergence rates compared to existing methods, highlighting their practical efficacy and potential in real-world applications.

## APPENDIX A

### RESTRICTED EIGENVALUE & SPARSE EIGENVALUE

We first introduce an equivalent definition of definition 8.

**Definition 19.** For three positive integers  $s_0, \vartheta, r$ , the largest and smallest localized restricted eigenvalues are defined as

$$\psi_{s_0, \vartheta, r}^+ = \sup \left\{ \frac{\mathbf{u}^\top \nabla^2 f(\Sigma) \mathbf{u}}{\mathbf{u}^\top \mathbf{u}} \mid (\mathbf{u}, \Sigma) \in \mathcal{I}(s_0, \vartheta, r) \right\};$$

$$\psi_{s_0, \vartheta, r}^- = \inf \left\{ \frac{\mathbf{u}^\top \nabla^2 f(\Sigma) \mathbf{u}}{\mathbf{u}^\top \mathbf{u}} \mid (\mathbf{u}, \Sigma) \in \mathcal{I}(s_0, \vartheta, r) \right\},$$

where  $\mathcal{I}(s_0, \vartheta, r)$  is defined as

$$\{(\mathbf{u}, \Sigma) \mid \mathcal{S}^* \subseteq \mathcal{J}, |\mathcal{J}| \leq s_0, \|\mathbf{u}_{\mathcal{J}}\|_1 \leq \vartheta \|\mathbf{u}_{\mathcal{J}^c}\|_1, \|\Sigma - \Sigma^*\|_F \leq r\},$$

which represents a local  $\ell_1$  cone.

We now proceed to introduce the relationships between SE and RE.

**Proposition 20.** For any  $\Sigma \in \mathcal{I}(s, \vartheta, r) \cap \mathcal{B}(\Sigma^*, r)$ , the following inequalities hold:

$$c_1 \psi_{s, \vartheta, r}^- \leq \rho_s^- \leq c_2 \psi_{s, \vartheta, r}^-,$$

$$c_3 \psi_{s, \vartheta, r}^+ \leq \rho_s^+ \leq c_4 \psi_{s, \vartheta, r}^+,$$

where  $c_1, c_2, c_3$  and  $c_4$  are constant.

For a detailed proof of Proposition 20, readers are referred to the foundational work in [68], which provides comprehensive mathematical derivations and discussions. This reference is omitted here for brevity.

## APPENDIX B

### COMPUTING THE NEWTON DIRECTION WITH ACTIVE SET STRATEGY

In this section, we detail the computation of the Newton Direction as applied within our optimization framework. For clarity, we omit the stage index  $k$  and focus on a single Newton iteration. The gradient and Hessian for  $f(\Sigma)$  are given by

$$\nabla f(\Sigma) = -\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma)) - \tau \Sigma^{-1},$$

and

$$\nabla^2 f(\Sigma) = \frac{1}{m} \sum_{i=1}^m (\mathbf{A}_i \otimes \mathbf{A}_i) + \tau (\Sigma^{-1} \otimes \Sigma^{-1}).$$

The direct computation of the Hessian matrix involves significant computational overhead, with a complexity of  $\mathcal{O}(md^4)$ , making it impractical for high-dimensional data.

However, the inherent symmetric structure of the Hessian enables certain computational optimizations. By utilizing a coordinate descent approach, it is possible to update all variables in  $\mathcal{O}(md^3)$  time. This methodology has demonstrated efficacy in addressing Lasso-type problems [81]–[83] with theoretical justifications [84].

In order to provide a more precise articulation of the issue, we reformulate (4) as follows:

$$\begin{aligned} \bar{f}(\Delta) &= f(\Sigma_t) + \text{tr}\left((\mathbf{Q} - \tau\mathbf{W})^\top \Delta\right) \\ &+ \frac{1}{2} \left( \frac{1}{m} \sum_{i=1}^m \text{tr}(\mathbf{A}_i \Delta \mathbf{A}_i \Delta) + \tau \text{tr}(\mathbf{W} \Delta \mathbf{W} \Delta) \right), \end{aligned} \quad (9)$$

where  $\mathbf{Q} = -\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma))$  and  $\mathbf{W} = \Sigma^{-1}$ .

In refining the Newton direction, we represent the current approximation as  $\mathbf{D}$  and the updated direction as  $\mathbf{D}'$ . For the coordinate descent update of the variable  $\Sigma_{ij}$  (assuming  $i < j$ ), the update is constructed to preserve symmetry as follows:

$$\mathbf{D}' = \mathbf{D} + \mu(e_i e_j^\top + e_j e_i^\top),$$

where  $\mu$  is the scalar update magnitude, and  $e_i, e_j$  are the standard basis vectors. The corresponding one-dimensional optimization problem associated with the equation for (5) is expressed as follows:

$$\arg\min_{\mu} \bar{f}(\mathbf{D} + \mu(e_i e_j^\top + e_j e_i^\top)) + 2\Lambda_{ij} |\Sigma_{ij} + D_{ij} + \mu|.$$

After substituting  $\mathbf{D}' = \mathbf{D} + \mu(e_i e_j^\top + e_j e_i^\top)$  for  $\Delta$  in (9), the function  $\bar{f}$  is refined to focus exclusively on terms dependent on  $\mu$ , with other terms omitted. The contributions to the function from three terms are:

- The linear term  $\text{tr}((\mathbf{Q} - \tau\mathbf{W})^\top \Delta)$  contributes  $2\mu(Q_{ij} - \tau W_{ij})$ ,
- The penalty term gives  $2\Lambda_{ij} |\Sigma_{ij} + D_{ij} + \mu|$ ,
- The quadratic term yields

$$\begin{aligned} &\frac{1}{m} \sum_{l=1}^m 4\mu (\mathbf{A}_l)_{\cdot i}^\top \mathbf{D} (\mathbf{A}_l)_{\cdot j} + 2\mu^2 \tau (W_{ij}^2 + W_{ii} W_{jj}) \\ &+ \frac{1}{m} \sum_{l=1}^m 2\mu^2 \left( (\mathbf{A}_l)_{ij}^2 + (\mathbf{A}_l)_{ii} (\mathbf{A}_l)_{jj} \right) + 4\mu \tau \mathbf{W}_{\cdot i}^\top \mathbf{D} \mathbf{W}_{\cdot j}. \end{aligned}$$

Combining all contributions yields a quadratic function of  $\mu$ , expressed as (10) and the closed-form solution for  $\mu$  is given by

$$\mu = -e + \mathcal{T}(e - g/h, \Lambda_{ij}/h), \quad (11)$$

where  $\mathcal{T}(u, v) = \text{sign}(u) \cdot \max\{|u| - v, 0\}$  is the soft-thresholding function, and

$$\begin{aligned} h &= \tau (W_{ij}^2 + W_{ii} W_{jj}) + \frac{\sum_{l=1}^m \left( (\mathbf{A}_l)_{ij}^2 + (\mathbf{A}_l)_{ii} (\mathbf{A}_l)_{jj} \right)}{m}, \\ g &= Q_{ij} - \tau W_{ij} + \tau \mathbf{W}_{\cdot i}^\top \mathbf{D} \mathbf{W}_{\cdot j} + \frac{\sum_{l=1}^m (\mathbf{A}_l)_{\cdot i}^\top \mathbf{D} (\mathbf{A}_l)_{\cdot j}}{m}, \\ e &= \Sigma_{ij} + D_{ij}. \end{aligned}$$

When  $i = j$ , the substitution of  $\mathbf{D}' = \mathbf{D} + \mu e_i e_i^\top$  yields the update formula for  $D_{ii}$  as derived from (11). The components of the update formula are defined as follows:

$$\begin{aligned} h &= \tau D_{ii}^2 + \frac{\sum_{l=1}^m (\mathbf{A}_l)_{ii}^2}{m}, \\ g &= Q_{ii} - \tau W_{ii} + \tau \mathbf{W}_{\cdot i}^\top \mathbf{D} \mathbf{W}_{\cdot i} + \frac{\sum_{l=1}^m (\mathbf{A}_l)_{\cdot i}^\top \mathbf{D} (\mathbf{A}_l)_{\cdot i}}{m}, \\ e &= \Sigma_{ii} + D_{ii}. \end{aligned}$$

The primary computational challenge arises in evaluating the expressions  $\mathbf{W}_{\cdot i}^\top \mathbf{D} \mathbf{W}_{\cdot j}$  and  $\sum_{l=1}^m (\mathbf{A}_l)_{\cdot i}^\top \mathbf{D} (\mathbf{A}_l)_{\cdot j}$ . To alleviate the computational burden, which typically necessitates  $\mathcal{O}(md^2)$  time, we introduce intermediary  $d \times d$  matrices defined as  $\mathbf{U}_l = \mathbf{D} \mathbf{A}_l$ ,  $\mathbf{V} = \mathbf{D} \mathbf{W}$ . This strategy facilitates the computation of  $\mathbf{W}_{\cdot i}^\top \mathbf{V}_{\cdot j}$  and  $\mathbf{W}_{\cdot j}^\top (\mathbf{A}_l)_{\cdot i}$  using  $\mathcal{O}(d)$  floating-point operations. Furthermore, the maintenance  $\mathbf{U}_l$  and  $\mathbf{V}$  requires the adjustment of  $\mathcal{O}(d)$  elements. For a comprehensive description of the methodology, please consult Algorithm 3.

#### A. Updating Only a Subset of Variables

We introduce an algorithm that leverages the active set update strategy to exploit the solution sparsity, thereby accelerating computation. The strategy involves two consecutive nested loops. In the outer loop, all coordinates are categorized into two sets based on a heuristic coordinate gradient thresholding rule (strong rule, [85]): the free set (active coordinates) and the fixed set (inactive coordinates). Subsequently, each iteration of the outer loop initiates an inner loop that performs coordinate optimization exclusively on the active set until convergence, while the inactive coordinates remain fixed at zero. Post each inner loop iteration, the algorithm employs a heuristic rule to reassess and update the active set, aiming to further reduce the objective value. This iterative process continues until the composition of the active set stabilizes across subsequent outer loop iterations. The efficacy of the active set strategy is well-documented in practical applications [82] and is underpinned by strong theoretical foundations [86].

$$\begin{aligned} &\frac{1}{2} \left( \frac{\sum_{l=1}^m \left( (\mathbf{A}_l)_{ij}^2 + (\mathbf{A}_l)_{ii} (\mathbf{A}_l)_{jj} \right)}{m} + \tau (W_{ij}^2 + W_{ii} W_{jj}) \right) \mu^2 + \\ &\left( Q_{ij} - \tau W_{ij} + \tau \mathbf{W}_{\cdot i}^\top \mathbf{D} \mathbf{W}_{\cdot j} + \frac{\sum_{l=1}^m (\mathbf{A}_l)_{\cdot i}^\top \mathbf{D} (\mathbf{A}_l)_{\cdot j}}{m} \right) \mu + 2\Lambda_{ij} |\Sigma_{ij} + D_{ij} + \mu| \end{aligned} \quad (10)$$

$$\hat{D}_{ij} = \begin{cases} \mathcal{T} \left( -\frac{Q_{ij}}{\tau W_{ii} W_{jj} + \sum_{l=1}^m ((\mathbf{A}_l)_{ij}^2 + (\mathbf{A}_l)_{ii} (\mathbf{A}_l)_{jj})}, \frac{\Lambda_{ij}}{\tau W_{ii} W_{jj} + \sum_{l=1}^m ((\mathbf{A}_l)_{ij}^2 + (\mathbf{A}_l)_{ii} (\mathbf{A}_l)_{jj})} \right) & \text{if } i \neq j, \\ -\Sigma_{ii} + \mathcal{T} \left( \Sigma_{ii} - \frac{Q_{ii} - \tau W_{ii}}{\tau W_{ii}^2 + \sum_{l=1}^m (\mathbf{A}_l)_{ii}^2}, \frac{\Lambda_{ii}}{\tau W_{ii}^2 + \sum_{l=1}^m (\mathbf{A}_l)_{ii}^2} \right) & \text{if } i = j. \end{cases} \quad (12)$$

**Algorithm 3:** Coordinate Descent Algorithms in Conjunction with the active set strategy.

---

**Input:**  $\nabla f(\Sigma)$ ,  $\Lambda$ ,  $\Sigma$ ,  $\{a_l\}_{l=1}^m$   
**1 Initialize**  $\mathbf{W} = \Sigma^{-1}$ ,  $\mathbf{A}_l = a_l a_l^\top$ ,  $\mathbf{D} = \mathbf{0}$ ,  $\mathbf{U}_l = \mathbf{0}$ ,  $\mathbf{V} = \mathbf{0}$   
**2 while not converged do**  
    // Partition the variables into fixed and free sets  
**3**  $\mathcal{C}_{\text{fixed}} = \{(i, j) \mid |\nabla f_{ij}(\Sigma)| \leq \Lambda_{ij} \text{ and } \Sigma_{ij} = 0\}$   
**4**  $\mathcal{C}_{\text{free}} = \{(i, j) \mid |\nabla f_{ij}(\Sigma)| > \Lambda_{ij} \text{ and } \Sigma_{ij} \neq 0\}$   
**5 for**  $(i, j) \in \mathcal{C}_{\text{free}}$  **do**  
**6**  $h =$   
     $\tau (W_{ij}^2 + W_{ii} W_{jj}) + \frac{\sum_{l=1}^m ((\mathbf{A}_l)_{ij}^2 + (\mathbf{A}_l)_{ii} (\mathbf{A}_l)_{jj})}{m}$   
**7**  $g = \nabla_{ij} f(\Sigma) + \tau \mathbf{W}_{\cdot i}^\top \mathbf{V}_{\cdot j} + \frac{\sum_{l=1}^m (\mathbf{A}_l)_{\cdot i}^\top (\mathbf{U}_l)_{\cdot j}}{m}$   
**8**  $e = \Sigma_{ij} + D_{ij}$   
**9**  $\mu = -e + \mathcal{T}(e - g/h, \Lambda_{jk}/h)$   
**10**  $D_{ij} = D_{ij} + \mu$   
**11**  $(\mathbf{U}_l)_{i \cdot} = (\mathbf{U}_l)_{i \cdot} + \mu (\mathbf{A}_l)_{j \cdot}$   
**12**  $(\mathbf{U}_l)_{j \cdot} = (\mathbf{U}_l)_{j \cdot} + \mu (\mathbf{A}_l)_{i \cdot}$   
**13**  $\mathbf{V}_{i \cdot} = \mathbf{V}_{i \cdot} + \mu \mathbf{W}_{j \cdot}$ ,  $\mathbf{V}_{j \cdot} = \mathbf{V}_{j \cdot} + \mu \mathbf{W}_{i \cdot}$   
**14 end**  
**15 end**  
**Output:**  $\mathbf{D}$

---

The free set and fixed set are respectively defined as follows:

$$\begin{aligned} \Sigma_{ij} \in \mathcal{C}_{\text{fixed}} & \text{ if } |\nabla f_{ij}(\Sigma)| \leq \Lambda_{ij} \text{ and } \Sigma_{ij} = 0, \\ \Sigma_{ij} \in \mathcal{C}_{\text{free}} & \text{ if } |\nabla f_{ij}(\Sigma)| > \Lambda_{ij} \text{ and } \Sigma_{ij} \neq 0. \end{aligned}$$

Further, we ascertain that a Newton update confined to the variables within the fixed set does not alter any coordinates in that set. For a detailed discussion on this analysis, we refer readers to the work by [82], [87].

*Remark 21.* The computation of the Newton direction is streamlined when  $\Sigma$  is diagonal, which is encountered in the initial Newton iteration, wherein the Algorithm is started with an identity matrix. This simple scenario has a succinct closed-form optimal solution presented in (12). On this occasion, it only takes  $\mathcal{O}1$  time to determine each variable, so the time complexity for solving the first Newton direction drops sharply to  $\mathcal{O}(md^2)$ . The core cause of this occurrence lies in the fact that, in this case, the Hessian matrix is diagonal, indicating that each one-variable sub-problem is detached from the others. This means that updating each variable individually once suffices to achieve the optimal solution.

## APPENDIX C

### PROOF OF STATISTICAL THEORY

In this appendix, we provide the proofs of all the statistical theoretical results in Section IV-C. In Subsection C-A, we

begin by Lemma 22 that connects the SE condition to the localized version of the sparse strong convexity/sparse strong smoothness. Subsequently, we introduce several preliminary lemmas that support Lemma 24. Following this, we present Lemma 25, which sets bounds on the estimation error for the general problem (??), followed by Lemma 26 that outlines the estimation error limited for approximate solutions derived using the MM-based algorithm. Subsection C-B elaborates on Theorem 12, leveraging Lemma 25 and Lemma 26 to illustrate the solution path's contraction characteristic. Subsection C-C focuses on key concentration inequalities, which are vital for deducing the explicit statistical convergence rate. In Subsections C-D and C-E, the document presents the proofs for Corollary 14 and Corollary 15, respectively.

#### A. Technical Lemmata

**Lemma 22.** Let  $f(\cdot)$  be a convex differentiable function, and let  $G_f(\Sigma_1, \Sigma_2)$  be defined as

$$G_f(\Sigma_1, \Sigma_2) = f(\Sigma_1) - f(\Sigma_2) - \langle \nabla f(\Sigma_2), \Sigma_1 - \Sigma_2 \rangle.$$

Define the symmetrized Bregman divergence for  $f(\cdot)$  as

$$\begin{aligned} G_f^s(\Sigma_1, \Sigma_2) &= G_f(\Sigma_1, \Sigma_2) + G_f(\Sigma_2, \Sigma_1) \\ &= \langle \nabla f(\Sigma_1) - \nabla f(\Sigma_2), \Sigma_1 - \Sigma_2 \rangle. \end{aligned}$$

For any  $\Sigma_1, \Sigma_2 \in \mathcal{B}(\Sigma^*, r)$  such that  $\max \{ \|(\Sigma_1)_{\bar{\mathcal{S}}^*}\|_0, \|(\Sigma_2)_{\bar{\mathcal{S}}^*}\|_0 \} \leq \tilde{s}$ , the following holds:

$$\begin{aligned} \frac{1}{2} \rho_{2s^*+2\tilde{s}}^- \|\Sigma_1 - \Sigma_2\|_2^2 &\leq G_f(\Sigma_1, \Sigma_2) \\ &\leq \frac{1}{2} \rho_{2s^*+2\tilde{s}}^+ \|\Sigma_1 - \Sigma_2\|_2^2, \end{aligned}$$

$$\begin{aligned} \rho_{2s^*+2\tilde{s}}^- \|\Sigma_1 - \Sigma_2\|_2^2 &\leq G_f^s(\Sigma_1, \Sigma_2) \\ &\leq \rho_{2s^*+2\tilde{s}}^+ \|\Sigma_1 - \Sigma_2\|_2^2. \end{aligned}$$

*Proof.* According to the mean value theorem and the convexity of the local region  $\mathcal{B}(\Sigma^*, r)$ , there exist parameters  $\theta_1, \theta_2, \theta_3 \in [0, 1]$  such that

$$\begin{aligned} \tilde{\Sigma} &= \theta_1 \Sigma_1 + (1 - \theta_1) \Sigma_2 \in \mathcal{B}(\Sigma^*, r), \\ \tilde{\Sigma}' &= \theta_2 \tilde{\Sigma} + (1 - \theta_2) \Sigma_2 \in \mathcal{B}(\Sigma^*, r), \\ \tilde{\Sigma}'' &= \theta_3 \Sigma_1 + (1 - \theta_3) \Sigma_2 \in \mathcal{B}(\Sigma^*, r), \end{aligned}$$

and satisfies  $\max \{ \|\tilde{\Sigma}\|_0, \|\tilde{\Sigma}'\|_0 \} \leq 2\tilde{s}$ . The function  $G_f(\Sigma_1, \Sigma_2)$  and the symmetrized Bregman divergence can be articulated as follows:

$$\begin{aligned} f(\Sigma_1) - f(\Sigma_2) - \langle \nabla f(\Sigma_2), \Sigma_1 - \Sigma_2 \rangle \\ = \langle \nabla f(\tilde{\Sigma}) - \nabla f(\Sigma_2), \Sigma_1 - \Sigma_2 \rangle = \|\Sigma_1 - \Sigma_2\|_{\nabla^2 f(\tilde{\Sigma}')}^2, \end{aligned}$$

and

$$\langle \nabla f(\Sigma_1) - \nabla f(\Sigma_2), \Sigma_1 - \Sigma_2 \rangle = \|\Sigma_1 - \Sigma_2\|_{\nabla^2 f(\tilde{\Sigma}'')}^2.$$

By applying the definition of the sparse eigenvalue, we arrive at the desired conclusion.  $\square$

**Lemma 23.** Consider  $\Sigma(\theta) = \Sigma^* + \theta(\Sigma - \Sigma^*)$  for  $\theta \in (0, 1]$ . Then, we have

$$G_f^s(\Sigma(\theta), \Sigma^*) \leq \theta G_f^s(\Sigma, \Sigma^*).$$

*Proof.* Define  $\varphi(\theta)$  as follows:

$$\begin{aligned} \varphi(\theta) &= G_f(\Sigma(\theta), \Sigma^*) \\ &= f(\Sigma(\theta)) - f(\Sigma^*) - \langle \nabla f(\Sigma^*), \Sigma(\theta) - \Sigma^* \rangle. \end{aligned}$$

The derivative of  $f(\Sigma(\theta))$  with respect to  $\theta$  is  $\langle \nabla f(\Sigma(\theta)), \Sigma - \Sigma^* \rangle$ . Thus

$$\varphi'(\theta) = \langle \nabla f(\Sigma(\theta)) - \nabla f(\Sigma^*), \Sigma - \Sigma^* \rangle.$$

Consequently, the symmetric Bregman divergence  $G_f^s(\Sigma(\theta), \Sigma^*)$  is

$$\begin{aligned} G_f^s(\Sigma(\theta), \Sigma^*) &= \langle \nabla f(\Sigma(\theta)) - \nabla f(\Sigma^*), \theta(\Sigma - \Sigma^*) \rangle \\ &= \theta \varphi'(\theta) \end{aligned}$$

for  $0 < \theta \leq 1$ . Given  $\varphi'1 = G_f^s(\Sigma, \Sigma^*)$ , and assuming the convexity of  $\varphi(\theta)$  based on the convexity of  $f(\cdot)$  and the linearity of  $\Sigma(\theta)$ ,  $\varphi'(\theta)$  is non-decreasing. Thus

$$G_f^s(\Sigma(\theta), \Sigma^*) = \theta \varphi'(\theta) \leq \theta \varphi'1 = \theta G_f^s(\Sigma, \Sigma^*),$$

completing the proof. For a more detailed proof of the convexity of  $\varphi(\theta)$ , see [71], [88].  $\square$

**Lemma 24.** Consider a set  $\mathcal{E}$  such that  $\mathcal{S}^* \subseteq \mathcal{E}$ . Given the condition  $\|\nabla f(\Sigma^*)\|_{\max} + \varepsilon \leq \|\Lambda_{\mathcal{E}}\|_{\min}$ , we have

$$\left\| (\tilde{\Sigma} - \Sigma^*)_{\bar{\mathcal{E}}} \right\|_1 \leq 5 \left\| (\tilde{\Sigma} - \Sigma^*)_{\mathcal{E}} \right\|_1.$$

*Proof.* Define  $\Omega = \nabla f(\tilde{\Sigma}) + \Lambda \odot \tilde{\Xi}$  and  $\tilde{\Delta} = \tilde{\Sigma} - \Sigma^*$ . By the mean value theorem, there exists a  $\theta \in [0, 1]$ , such that

$$\nabla f(\tilde{\Sigma}) - \nabla f(\Sigma^*) = \text{mat} \left( \mathbf{H}(\theta) \text{vec}(\tilde{\Delta}) \right),$$

where  $\mathbf{H}(\theta) = \nabla^2 f(\theta \Sigma^* + (1 - \theta) \tilde{\Sigma})$ . Then we have

$$\begin{aligned} &\langle \nabla f(\tilde{\Sigma}) + \Lambda \odot \tilde{\Xi}, \tilde{\Delta} \rangle \\ &= \langle \nabla f(\Sigma^*) + \text{mat}(\mathbf{H}(\theta) \text{vec}(\tilde{\Delta})) + \Lambda \odot \tilde{\Xi}, \tilde{\Delta} \rangle \\ &\leq \|\Omega\|_{\max} \left\| \tilde{\Delta} \right\|_1. \end{aligned}$$

Given  $\left\| \tilde{\Delta} \right\|_{H(\theta)}^2 \geq 0$ , it follows that:

$$0 \leq \|\Omega\|_{\max} \left\| \tilde{\Delta} \right\|_1 - \underbrace{\langle \nabla f(\Sigma^*), \tilde{\Delta} \rangle}_{\text{I}} - \underbrace{\langle \Lambda \odot \tilde{\Xi}, \tilde{\Delta} \rangle}_{\text{II}}. \quad (13)$$

For term I, separating the support of  $\nabla f(\Sigma^*)$  and  $\tilde{\Delta}$  to  $\mathcal{E}$  and  $\bar{\mathcal{E}}$ , and then using the matrix Hölder's inequality, we obtain

$$\begin{aligned} \text{I} &= \langle (\nabla f(\Sigma^*))_{\mathcal{E}}, \tilde{\Delta}_{\mathcal{E}} \rangle + \langle (\nabla f(\Sigma^*))_{\bar{\mathcal{E}}}, \tilde{\Delta}_{\bar{\mathcal{E}}} \rangle \\ &\geq -\|(\nabla f(\Sigma^*))_{\mathcal{E}}\|_{\text{F}} \left\| \tilde{\Delta}_{\mathcal{E}} \right\|_{\text{F}} - \|(\nabla f(\Sigma^*))_{\bar{\mathcal{E}}}\|_{\max} \left\| \tilde{\Delta}_{\bar{\mathcal{E}}} \right\|_1 \\ &\geq -\|(\nabla f(\Sigma^*))_{\mathcal{E}}\|_{\text{F}} \left\| \tilde{\Delta} \right\|_{\text{F}} - \|(\nabla f(\Sigma^*))\|_{\max} \left\| \tilde{\Delta} \right\|_1. \end{aligned}$$

For term II, separating the support of  $\Lambda \odot \tilde{\Xi}$  and  $\tilde{\Delta}$  to  $\mathcal{E}$  and  $\bar{\mathcal{E}}$ , and then using the matrix Hölder's inequality, we obtain

$$\begin{aligned} \text{II} &= \langle (\Lambda \odot \tilde{\Xi})_{\mathcal{E}}, \tilde{\Delta}_{\mathcal{E}} \rangle + \langle (\Lambda \odot \tilde{\Xi})_{\bar{\mathcal{E}}}, \tilde{\Delta}_{\bar{\mathcal{E}}} \rangle \\ &\geq -\|\Lambda_{\mathcal{E}}\|_{\max} \left\| \tilde{\Delta}_{\mathcal{E}} \right\|_1 + \|\Lambda_{\bar{\mathcal{E}}}\|_{\min} \left\| \tilde{\Delta}_{\bar{\mathcal{E}}} \right\|_1. \end{aligned}$$

Plugging the above results into 13 yields

$$\begin{aligned} \left\| \tilde{\Delta}_{\bar{\mathcal{E}}} \right\|_1 &\leq \frac{\lambda + \|(\nabla f(\Sigma^*))\|_{\max} + \omega_{\Lambda}(\tilde{\Sigma})}{\|\Lambda_{\bar{\mathcal{E}}}\|_{\min} - (\|(\nabla f(\Sigma^*))\|_{\max} + \omega_{\Lambda}(\tilde{\Sigma}))} \left\| \tilde{\Delta}_{\mathcal{E}} \right\|_1 \\ &\stackrel{(i)}{\leq} 5 \left\| \tilde{\Delta}_{\mathcal{E}} \right\|_1, \end{aligned}$$

where (i) is from  $\|\Lambda_{\bar{\mathcal{E}}}\|_{\min} \geq \frac{\lambda}{2}$ , Assumption 10 and  $\omega_{\Lambda}(\tilde{\Sigma}) \leq \varepsilon$ .  $\square$

**Lemma 25.** Suppose that Assumption 3, 8, 9 and 10 hold. Consider the general problem in (??). Let there exists a set  $\mathcal{E}$  such that

$$\mathcal{S}^* \subseteq \mathcal{E}, |\mathcal{E}| \leq 2s^*, \text{ and } \|\Lambda_{\bar{\mathcal{E}}}\|_{\min} \geq \frac{\lambda}{2}.$$

Then the  $\varepsilon$ -optimal solution  $\tilde{\Sigma}$  satisfies

$$\begin{aligned} \left\| \tilde{\Sigma} - \Sigma^* \right\|_{\text{F}} &\leq \frac{1}{\rho_{2s^*+2\tilde{s}}} \left( \|(\nabla f(\Sigma^*))_{\mathcal{E}}\|_{\text{F}} + \varepsilon \sqrt{|\mathcal{E}|} + \|\Lambda_{\mathcal{S}^*}\|_{\text{F}} \right) \\ &\leq \frac{2 + \sqrt{2}}{2\rho_{2s^*+2\tilde{s}}} \lambda \sqrt{s^*}. \end{aligned}$$

*Proof.* Given the Assumption 10, it follows that  $\lambda \geq 2(\|\nabla f(\Sigma^*)\|_{\max} + \varepsilon)$ . We introduce an intermediate estimator  $\tilde{\Sigma}^* = \Sigma^* + \theta(\tilde{\Sigma} - \Sigma^*)$ , where  $\theta$  is chosen to ensure  $\|\tilde{\Sigma}^* - \Sigma^*\|_{\text{F}} = r$  if  $\|\tilde{\Sigma} - \Sigma^*\|_{\text{F}} > r$ ; otherwise  $\theta = 1$ . This construction guarantees that  $\|\tilde{\Sigma}^* - \Sigma^*\|_{\text{F}} \leq r$ . By leveraging Lemma 24, we deduce that the approximate solution is contained within the  $\ell_1$ -cone. Given the structure of  $\tilde{\Sigma}^*$ , we find that  $\tilde{\Sigma}^* - \Sigma^* = \theta(\tilde{\Sigma} - \Sigma^*)$ . This leads to the following inequality

$$\left\| (\tilde{\Sigma}^* - \Sigma^*)_{\mathcal{E}} \right\|_1 \leq 5 \left\| (\tilde{\Sigma}^* - \Sigma^*)_{\bar{\mathcal{E}}} \right\|_1.$$

By integrating the aforementioned inequality with the premise that  $|\mathcal{E}| \leq 2s^*$ , it follows that  $\tilde{\Sigma}^*$  resides within the local  $\ell_1$ -cone. Furthermore, by synthesizing Definition 19, Proposition 20, and Lemma 22, one can deduce the presence of localized restricted strong convexity, i.e.

$$\rho_{2s^*+2\tilde{s}} \left\| \tilde{\Sigma}^* - \Sigma^* \right\|_{\text{F}}^2 \leq G_f^s(\tilde{\Sigma}^*, \Sigma^*). \quad (14)$$



Using Lemma 23, we can bound the right-hand side of the above inequality, such as

$$\begin{aligned} G_f^s(\tilde{\Sigma}^*, \Sigma^*) &\leq \theta G_f^s(\tilde{\Sigma}, \Sigma^*) \\ &= \theta \langle \nabla f(\tilde{\Sigma}) - \nabla f(\Sigma^*), \tilde{\Sigma} - \Sigma^* \rangle. \end{aligned} \quad (15)$$

Plugging (15) back into (14) yields

$$\begin{aligned} \rho_{2s^*+2\bar{s}}^- \|\tilde{\Sigma}^* - \Sigma^*\|_F^2 &\leq \theta \langle \nabla f(\tilde{\Sigma}) - \nabla f(\Sigma^*), \tilde{\Delta} \rangle \\ &= \theta \underbrace{\langle \nabla f(\tilde{\Sigma}) + \Lambda \odot \tilde{\Xi}, \tilde{\Delta} \rangle}_{\text{I}} \\ &\quad - \theta \underbrace{\langle \nabla f(\Sigma^*), \tilde{\Delta} \rangle}_{\text{II}} - \theta \underbrace{\langle \Lambda \odot \tilde{\Xi}, \tilde{\Delta} \rangle}_{\text{III}}, \end{aligned} \quad (16)$$

where  $\tilde{\Delta} = \tilde{\Sigma} - \Sigma^*$ ,  $\tilde{\Xi}^{(t+1)} \in \partial \|\Sigma^{(t+1)}\|_1$ . Then we establish limits for terms I, II and III, respectively.

Define  $\Omega = \nabla f(\tilde{\Sigma}) + \Lambda \odot \tilde{\Xi}$ . For term I, we partition the support of  $\Omega$  and  $\tilde{\Delta}$  into the set  $\mathcal{E}$  and its complement  $\bar{\mathcal{E}}$ . By employing the matrix Hölder's inequality, we derive

$$\begin{aligned} \text{I} &= \langle \Omega_{\mathcal{E}}, \tilde{\Delta}_{\mathcal{E}} \rangle + \langle \Omega_{\bar{\mathcal{E}}}, \tilde{\Delta}_{\bar{\mathcal{E}}} \rangle \\ &\leq \|\Omega_{\mathcal{E}}\|_F \|\tilde{\Delta}_{\mathcal{E}}\|_F + \|\Omega_{\bar{\mathcal{E}}}\|_{\max} \|\tilde{\Delta}_{\bar{\mathcal{E}}}\|_1 \\ &\leq \sqrt{|\mathcal{E}|} \|\Omega_{\mathcal{E}}\|_{\max} \|\tilde{\Delta}_{\mathcal{E}}\|_F + \|\Omega\|_{\max} \|\tilde{\Delta}_{\bar{\mathcal{E}}}\|_1 \\ &\stackrel{(i)}{\leq} \varepsilon \sqrt{|\mathcal{E}|} \|\tilde{\Delta}_{\mathcal{E}}\|_F + \varepsilon \|\tilde{\Delta}_{\bar{\mathcal{E}}}\|_1, \end{aligned}$$

where (i) is from  $\|\Omega\|_{\max} \leq \varepsilon$  by Definition 2. For term II, we divide the support of  $\nabla f(\Sigma^*)$  and  $\tilde{\Delta}$  into the set  $\mathcal{E}$  and  $\bar{\mathcal{E}}$ . Applying the matrix Hölder's inequality, we obtain

$$\begin{aligned} \text{II} &= \langle (\nabla f(\Sigma^*))_{\mathcal{E}}, \tilde{\Delta}_{\mathcal{E}} \rangle + \langle (\nabla f(\Sigma^*))_{\bar{\mathcal{E}}}, \tilde{\Delta}_{\bar{\mathcal{E}}} \rangle \\ &\geq -\|(\nabla f(\Sigma^*))_{\mathcal{E}}\|_F \|\tilde{\Delta}_{\mathcal{E}}\|_F - \|(\nabla f(\Sigma^*))_{\bar{\mathcal{E}}}\|_{\max} \|\tilde{\Delta}_{\bar{\mathcal{E}}}\|_1. \end{aligned}$$

For term III, we separate the support of  $\Lambda \odot \tilde{\Xi}$  and  $\tilde{\Delta}$  into the set  $\mathcal{S}^*$  and  $\bar{\mathcal{S}}^*$ . And then applying the matrix Hölder's inequality, we obtain

$$\begin{aligned} \text{III} &= \langle (\Lambda \odot \tilde{\Xi})_{\mathcal{S}^*}, \tilde{\Delta}_{\mathcal{S}^*} \rangle + \langle (\Lambda \odot \tilde{\Xi})_{\bar{\mathcal{S}}^*}, \tilde{\Delta}_{\bar{\mathcal{S}}^*} \rangle \\ &\stackrel{(i)}{=} \langle (\Lambda \odot \tilde{\Xi})_{\mathcal{S}^*}, \tilde{\Delta}_{\mathcal{S}^*} \rangle + \langle \Lambda_{\bar{\mathcal{S}}^*}, |\tilde{\Delta}_{\bar{\mathcal{S}}^*}| \rangle \\ &\geq -\|\Lambda_{\mathcal{S}^*}\|_F \|\tilde{\Delta}_{\mathcal{S}^*}\|_F + \langle \Lambda_{\bar{\mathcal{E}}}, |\tilde{\Delta}_{\bar{\mathcal{E}}}| \rangle \\ &\stackrel{(ii)}{\geq} -\|\Lambda_{\mathcal{S}^*}\|_F \|\tilde{\Delta}_{\mathcal{E}}\|_F + \|\Lambda_{\bar{\mathcal{E}}}\|_{\min} \|\tilde{\Delta}_{\bar{\mathcal{E}}}\|_1, \end{aligned}$$

where (i) is from

$$\langle (\Lambda \odot \tilde{\Xi})_{\bar{\mathcal{S}}^*}, \tilde{\Delta}_{\bar{\mathcal{S}}^*} \rangle = \langle \Lambda_{\bar{\mathcal{S}}^*}, |\tilde{\Delta}_{\bar{\mathcal{S}}^*}| \rangle = \langle \Lambda_{\bar{\mathcal{S}}^*}, |\tilde{\Delta}_{\bar{\mathcal{S}}^*}| \rangle,$$

and (ii) is from

$$\begin{aligned} \langle \Lambda_{\bar{\mathcal{E}}}, |\tilde{\Delta}_{\bar{\mathcal{E}}}| \rangle &= \sum_{(i,j) \in \bar{\mathcal{E}}} \Lambda_{ij} |\tilde{\Delta}_{ij}| \geq \|\Lambda_{\bar{\mathcal{E}}}\|_{\min} \sum_{(i,j) \in \bar{\mathcal{E}}} |\tilde{\Delta}_{ij}| \\ &= \|\Lambda_{\bar{\mathcal{E}}}\|_{\min} \|\tilde{\Delta}_{\bar{\mathcal{E}}}\|_1, \end{aligned}$$

and

$$\|\tilde{\Delta}_{\mathcal{S}^*}\|_F \leq \|\tilde{\Delta}_{\mathcal{E}}\|_F.$$

Combining the above results into (16) yields

$$\begin{aligned} \rho_{2s^*+2\bar{s}}^- \|\tilde{\Sigma}^* - \Sigma^*\|_F^2 &\leq \left( \|\Lambda_{\mathcal{S}^*}\|_F + \|(\nabla f(\Sigma^*))_{\mathcal{E}}\|_F + \varepsilon \sqrt{|\mathcal{E}|} \right) \times \theta \|\tilde{\Delta}_{\mathcal{E}}\|_F \\ &\quad - \theta (\|\Lambda_{\bar{\mathcal{E}}}\|_{\min} - (\|(\nabla f(\Sigma^*))_{\bar{\mathcal{E}}}\|_{\max} + \varepsilon)) \|\tilde{\Delta}_{\bar{\mathcal{E}}}\|_1 \\ &\stackrel{(i)}{\leq} \left( \|\Lambda_{\mathcal{S}^*}\|_F + \|(\nabla f(\Sigma^*))_{\mathcal{E}}\|_F + \varepsilon \sqrt{|\mathcal{E}|} \right) \times \theta \left\| (\tilde{\Sigma} - \Sigma^*)_{\mathcal{E}} \right\|_F \\ &\leq \left( (\|(\nabla f(\Sigma^*))_{\mathcal{E}}\|_{\max} + \varepsilon) \sqrt{|\mathcal{E}|} + \|\Lambda_{\mathcal{S}^*}\|_{\max} \sqrt{|\mathcal{S}^*|} \right) \\ &\quad \times \theta \left\| (\tilde{\Sigma} - \Sigma^*)_{\mathcal{E}} \right\|_F \\ &\leq \left( (\|(\nabla f(\Sigma^*))_{\mathcal{E}}\|_{\max} + \varepsilon) \sqrt{2s^*} + \lambda \sqrt{s^*} \right) \\ &\quad \times \theta \left\| (\tilde{\Sigma} - \Sigma^*)_{\mathcal{E}} \right\|_F \\ &\leq \frac{2+\sqrt{2}}{2} \lambda \sqrt{s^*} \times \theta \underbrace{\left\| (\tilde{\Sigma} - \Sigma^*)_{\mathcal{E}} \right\|_F}_{\text{IV}}, \end{aligned}$$

where (i) is due to the fact that  $\|\Lambda_{\bar{\mathcal{E}}}\|_{\min} \geq \frac{\lambda}{2} \geq \|(\nabla f(\Sigma^*))_{\bar{\mathcal{E}}}\|_{\max} + \varepsilon$ .

For IV, we have  $\theta \|(\tilde{\Sigma} - \Sigma^*)_{\mathcal{E}}\|_F = \|(\tilde{\Sigma}^* - \Sigma^*)_{\mathcal{E}}\|_F$ . Consequently, we derive

$$\rho_{2s^*+2\bar{s}}^- \|\tilde{\Sigma}^* - \Sigma^*\|_F \leq \frac{2+\sqrt{2}}{2} \lambda \sqrt{s^*},$$

which serves as a contraction in relation to the construction of  $\tilde{\Sigma}^*$ . This leads to the conclusion that  $\tilde{\Sigma}^* = \tilde{\Sigma}$ . Therefore, the desired bound is satisfied for  $\tilde{\Sigma}$ .  $\square$

**Lemma 26.** Suppose that Assumptions 4 and 8 hold. Consider the problem in (??). Define the set  $\mathcal{E}^k = \mathcal{S}^* \cup \mathcal{S}^k$ , where  $\mathcal{S}^k = \{(i, j) \mid \Lambda_{ij}^{k-1} \leq p'_\lambda(u)\}$  with  $u = c\lambda$  and  $c = \frac{2+\sqrt{2}}{2\rho_{2s^*+2\bar{s}}^-} \in (0, \alpha)$ , such that  $p'_\lambda(c\lambda) \geq \frac{\lambda}{2}$ . This can always hold due to Assumption 1. Then for  $k \geq 1$ , we establish that: (1)  $\lambda \geq 2(\|(\nabla f(\Sigma^*))_{\max} + \varepsilon)$ , (2)  $|\mathcal{E}^k| \leq 2s^*$ , (3)  $\|\Lambda_{\mathcal{E}^k}^{k-1}\|_{\min} \geq \frac{\lambda}{2}$ , and

$$\begin{aligned} \|\tilde{\Sigma}^k - \Sigma^*\|_F &\leq \frac{1}{\rho_{2s^*+2\bar{s}}^-} \left( \|(\nabla f(\Sigma^*))_{\mathcal{E}^k}\|_F + \varepsilon \sqrt{|\mathcal{E}^k|} \right) \\ &\quad + \frac{1}{\rho_{2s^*+2\bar{s}}^-} \|\Lambda_{\mathcal{S}^*}^{k-1}\|_F \\ &\leq \frac{2+\sqrt{2}}{2\rho_{2s^*+2\bar{s}}^-} \lambda \sqrt{s^*}. \end{aligned}$$

*Proof.* We first show that  $|\mathcal{E}^k| \leq 2s^*$  for all  $k$  by induction on  $k$ .

**Base Case** ( $k = 1$ ):

For  $k = 1$ , we have  $\Lambda_{ij}^{(0)} = \lambda$  for  $i \neq j$ , where  $\lambda$  is a threshold parameter satisfying  $\lambda \geq p'_\lambda(u)$ . It follows that  $|\mathcal{S}^1| \leq s^*$ . And we see  $\mathcal{E}^1 = \mathcal{S}^* \cup \mathcal{S}^1$ , which implies  $|\mathcal{E}^1| \leq 2s^*$  holds.

**Inductive Step** ( $k \geq 2$ ):

Assume for some  $k \geq 2$ ,  $|\mathcal{E}^{k-1}| \leq 2s^*$  holds true. We aim to show that  $|\mathcal{E}^k| \leq 2s^*$ . For any  $(i, j) \in \mathcal{S}^k$ , we obtain  $|\tilde{\Sigma}_{ij}^{k-1}| \geq u$  and further have

$$\begin{aligned} \sqrt{|\mathcal{S}^k \setminus \mathcal{S}^*|} &\leq \sqrt{\sum_{(i,j) \in \mathcal{S}^k \setminus \mathcal{S}^*} \left(u^{-1} \tilde{\Sigma}_{ij}^{k-1}\right)^2} \\ &= u^{-1} \left\| \tilde{\Sigma}_{\mathcal{S}^k \setminus \mathcal{S}^*}^{k-1} \right\|_F \\ &= u^{-1} \left\| \left( \tilde{\Sigma}^{k-1} - \Sigma^* \right)_{\mathcal{S}^k \setminus \mathcal{S}^*} \right\|_F \\ &\leq u^{-1} \left\| \tilde{\Sigma}^{k-1} - \Sigma^* \right\|_F. \end{aligned} \quad (17)$$

For any  $(i, j) \in \mathcal{S}^{k-1}$ , it follows that  $\Lambda_{ij}^{k-2} = p'_\lambda \left( \tilde{\Sigma}_{ij}^{k-2} \right) \geq p'_\lambda(u) \geq \frac{\lambda}{2}$ . This chain of inequalities indicates that

$$\left\| \Lambda_{\mathcal{E}^{k-1}}^{k-2} \right\|_{\min} \geq \left\| \Lambda_{\mathcal{S}^{k-1}}^{k-2} \right\|_{\min} \geq p'_\lambda(u) \geq \frac{\lambda}{2}.$$

Furthermore, it is established that  $|\mathcal{E}^{k-1}| \leq 2s^*$  and  $\mathcal{S}^* \subseteq \mathcal{E}^{k-1}$ . By applying Lemma 25 with  $\tilde{\Sigma} = \tilde{\Sigma}^{k-1}$ ,  $\mathcal{E} = \mathcal{E}^{k-1}$ , and  $\Lambda_{\mathcal{S}^*} = \Lambda_{\mathcal{S}^*}^{k-2}$  yields

$$\left\| \tilde{\Sigma}^{k-1} - \Sigma^* \right\|_F \leq \frac{2 + \sqrt{2}}{2\rho_{2s^*+2\tilde{s}}} \lambda \sqrt{s^*}.$$

Substituting this result into the inequality (17) yields

$$\sqrt{|\mathcal{S}^k \setminus \mathcal{S}^*|} \leq \frac{2 + \sqrt{2}}{2u\rho_{2s^*+2\tilde{s}}} \lambda \sqrt{s^*} = \sqrt{s^*}.$$

Thus, we have

$$|\mathcal{E}^k| = |\mathcal{S}^* \cup (\mathcal{S}^k \setminus \mathcal{S}^*)| = |\mathcal{S}^*| + |\mathcal{S}^k \setminus \mathcal{S}^*| \leq 2s^*,$$

completing the induction process.

Then according to the definition of  $\mathcal{E}^k$  and  $\mathcal{S}^k$ , it follows that

$$\left\| \Lambda_{\mathcal{E}^k}^{k-1} \right\|_{\min} \geq \left\| \Lambda_{\mathcal{S}^k}^{k-1} \right\|_{\min} \geq p'_\lambda(u) \geq \frac{\lambda}{2}.$$

By employing Lemma 25 with  $\tilde{\Sigma} = \tilde{\Sigma}^k$ ,  $\mathcal{E} = \mathcal{E}^k$ , and  $\Lambda_{\mathcal{S}^*} = \Lambda_{\mathcal{S}^*}^{k-1}$ , the  $\varepsilon$ -optimal solution  $\tilde{\Sigma}^k$  to (??) satisfies

$$\begin{aligned} &\left\| \tilde{\Sigma}^k - \Sigma^* \right\|_F \\ &\leq \frac{1}{\rho_{2s^*+2\tilde{s}}} \left( \left\| (\nabla f(\Sigma^*))_{\mathcal{E}^k} \right\|_F + \varepsilon \sqrt{|\mathcal{E}^k|} + \left\| \Lambda_{\mathcal{S}^*}^{k-1} \right\|_F \right) \\ &\leq \frac{2 + \sqrt{2}}{2\rho_{2s^*+2\tilde{s}}} \lambda \sqrt{s^*}. \end{aligned}$$

## B. Proof of Theorem 12

*Proof.* Based on Lemma 26, we have

$$\begin{aligned} \left\| \tilde{\Sigma}^k - \Sigma^* \right\|_F &\leq \underbrace{\frac{1}{\rho_{2s^*+2\tilde{s}}} \left( \left\| (\nabla f(\Sigma^*))_{\mathcal{E}^k} \right\|_F + \varepsilon \sqrt{|\mathcal{E}^k|} \right)}_{\text{I}} \\ &\quad + \underbrace{\frac{1}{\rho_{2s^*+2\tilde{s}}} \left\| \Lambda_{\mathcal{S}^*}^{k-1} \right\|_F}_{\text{II}}. \end{aligned} \quad (18)$$

Then, we proceed to establish bounds for the term **I** and **II**, respectively.

For term **I**, dividing the support set into  $\mathcal{S}^*$  and  $\mathcal{E}^{(q)} \setminus \mathcal{S}^*$ , we obtain

$$\begin{aligned} \text{I} &\leq \left\| (\nabla f(\Sigma^*))_{\mathcal{S}^*} \right\|_F + \varepsilon \sqrt{s^*} \\ &\quad + \left( \left\| \nabla f(\Sigma^*) \right\|_{\max} + \varepsilon \right) \sqrt{|\mathcal{E}^k \setminus \mathcal{S}^*|} \\ &\leq \left\| (\nabla f(\Sigma^*))_{\mathcal{S}^*} \right\|_F + \varepsilon \sqrt{s^*} + \frac{\lambda}{2u} \left\| \tilde{\Sigma}^{k-1} - \Sigma^* \right\|_F, \end{aligned}$$

where the second inequality is due to

$$\sqrt{|\mathcal{E}^k \setminus \mathcal{S}^*|} = \sqrt{|\mathcal{S}^k \setminus \mathcal{S}^*|} \leq u^{-1} \left\| \tilde{\Sigma}^{k-1} - \Sigma^* \right\|_F,$$

which follows from the inequality (17).

By Assumptions 1 and 9, for any  $\Sigma$ , if  $|\Sigma_{ij} - \Sigma_{ij}^*| \geq u$ , then  $p'_\lambda(\Sigma_{ij}) \leq \lambda \leq \lambda u^{-1} |\Sigma_{ij} - \Sigma_{ij}^*|$ ; otherwise,  $p'_\lambda(\Sigma_{ij}) \leq p'_\lambda(|\Sigma_{ij}^*| - u) = 0$ . Therefore, for term **II**, we have

$$\text{II} \leq \lambda u^{-1} \left\| \tilde{\Sigma}_{\mathcal{S}^*}^{k-1} - \Sigma_{\mathcal{S}^*}^* \right\|_F \leq \lambda u^{-1} \left\| \tilde{\Sigma}^{k-1} - \Sigma^* \right\|_F.$$

Substituting the above results into (18) yields

$$\begin{aligned} &\left\| \tilde{\Sigma}^k - \Sigma^* \right\|_F \\ &\leq \frac{1}{\rho_{2s^*+2\tilde{s}}} \left( \left\| (\nabla f(\Sigma^*))_{\mathcal{S}^*} \right\|_F + \varepsilon \sqrt{s^*} \right) + \delta \left\| \tilde{\Sigma}^{k-1} - \Sigma^* \right\|_F, \end{aligned}$$

where  $\delta = \frac{3\lambda}{2u\rho_{2s^*+2\tilde{s}}} = \frac{3}{2+\sqrt{2}} \in (0, 1)$ .  $\square$

## C. Concentration Inequality

**Lemma 27** (Hanson-Wright Inequality). *Let  $\mathbf{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$  be a random vector with independent components  $x_i$  which satisfy  $\mathbb{E}(x_i) = 0$  and  $\|x_i\|_{\psi_2} \leq L$ . Let  $\mathbf{B}$  be an  $n \times n$  matrix. Then for any  $t > 0$ ,*

$$\begin{aligned} &\mathbb{P} \{ |\mathbf{x}^\top \mathbf{B} \mathbf{x} - \mathbb{E}(\mathbf{x}^\top \mathbf{B} \mathbf{x})| > t \} \\ &\leq 2 \exp \left( -c' \min \left( \frac{t^2}{L^4 \|\mathbf{B}\|_F^2}, \frac{t}{L^2 \|\mathbf{B}\|} \right) \right). \end{aligned} \quad (19)$$

**Lemma 28** (Lemma D.1 in [88]). *Let  $\mathbf{x}$  be a sub-Gaussian random vector with zero mean and covariance  $\Sigma^*$  and  $\{\mathbf{x}_i\}_{i=1}^n$  be a collection of i.i.d. samples from  $\mathbf{x}$ . There exists some constant  $c_1, c_2$ , and  $t_0$  such that for all  $t$  with  $0 < t < t_0$ , the sample covariance matrix  $\mathbf{S}$  satisfies the following tail-bound*

$$\mathbb{P}(|\Sigma_{ij}^* - S_{ij}| > t) \leq c_1 \exp(-c_2 n t^2).$$

**Lemma 29.** *Under Assumptions 9 and the same conditions in Lemma 28, the columns of  $\tilde{\mathbf{A}}$  are normalized such that  $\max_l \|\tilde{\mathbf{A}}_{\cdot l}\|_2 \leq \sqrt{m}$ , there exists some constant  $c_1$  such that*

$$\mathbb{P} \left( \left\| \frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma^*)) \right\|_{\max} \leq \lambda \right) \geq 1 - \frac{c_1}{d}.$$

*Proof.* We begin by establishing a bound on the probability that  $\left\| \frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma^*)) \right\|_{\max} > t$  using a union bound approach.

First, consider the bound  $\mathbb{P}(\|\frac{1}{m} \sum_{i=1}^m \eta'_i \mathbf{a}_i \mathbf{a}_i^\top\|_{\max} > t)$ , where  $\eta'_i = \mathbf{a}_i^\top (\boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}^*) \mathbf{a}_i + \eta_i$ . By applying the Hanson-Wright inequality (19), we establish

$$\begin{aligned} & \mathbb{P}(\mathbf{a}_i^\top (\boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}^*) \mathbf{a}_i > t) \\ &= \mathbb{P}(|\langle \mathbf{a}_i \mathbf{a}_i^\top, (\boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}^*) \rangle| > t) \\ &\leq 2 \exp\left(-c' \min\left(\frac{t^2}{L^4 \|\boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}^*\|_{\text{F}}^2}, \frac{t}{L^2 \|\boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}^*\|}\right)\right), \end{aligned}$$

where  $c'$  is a positive constant, and  $L$  is a scaling factor. This indicates that  $\langle \mathbf{a}_i \mathbf{a}_i^\top, \boldsymbol{\Sigma}_N - \boldsymbol{\Sigma}^* \rangle$  is a sub-exponential random variable. Additionally, given that  $\eta_i$  follows a sub-exponential distribution,  $\eta'_i$  is also a sub-exponential random variable. By summing over individual probabilities, we obtain

$$\begin{aligned} & \mathbb{P}\left(\left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))\right\|_{\max} > t\right) \\ &= \mathbb{P}\left(\left\|\frac{1}{m} \sum_{i=1}^m \eta'_i \mathbf{a}_i \mathbf{a}_i^\top\right\|_{\max} > t\right) \\ &\leq \sum_{l=1}^{d^2} \mathbb{P}\left(\frac{1}{m} |\tilde{\mathbf{A}}_{\cdot l}^\top \cdot \boldsymbol{\eta}'| > t\right), \end{aligned} \quad (20)$$

Define  $\zeta_l = \tilde{\mathbf{A}}_{\cdot l}^\top \cdot \boldsymbol{\eta}'$ . Given that  $\eta'_i$  is sub-exponential  $(0, \xi^2)$  for  $i = 1, \dots, m$ , we apply Lemma 28 and employ concentration inequalities

$$\mathbb{E}(\exp\{t_0 \zeta_l\} + \exp\{-t_0 \zeta_l\}) \leq 2 \exp\left\{nm^{-2} \|\tilde{\mathbf{A}}_{\cdot l}\|^2 \xi^2 t_0^2 / 2\right\},$$

which implies

$$\mathbb{P}(|\zeta_l| \geq t) \exp\{t_0 t\} \leq 2 \exp\left\{nm^{-2} \|\tilde{\mathbf{A}}_{\cdot l}\|^2 \xi^2 t_0^2 / 2\right\}.$$

Selecting  $t_0 = t \left(nm^{-2} \|\tilde{\mathbf{A}}_{\cdot l}\|_2^2 \xi^2\right)^{-1}$  yields that

$$\mathbb{P}(|\zeta_l| \geq t) \leq 2 \exp\left\{-nt^2 \cdot \left(2\xi^2 \|\tilde{\mathbf{A}}_{\cdot l}\|_2^2 / m^2\right)^{-1}\right\}.$$

Plugging it into (20) results

$$\begin{aligned} & \mathbb{P}\left(\left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))\right\|_{\max} > t\right) \\ &\stackrel{(i)}{\leq} 2d^2 \exp\left\{-nmt^2 \cdot \left(2\xi^2 \max_l \left\{\|\tilde{\mathbf{A}}_{\cdot l}\|_2^2 / m\right\}\right)^{-1}\right\} \\ &= c_1 \exp\{-c_2 nmt^2 + 2 \log d\}, \end{aligned}$$

where (i) is from the column of  $\tilde{\mathbf{A}}$  are normalized such that  $\max_l \|\tilde{\mathbf{A}}_{\cdot l}\|_2 \leq \sqrt{m}$ , and  $c_1, c_2$  are constants. Then taking  $\lambda = \sqrt{\frac{3 \log d}{c_3 mn}} \asymp \sqrt{\frac{\log d}{mn}}$ , we obtain

$$\begin{aligned} & \mathbb{P}\left(\left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))\right\|_{\max} \leq \lambda\right) \\ &\geq 1 - c_1 \exp(-c_2 nm \lambda^2 + 2 \log d) \\ &= 1 - \frac{c_1}{d}. \end{aligned}$$

**Lemma 30.** Under the same conditions as in Lemma 29, the following result holds

$$\left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\right\|_{\text{F}} = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right).$$

*Proof.* Following the analysis presented in Lemma 29 analysis, consider any  $M$  such that  $0 < M \sqrt{\frac{1}{mn}}$ , we have

$$\begin{aligned} & \mathbb{P}\left(\left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\right\|_{\max} > M \sqrt{\frac{1}{mn}}\right) \\ &\leq c_1 s^* \exp(-c_2 M) \\ &= c_1 \exp(-c_2 M + \log s^*). \end{aligned}$$

Setting  $M$  such that  $\sqrt{\frac{2 \log s^*}{c_3}} < M$  and letting  $M \rightarrow \infty$ , we obtain

$$\lim_{M \rightarrow \infty} \sup_m \mathbb{P}\left(\left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\right\|_{\max} > M \sqrt{\frac{1}{mn}}\right) = 0.$$

The proof is completed by applying

$$\begin{aligned} & \left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\right\|_{\text{F}} \\ &\leq \sqrt{s^*} \left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))_{\mathcal{S}^*}\right\|_{\max}. \end{aligned}$$

□

#### D. Proof of Corollary 14

*Proof.* We begin by bound the gradient norm of  $f(\boldsymbol{\Sigma})$  at  $\boldsymbol{\Sigma}^*$ :

$$\begin{aligned} \|\nabla f(\boldsymbol{\Sigma}^*)\|_{\max} &\leq \left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))\right\|_{\max} \\ &\quad + \tau \left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\max}. \end{aligned}$$

Assuming  $\lambda \asymp \sqrt{\frac{\log d}{mn}}$  and  $\tau \lesssim \sqrt{\frac{1}{mn}} \left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\max}^{-1}$ , and invoking Lemma 29, we establish that  $\lambda \geq 2(\|\nabla f(\boldsymbol{\Sigma}^*)\|_{\max} + \varepsilon)$  holds w.h.p.

Applying Lemma 26 with  $k = 1$ , we obtain

$$\left\|\tilde{\boldsymbol{\Sigma}}^1 - \boldsymbol{\Sigma}^*\right\|_{\text{F}} \leq \frac{2 + \sqrt{2}}{2\rho_{2s^*+2\tilde{s}}} \lambda \sqrt{s^*}.$$

Under the condition  $\lambda \asymp \sqrt{\frac{\log d}{mn}}$ , the Frobenius norm difference is asymptotically bounded as  $\left\|\tilde{\boldsymbol{\Sigma}}^1 - \boldsymbol{\Sigma}^*\right\|_{\text{F}} \lesssim \sqrt{\frac{s^* \log d}{mn}}$  w.h.p. □

#### E. Proof of Corollary 15

*Proof.* We begin by bound the gradient norm of  $f(\boldsymbol{\Sigma})$  at  $\boldsymbol{\Sigma}^*$ :

$$\begin{aligned} \|\nabla f(\boldsymbol{\Sigma}^*)\|_{\max} &\leq \left\|\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\boldsymbol{\Sigma}^*))\right\|_{\max} \\ &\quad + \tau \left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\max}. \end{aligned}$$

Under Assumptions 3 and 9 condition, and with parameters  $\lambda$  and  $\tau$  scaled as  $\lambda \asymp \sqrt{\frac{\log d}{mn}}$  and  $\tau \lesssim \sqrt{\frac{1}{mn}} \left\|(\boldsymbol{\Sigma}^*)^{-1}\right\|_{\max}^{-1}$ , then by Lemma 29,  $\lambda \geq 2(\|\nabla f(\boldsymbol{\Sigma}^*)\|_{\max} + \varepsilon)$  holds w.h.p. □

Next, applying Theorem 12, we derive

$$\begin{aligned}
& \|\tilde{\Sigma}^1 - \Sigma^*\|_F \\
& \leq \frac{1}{\rho_{2s^*+2\tilde{s}}} \left( \|(\nabla f(\Sigma^*))_{S^*}\|_F + \varepsilon\sqrt{s^*} \right) + \delta \|\tilde{\Sigma}^{k-1} - \Sigma^*\|_F \\
& \leq \frac{1}{\rho_{2s^*+2\tilde{s}}} \left( \|(\nabla f(\Sigma^*))_{S^*}\|_F + \varepsilon\sqrt{s^*} \right) + \delta^{k-1} \|\tilde{\Sigma}^1 - \Sigma^*\|_F \\
& \leq \frac{1}{\rho_{2s^*+2\tilde{s}}} \left( \|(\nabla f(\Sigma^*))_{S^*}\|_F + \varepsilon\sqrt{s^*} \right) + \delta^{k-1} \frac{2+\sqrt{2}}{2\rho_{2s^*+2\tilde{s}}} \lambda\sqrt{s^*},
\end{aligned}$$

where the last inequality is due to  $\|\tilde{\Sigma}^1 - \Sigma^*\|_F \leq \frac{2+\sqrt{2}}{2\rho_{2s^*+2\tilde{s}}} \lambda\sqrt{s^*}$ , which follows from Lemma 26 with  $k=1$ .

Finally, considering the norm of the projected gradient within the support set  $S^*$ , we obtain

$$\begin{aligned}
& \|(\nabla f(\Sigma^*))_{S^*}\|_F \\
& = \left\| \left( -\frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma^*)) - \tau(\Sigma^*)^{-1} \right)_{S^*} \right\|_F \\
& \leq \left\| \frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma^*))_{S^*} \right\|_F + \tau \left\| (\Sigma^*)^{-1} \right\|_F \\
& \leq \left\| \frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma^*))_{S^*} \right\|_F + \tau \left\| (\Sigma^*)^{-1} \right\|_F
\end{aligned}$$

By Lemma 30,  $\left\| \frac{1}{m} \mathcal{A}^*(\mathbf{y} - \mathcal{A}(\Sigma^*))_{S^*} \right\|_F = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right)$ .

If  $\tau \lesssim \sqrt{\frac{1}{mn}} \left\| (\Sigma^*)^{-1} \right\|_{\max}^{-1}$ , then  $\|(\nabla f(\Sigma^*))_{S^*}\|_F = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right)$ .

Furthermore, if  $K \geq 1 + \frac{\log(\lambda\sqrt{mn})}{\log \delta^{-1}} \gtrsim \log(\lambda\sqrt{mn}) \gtrsim \log \log d$ , we derive

$$\delta^{K-1} \lambda\sqrt{s^*} \leq \frac{1}{\lambda\sqrt{mn}} \lambda\sqrt{s^*} \leq \sqrt{\frac{s^*}{mn}}.$$

By integrating all findings with the condition  $\varepsilon \lesssim \sqrt{\frac{1}{mn}}$ , we conclude that  $\|\tilde{\Sigma}^k - \Sigma^*\|_F = \mathcal{O}_p\left(\sqrt{\frac{s^*}{mn}}\right)$ .  $\square$

## APPENDIX D

### PROOF OF COMPUTATIONAL THEORY 18

In this appendix, we first show  $f(\Sigma)$  exhibits properties of the local strong convexity, local strong smoothness and local restricted Hessian smoothness. Then we demonstrate the sparsity of the solution is preserved within the vicinity of the true model parameter  $\Sigma^*$ , provided that the starting point is a sparse solution. Following this, it is established that proximal Newton updates achieve a quadratic rate of convergence towards a local minimizer at each step within each stage, assuming the initialization occurs within a suitably refined sparse region. Subsequently, Lemmas 38 and 35 confirm that each subproblem experiences a substantial reduction. Furthermore, we outline the necessary number of iterations required at each convex relaxation stage to satisfy the approximate KKT conditions.

### A. Proof of Lemma 16

*Proof.* We first verify the convexity of the set  $\mathcal{B}\left(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}}{4\tau\kappa}\right)$ . For any  $\Sigma_1, \Sigma_2 \in \mathcal{B}\left(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}}{4\tau\kappa}\right)$ , the linearity of the space implies that any convex combination of these points also resides within the set due to the properties of the norm and the definition of the set.

To establish the sparse strong convexity of  $f(\Sigma)$  within this set, consider two arbitrary points  $\Sigma_3, \Sigma_4$  and convex combination  $\Sigma_\rho = \rho\Sigma_3 + (1-\rho)\Sigma_4$  for  $\rho \in [0, 1]$ . The function  $f(\Sigma)$  can be expanded using a Taylor series approximation around  $\Sigma_4$ :

$$\begin{aligned}
f(\Sigma_3) &= f(\Sigma_4) + \langle \nabla f(\Sigma_4), \Sigma_3 - \Sigma_4 \rangle \\
&\quad + \frac{1}{2} \text{vec}^\top(\Sigma_3 - \Sigma_4) \nabla^2 f(\Sigma_\rho) \text{vec}(\Sigma_3 - \Sigma_4).
\end{aligned}$$

The next step is to estimate the lower bound of  $\lambda_{\min}(\nabla^2 f(\Sigma_\rho))$ . Using the fact that  $\lambda_{\min}(\mathbf{X}^{-1}) = \|\mathbf{X}\|_2^{-1}$  and  $\|\mathbf{X} \otimes \mathbf{Y}\|_2 = \|\mathbf{X}\|_2 \|\mathbf{Y}\|_2$  for any matrices  $\mathbf{X}, \mathbf{Y}$ , we deduce

$$\begin{aligned}
& \lambda_{\min}(\nabla^2 f(\Sigma_\rho)) \\
& = \lambda_{\min}\left(\sum_{i=1}^m \mathbf{A}_i \otimes \mathbf{A}_i^\top + \tau \Sigma_\rho^{-1} \otimes \Sigma_\rho^{-1}\right) \\
& \geq \lambda_{\min}\left(\sum_{i=1}^m \mathbf{A}_i \otimes \mathbf{A}_i^\top\right) + \tau \lambda_{\min}(\Sigma_\rho^{-1} \otimes \Sigma_\rho^{-1}) \\
& \stackrel{(i)}{\geq} \rho_{2s^*+2\tilde{s}} + \tau \|\Sigma_\rho\|_2^{-2},
\end{aligned}$$

where (i) is from Assumption 7. One has

$$\begin{aligned}
\|\Sigma_\rho\|_2 &= \|\Sigma^* - \Sigma^* + \rho\Sigma_1 + (1-\rho)\Sigma_2\|_2 \\
&\leq \|\Sigma^*\|_2 + \rho\|\Sigma_1 - \Sigma^*\|_2 + (1-\rho)\|\Sigma_2 - \Sigma^*\|_2 \\
&\leq \|\Sigma^*\|_2 + \rho\|\Sigma_1 - \Sigma^*\|_F + (1-\rho)\|\Sigma_2 - \Sigma^*\|_F \\
&\stackrel{(i)}{\leq} \kappa + \frac{\rho_{2s^*+2\tilde{s}}}{4\tau\kappa} \leq \kappa + \frac{\kappa\rho_{2s^*+2\tilde{s}}}{4\tau},
\end{aligned}$$

where (i) holds due to  $\|\Sigma^*\|_2 \leq \kappa$  and  $\Sigma_1, \Sigma_2 \in \mathcal{B}\left(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}}{4\tau\kappa}\right)$ , the last inequality holds due to  $\kappa \geq 1$ . Thus,

$f(\Sigma)$  is shown to be strongly convex over  $\mathcal{B}\left(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}}{4\tau\kappa}\right)$  with a convexity parameter of  $\left(\rho_{2s^*+2\tilde{s}} + \frac{16\tau^3}{\kappa^2(4\tau + \rho_{2s^*+2\tilde{s}})^2}\right)$ .

Next, we establish the strong smoothness of  $f(\Sigma)$ , which involves setting an upper bound for  $\lambda_{\max}(\nabla^2 f(\Sigma_\rho))$ . By the fact that  $\lambda_{\max}(\mathbf{X}^{-1}) = \lambda_{\min}^{-1}(\mathbf{X})$  and  $\lambda_{\min}(\mathbf{X} \otimes \mathbf{Y})$  for any matrices  $\mathbf{X}, \mathbf{Y}$ . Therefore,

$$\begin{aligned}
& \lambda_{\max}(\nabla^2 f(\Sigma_\rho)) \\
& = \lambda_{\max}\left(\sum_{i=1}^m \mathbf{A}_i \otimes \mathbf{A}_i^\top + \tau \Sigma_\rho^{-1} \otimes \Sigma_\rho^{-1}\right) \\
& \leq \lambda_{\max}\left(\sum_{i=1}^m \mathbf{A}_i \otimes \mathbf{A}_i^\top\right) + \tau \lambda_{\max}(\Sigma_\rho^{-1} \otimes \Sigma_\rho^{-1}) \\
& \leq \rho_{2s^*+2\tilde{s}}^+ + \tau \lambda_{\min}^{-2}(\Sigma_\rho).
\end{aligned}$$

One has

$$\begin{aligned}
& \lambda_{\min}(\Sigma_\rho) \\
&= \lambda_{\min}(\Sigma^* - \Sigma^* + \rho \Sigma_1 + (1-\rho) \Sigma_2) \\
&\geq \lambda_{\min}(\Sigma^*) + \rho \lambda_{\min}(\Sigma_1 - \Sigma^*) + (1-\rho) \lambda_{\min}(\Sigma_2 - \Sigma^*) \\
&\geq \lambda_{\min}(\Sigma^*) - \rho \lambda_{\max}(\Sigma_1 - \Sigma^*) - (1-\rho) \lambda_{\max}(\Sigma_2 - \Sigma^*) \\
&\geq \lambda_{\min}(\Sigma^*) - \rho \|\Sigma_1 - \Sigma^*\|_F - (1-\rho) \|\Sigma_2 - \Sigma^*\|_F \\
&\geq \frac{1}{\kappa} - \frac{\rho_{2s^*+2\tilde{s}}}{4\tau\kappa},
\end{aligned}$$

where the last inequality holds due to  $\lambda_{\min}(\Sigma^*) \geq \frac{1}{\kappa}$  and  $\Sigma_1, \Sigma_2 \in \mathcal{B}\left(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}}{4\tau\kappa}\right)$ . Thus,  $f(\Sigma)$  is proven to be  $\left(\rho_{2s^*+2\tilde{s}}^+ + \frac{16\tau^3\kappa^2}{(4\tau - \rho_{2s^*+2\tilde{s}})^2}\right)$ -smooth over  $\mathcal{B}\left(\Sigma^*, \frac{\rho_{2s^*+2\tilde{s}}}{4\tau\kappa}\right)$ .  $\square$

### B. Preliminary Lemmas

**Lemma 31** (Local Restricted Hessian Smoothness). *Recall  $\tilde{s}$  introduced in Assumption 7. We assert the existence of constant  $L_{2s^*+2\tilde{s}}$  and  $r$  such that for any  $\Sigma, \Sigma' \in \mathcal{B}(\Sigma^*, r)$  with  $\|\Sigma_{\tilde{S}}\|_0 \leq \tilde{s}$  and  $\|\Sigma'_{\tilde{S}}\|_0 \leq \tilde{s}$ , the following inequality is satisfied:*

$$\sup_{\mathbf{U} \in \Omega, \|\mathbf{U}\|_F=1} \|\mathbf{U}\|_{\nabla^2 f(\Sigma') - \nabla^2 f(\Sigma)}^2 \leq L_{2s^*+2\tilde{s}} \|\Sigma - \Sigma'\|_F^2,$$

where  $\Omega = \{\mathbf{U} \mid \text{supp}(\mathbf{U}) \subseteq (\text{supp}(\Sigma) \cup \text{supp}(\Sigma'))\}$ .

*Proof.* To establish the Lipschitz continuity of  $\nabla^2 f(\Sigma)$ , consider the third derivative  $\nabla^3 f(\Sigma)$ , which is expressed as  $-2\tau \Sigma^{-1} \otimes \Sigma^{-1} \otimes \Sigma^{-1}$ . Employing a proof technique analogous to that utilized in Lemma 16, we have

$$\lambda_{\max}(|\nabla^3 f(\Sigma)|) \leq 2\tau \lambda_{\min}^{-3}(\Sigma) \leq 2\tau \kappa^3.$$

This finding allows us to conveniently define the Lipschitz constant for the Hessian matrix, represented as  $L_{2s^*+2\tilde{s}}$ , to be  $2\tau \kappa^3$ .  $\square$

**Lemma 32.** *Under the same condition of Lemma 26, we have the following basic inequality*

$$\left\langle \nabla f(\tilde{\Sigma}^1) - \nabla f(\Sigma^*), \tilde{\Sigma}^1 - \Sigma^* \right\rangle \leq \frac{c_1 \lambda^2 s^*}{\rho_{2s^*+2\tilde{s}}}.$$

*Proof.* Applying Lemma 26 with  $k=1$ , we obtain

$$\|\tilde{\Sigma}^1 - \Sigma^*\|_F \leq \frac{2 + \sqrt{2}}{2\rho_{2s^*+2\tilde{s}}} \lambda \sqrt{s^*}. \quad (21)$$

Moreover, by invoking Lemma 24 yields that

$$\begin{aligned}
\|\tilde{\Sigma}^1 - \Sigma^*\|_1 &\leq \left\| (\tilde{\Sigma}^1 - \Sigma^*)_{\tilde{\mathcal{E}}^1} \right\|_1 + \left\| (\tilde{\Sigma}^1 - \Sigma^*)_{\mathcal{E}^1} \right\|_1 \\
&\leq 6 \left\| (\tilde{\Sigma}^1 - \Sigma^*)_{\mathcal{E}^1} \right\|_1,
\end{aligned}$$

where  $\mathcal{E}^1$  can be taken as  $\mathcal{S}^*$ . Combining these results with (21), we conclude

$$\left\| (\tilde{\Sigma}^1 - \Sigma^*)_{\mathcal{S}^*} \right\|_1 \leq \sqrt{s^*} \|\tilde{\Sigma}^1 - \Sigma^*\|_F \leq \frac{2 + \sqrt{2}}{2\rho_{2s^*+2\tilde{s}}} \lambda s^*.$$

Therefore, we obtain

$$\|\tilde{\Sigma}^1 - \Sigma^*\|_1 \leq 6 \left\| (\tilde{\Sigma}^1 - \Sigma^*)_{\mathcal{S}^*} \right\|_1 \leq \frac{3(2 + \sqrt{2})}{\rho_{2s^*+2\tilde{s}}} \lambda s^*.$$

Since  $\tilde{\Sigma}^1$  is a  $\varepsilon$ -optimal solution, we obtain

$$\begin{aligned}
& \left\langle \nabla f(\tilde{\Sigma}^1) - \nabla f(\Sigma^*), \tilde{\Sigma}^1 - \Sigma^* \right\rangle \\
&\leq \left\| \nabla f(\tilde{\Sigma}^1) + \Lambda \odot \tilde{\Xi}^1 \right\|_{\max} \|\tilde{\Sigma}^1 - \Sigma^*\|_1 \\
&\quad + \left\| \Lambda \odot \tilde{\Xi}^1 + \nabla f(\Sigma^*) \right\|_{\max} \|\tilde{\Sigma}^1 - \Sigma^*\|_1 \\
&\leq c_0 \lambda \|\tilde{\Sigma}^1 - \Sigma^*\|_1 \leq \frac{c_1 \lambda^2 s^*}{\rho_{2s^*+2\tilde{s}}},
\end{aligned}$$

for some constants  $c_0$  and  $c_1$ .  $\square$

**Lemma 33.** *Given  $\omega_{\Lambda^{k-1}}(\tilde{\Sigma}^k) \leq \frac{\lambda}{8}$ , we have that for all  $t \geq 1$  at the  $(k+1)$ -th stage,  $\omega_{\Lambda^k}(\Sigma_t^{(k+1)}) \leq \frac{\lambda}{4}$  and*

$$F_{\Lambda^k}(\Sigma_t^{(k+1)}) \leq F_{\Lambda^k}(\Sigma^*) + \frac{\lambda}{4} \|\Sigma_t^k - \Sigma^*\|_1.$$

*Proof.* We begin by establishing the foundational inequality at the  $(k+1)$ -th stage. Note that  $\Sigma_0^{(k+1)} = \tilde{\Sigma}^k$ . By definition, we have

$$\omega_{\Lambda^k}(\Sigma_0) = \min_{\Xi \in \partial \|\Sigma_0\|_1} \left\| \nabla f(\Sigma_0) + \Lambda^k \odot \Xi \right\|_{\max}.$$

Using the triangle inequality, we can expand the above expression as follows:

$$\begin{aligned}
\omega_{\Lambda^k}(\Sigma_0) &\leq \min_{\Xi \in \partial \|\Sigma_0\|_1} \left\| \nabla f(\Sigma_0) + \Lambda^{k-1} \odot \Xi \right\|_{\max} \\
&\quad + \left\| (\Lambda^k - \Lambda^{k-1}) \odot \Xi \right\|_{\max} \\
&\stackrel{(i)}{\leq} \omega_{\Lambda^{k-1}}(\Sigma_0) + \left\| \Lambda^k - \Lambda^{k-1} \right\|_{\max} \\
&\stackrel{(ii)}{\leq} \frac{\lambda}{8} + \frac{\lambda}{8} \leq \frac{\lambda}{4},
\end{aligned}$$

where (i) is from the definition of the approximate KKT condition and  $\Xi$ , and (ii) is from  $\omega_{\Lambda^{k-1}}(\Sigma_0) = \omega_{\Lambda^{k-1}}(\tilde{\Sigma}^k) \leq \frac{\lambda}{8}$

and  $\left\| \Lambda^k - \Lambda^{k-1} \right\|_{\max} \leq \frac{\lambda}{8}$ .

Continuing, for any  $t \geq 0$ , consider

$$\Xi_t = \arg \min_{\Xi \in \partial \|\Sigma_t\|_1} \left\| \nabla f(\Sigma_t) + \Lambda^k \odot \Xi \right\|_{\max}.$$

By the convexity of  $F$ , we have

$$\begin{aligned}
& F_{\Lambda^k}(\Sigma^*) \\
&\geq F_{\Lambda^k}(\Sigma_t) - \left\langle \nabla f(\Sigma_t) + \Lambda^k \odot \Xi_t, \Sigma_t - \Sigma^* \right\rangle \\
&\geq F_{\Lambda^k}(\Sigma_t) - \left\| \nabla f(\Sigma_t) + \Lambda^k \odot \Xi_t \right\|_{\max} \|\Sigma_t - \Sigma^*\|_1 \\
&\stackrel{(i)}{\geq} F_{\Lambda^k}(\Sigma_t) - \frac{\lambda}{4} \|\Sigma_t - \Sigma^*\|_1,
\end{aligned}$$

where (ii) is from that  $\left\| \nabla f(\Sigma_t) + \Lambda^k \odot \Xi_t \right\|_{\max} \leq \frac{\lambda}{4}$ .  $\square$

$$\begin{aligned}
& \eta_t \langle \nabla f(\Sigma_t), \Delta \Sigma_t \rangle + \frac{\eta_t^2}{2} \|\Delta \Sigma_t\|_{\nabla^2 f(\Sigma_t)}^2 + \left\| \Lambda^{k-1} \odot (\Sigma_t + \eta_t \Delta \Sigma_t) \right\|_1 \\
& \geq \langle \nabla f(\Sigma_t), \Delta \Sigma_t \rangle + \frac{1}{2} \|\Delta \Sigma_t\|_{\nabla^2 f(\Sigma_t)}^2 + \left\| \Lambda^{k-1} \odot (\Sigma_t + \Delta \Sigma_t) \right\|_1
\end{aligned} \tag{22}$$

$$\begin{aligned}
& \eta_t \langle \nabla f(\Sigma_t), \Delta \Sigma_t \rangle + \frac{\eta_t^2}{2} \|\Delta \Sigma_t\|_{\nabla^2 f(\Sigma_t)}^2 + \eta_t \left\| \Lambda^{k-1} \odot (\Sigma_t + \eta_t \Delta \Sigma_t) \right\|_1 + (1 - \eta_t) \left\| \Lambda^{k-1} \odot \Sigma_t \right\|_1 \\
& \geq \langle \nabla f(\Sigma_t), \Delta \Sigma_t \rangle + \frac{1}{2} \|\Delta \Sigma_t\|_{\nabla^2 f(\Sigma_t)}^2 + \left\| \Lambda^{k-1} \odot (\Sigma_t + \Delta \Sigma_t) \right\|_1
\end{aligned} \tag{23}$$

**Lemma 34.** Suppose  $\|(\Sigma_t)_{\bar{S}}\|_0 \leq \tilde{s}$  and  $\omega_{\Lambda^k}(\Sigma_t) \leq \frac{\lambda}{4}$ . Then, there exists a generic constant  $c_1$  such that

$$\|\Sigma_t - \Sigma^*\|_F \leq \frac{c_1 \lambda \sqrt{s^*}}{\rho_{2s^*+2\tilde{s}}}.$$

*Proof.* The proof is analogous to that of Lemma 26. Therefore, we omit it for brevity. For more details, we refer readers to [71].  $\square$

**Lemma 35.** Recall  $\Sigma_{t+\frac{1}{2}}$  defined in (5) and  $\delta_t$  as in (6). Denote  $\Delta \Sigma_t = \Sigma_{t+\frac{1}{2}} - \Sigma_t$ . Then we have

$$\delta_t \leq -\|\Delta \Sigma_t\|_{\nabla^2 f(\Sigma_t)}^2.$$

*Proof.* We note  $\Delta \Sigma_t$  is the solution for  $\bar{F}(\Sigma | \Sigma_t, \Lambda^{k-1})$ . For any  $\eta_t \in (0, 1]$ , we have formula (22). By the convexity of  $\ell_1$ -norm, we have formula (23). Rearranging the terms, canceling the  $(1 - \eta_t)$  factor from both sides and let  $\eta_t \rightarrow 1$ , we obtain the desired inequality  $\delta_t \leq -\|\Delta \Sigma_t\|_{\nabla^2 f(\Sigma_t)}^2$ .  $\square$

### C. Proof of Theorem 18

**Lemma 36** (Sparsity Preserving Lemma). Suppose that Assumptions 8 and 9 hold. Given  $\Sigma_t^k \in \mathcal{B}(\Sigma^*, r)$  and  $\left\| \left( \Sigma_t^k \right)_{\bar{S}} \right\|_0 \leq \tilde{s}$ , there exists a generic constant  $c_1$  such that

$$\left\| \left( \Sigma_{t+1}^k \right)_{\bar{S}} \right\|_0 \leq \tilde{s} \quad \text{and} \quad \left\| \Sigma_{t+1}^k - \Sigma^* \right\|_F \leq \frac{c_1 \lambda \sqrt{s^*}}{\rho_{2s^*+2\tilde{s}}}.$$

*Proof.* To simplify notation, we omit the stage index  $k$ . Considering  $\Sigma_{t+1}$  as the proximal Newton update, the system satisfies the following equation:

$$\text{mat}(\nabla^2 f(\Sigma_t) \text{vec}(\Sigma_{t+1} - \Sigma_t)) + \nabla f(\Sigma_t) + \Lambda \odot \Xi_{t+1} = \mathbf{0},$$

where  $\Xi_{t+1} \in \partial \|\Sigma_{t+1}\|_1$ .

From Lemma 25, under identical conditions, it holds that  $\|\Lambda_{\bar{\mathcal{E}}}\|_{\min} \geq \frac{\lambda}{2}$  and  $|\mathcal{E}| \leq 2s^*$  for some set  $\mathcal{E} \supseteq \mathcal{S}^*$ . In the following, we decompose

$$\text{mat}(\nabla^2 f(\Sigma_t) \text{vec}(\Sigma_{t+1} - \Sigma_t)) + \nabla f(\Sigma_t)$$

into four parts:

- $\mathbf{V}_1 := \text{mat}(\nabla^2 f(\Sigma_t) \text{vec}(\Sigma_{t+1} - \Sigma^*)),$
- $\mathbf{V}_2 := \text{mat}(\nabla^2 f(\Sigma_t) \text{vec}(\Sigma^* - \Sigma_t)),$
- $\mathbf{V}_3 := \nabla f(\Sigma_t) - \nabla f(\Sigma^*),$
- $\mathbf{V}_4 := \nabla f(\Sigma^*).$

1 For  $\mathbf{V}_1$ , we define set

$$\mathcal{H}_1 = \left\{ (i, j) \in \bar{\mathcal{E}} \mid |(\mathbf{V}_1)_{ij}| \geq \frac{\lambda}{4} \right\}.$$

Referring to Lemma 33, we establish that

$$F_{\Lambda}(\Sigma_{t+1}) \leq F_{\Lambda}(\Sigma^*) + \frac{\lambda}{4} \|\Sigma_{t+1} - \Sigma^*\|_1.$$

This relationship leads to the following inequality:

$$\begin{aligned}
& f(\Sigma_{t+1}) - f(\Sigma^*) \\
& \leq \lambda (\|\Sigma^*\|_1 - \|\Sigma_{t+1}\|_1) + \frac{\lambda}{4} \|\Sigma_{t+1} - \Sigma^*\|_1 \\
& = \lambda (\|\Sigma_{\mathcal{E}}^*\|_1 - \|(\Sigma_{t+1})_{\mathcal{E}}\|_1 - \|(\Sigma_{t+1})_{\bar{\mathcal{E}}}\|_1) + \frac{\lambda}{4} \|\Sigma_{t+1} - \Sigma^*\|_1 \\
& \leq \frac{5\lambda}{4} \|(\Sigma_{t+1})_{\mathcal{E}} - \Sigma_{\mathcal{E}}^*\|_1 - \frac{3\lambda}{4} \|(\Sigma_{t+1})_{\bar{\mathcal{E}}} - \Sigma_{\bar{\mathcal{E}}}^*\|_1,
\end{aligned} \tag{24}$$

where the equality holds since  $\Sigma_{\bar{\mathcal{E}}}^* = 0$ . On the other hand, we have

$$\begin{aligned}
& f(\Sigma_{t+1}) - f(\Sigma^*) \\
& \stackrel{(i)}{\geq} \langle \nabla f(\Sigma^*), \Sigma_{t+1} - \Sigma^* \rangle \\
& \geq -\|\nabla f(\Sigma^*)\|_{\max} \|\Sigma_{t+1} - \Sigma^*\|_1 \\
& \stackrel{(ii)}{\geq} -\frac{\lambda}{4} \|\Sigma_{t+1} - \Sigma^*\|_1 \\
& = -\frac{\lambda}{4} \|(\Sigma_{t+1})_{\mathcal{E}} - \Sigma_{\mathcal{E}}^*\|_1 - \frac{\lambda}{4} \|(\Sigma_{t+1})_{\bar{\mathcal{E}}} - \Sigma_{\bar{\mathcal{E}}}^*\|_1,
\end{aligned} \tag{25}$$

where (i) is from the convexity of  $f$  and (ii) is from Assumption 9. Combining (24) and (25), we conclude

$$\|(\Sigma_{t+1})_{\bar{\mathcal{E}}} - \Sigma_{\bar{\mathcal{E}}}^*\|_1 \leq 3 \|(\Sigma_{t+1})_{\mathcal{E}} - \Sigma_{\mathcal{E}}^*\|_1.$$

Consider a subset  $\mathcal{S}' \subset \mathcal{H}_1$  with  $|\mathcal{S}'| = s' \leq \tilde{s}$ . Select a vector  $\mathbf{v} \in \mathbb{R}^{d^2}$  such that  $\|\mathbf{v}\|_{\max} = 1$  and  $\|\mathbf{v}\|_0 = s'$ , satisfying the condition  $s' \frac{\lambda}{4} \leq \mathbf{v}^\top \nabla^2 f(\Sigma_t) \text{vec}(\Sigma_{t+1} - \Sigma^*)$ . Then, we



have

$$\begin{aligned}
s' \frac{\lambda}{4} &\leq \mathbf{v}^\top \nabla^2 f(\boldsymbol{\Sigma}_t) \text{vec}(\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*) \\
&\leq \left\| \mathbf{v}^\top (\nabla^2 f(\boldsymbol{\Sigma}_t))^{\frac{1}{2}} \right\|_2 \left\| (\nabla^2 f(\boldsymbol{\Sigma}_t))^{\frac{1}{2}} \text{vec}(\boldsymbol{\Sigma}_{t+1} - \boldsymbol{\Sigma}^*) \right\|_2 \\
&\stackrel{(i)}{\leq} c_1 \sqrt{\rho_{2s^*+2\tilde{s}}^+ \rho_{s'}^+} \|\mathbf{v}\|_2 \|\boldsymbol{\Sigma}^* - \boldsymbol{\Sigma}_{t+1}\|_F \\
&\stackrel{(ii)}{\leq} c_1 \sqrt{s' \rho_{2s^*+2\tilde{s}}^+ \rho_{s'}^+} \|\boldsymbol{\Sigma}^* - \boldsymbol{\Sigma}_{t+1}\|_F \\
&\stackrel{(iii)}{\leq} \frac{c_1 \sqrt{s' \rho_{2s^*+2\tilde{s}}^+ \rho_{s'}^+}}{\rho_{2s^*+2\tilde{s}}^-} \lambda \sqrt{s^*},
\end{aligned}$$

where (i) is from SE condition, (ii) is from the definition of  $\mathbf{v}$ , and (iii) is from Lemma 34. From the above inequalities, it follows that

$$s' \leq \frac{c_1 \rho_{2s^*+2\tilde{s}}^+ \rho_{s'}^+ s^*}{(\rho_{2s^*+2\tilde{s}}^-)^2} \leq c_1 \varpi_{2s^*+2\tilde{s}}^2 s^*,$$

where  $\varpi_{2s^*+2\tilde{s}} = \rho_{2s^*+2\tilde{s}}^+ / \rho_{2s^*+2\tilde{s}}^-$  is the condition number. Given that  $s' = |\mathcal{S}'|$  achieves the maximum possible value such that  $s' \leq \tilde{s}$  for any subset  $\mathcal{S}'$  of  $\mathcal{H}_1$ . We conclude  $\mathcal{S}' = \mathcal{H}_1$  to attain the maximum and thus  $|\mathcal{H}_1| = s' \leq c_1 \varpi_{2s^*+2\tilde{s}}^2 s^*$ .

(2) For  $\mathbf{V}_2$ , we define set

$$\mathcal{H}_2 = \left\{ (i, j) \in \bar{\mathcal{E}} \mid |(\mathbf{V}_2)_{ij}| \geq \frac{\lambda}{4} \right\}.$$

Similar to the argument of  $\mathcal{H}_1$ , we have  $|\mathcal{H}_2| \leq c_2 \varpi_{2s^*+2\tilde{s}}^2 s^*$ .

(3) For  $\mathbf{V}_3$ , we define set

$$\mathcal{H}_3 = \left\{ (i, j) \in \bar{\mathcal{E}} \mid |(\mathbf{V}_3)_{ij}| \geq \frac{\lambda}{4} \right\}.$$

Consider a vector  $\mathbf{V} \in \mathbb{R}^{d \times d}$  such that  $\|\mathbf{V}\|_{\max} = 1$ ,  $\|\mathbf{V}\|_0 = |\mathcal{H}_3|$ , and

$$\begin{aligned}
&\langle \mathbf{V}, \nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*) \rangle \\
&= \sum_{(i', j') \in \mathcal{H}_3} \mathbf{V}_{i'j'} (\nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*))_{i'j'} \\
&= \sum_{(i', j') \in \mathcal{H}_3} (\nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*))_{ij} \\
&\geq \frac{\lambda}{4}.
\end{aligned} \tag{28}$$

Then we have

$$\begin{aligned}
&\langle \mathbf{V}, \nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*) \rangle \\
&\leq \|\mathbf{V}\|_F \|\nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*)\|_F \\
&\stackrel{(i)}{\leq} \sqrt{|\mathcal{H}_3|} \|\nabla f(\boldsymbol{\Sigma}_t) - \nabla f(\boldsymbol{\Sigma}^*)\|_F \\
&\stackrel{(ii)}{\leq} \rho_{2s^*+2\tilde{s}}^+ \sqrt{|\mathcal{H}_3|} \|\boldsymbol{\Sigma}_t - \boldsymbol{\Sigma}^*\|_F,
\end{aligned} \tag{29}$$

where (i) is from the definition of  $\mathbf{V}$ , and (ii) is from the mean value theorem and analogous argument for  $\mathcal{H}_3$ .

Combining (28) and (29), we have

$$\begin{aligned}
\lambda |\mathcal{H}_3| &\leq 4 \rho_{2s^*+2\tilde{s}}^+ \sqrt{|\mathcal{H}_3|} \|\boldsymbol{\Sigma}_t - \boldsymbol{\Sigma}^*\|_F \\
&\stackrel{(i)}{\leq} 8 \lambda \varpi_{2s^*+2\tilde{s}} \sqrt{3s^* |\mathcal{H}_3|},
\end{aligned}$$

where (i) is from Lemma 34. This implies  $|\mathcal{H}_3| \leq c_3 \varpi_{2s^*+2\tilde{s}}^2 s^*$ .

(4) For  $\mathbf{V}_4$ , we define set

$$\mathcal{H}_4 = \left\{ (i, j) \in \bar{\mathcal{E}} \mid |(\mathbf{V}_4)_{ij}| \geq \frac{\lambda}{4} \right\}.$$

By Assumption 9, we have

$$\begin{aligned}
0 \leq |\mathcal{H}_4| &\leq \sum_{(i,j) \in \bar{\mathcal{E}}} \frac{4}{\lambda} |(\nabla f(\boldsymbol{\Sigma}^*))_{ij}| \cdot \mathbb{I} \left( |(\nabla f(\boldsymbol{\Sigma}^*))_{ij}| > \frac{\lambda}{4} \right) \\
&= \sum_{(i,j) \in \bar{\mathcal{E}}} \frac{4}{\lambda} |(\nabla f(\boldsymbol{\Sigma}^*))_{ij}| \cdot 0 = 0.
\end{aligned}$$

Combining the results for Set  $\mathcal{H}_1 \sim \mathcal{H}_4$ , we have that there exists some constant  $c_0$  such that

$$\left\| \left( \boldsymbol{\Sigma}_{t+\frac{1}{2}} \right)_{\bar{\mathcal{S}}^*} \right\|_0 \leq c_0 \kappa_{2s^*+2\tilde{s}}^2 s^* \leq \tilde{s}.$$

□

**Lemma 37.** Suppose Assumptions 8 and 9 hold. If  $\boldsymbol{\Sigma}^{(t)} \in \mathcal{B}(\boldsymbol{\Sigma}^*, r)$  and  $\|\boldsymbol{\Sigma}_{\bar{\mathcal{S}}^*}^{(t+1)}\|_0 \leq \tilde{s}$ , then for each stage  $k \geq 2$ , we have

$$\left\| \boldsymbol{\Sigma}_{t+1}^k - \hat{\boldsymbol{\Sigma}}^k \right\|_F \leq \frac{\tau \kappa^3}{\rho_{2s^*+2\tilde{s}}^-} \left\| \boldsymbol{\Sigma}_t^k - \hat{\boldsymbol{\Sigma}}^k \right\|_F^2.$$

*Proof.* First, we reformulate the prox-Newton update (5) as formula (26). By the Lemma 36, we have  $\left\| \left( \boldsymbol{\Sigma}_{t+1}^k \right)_{\bar{\mathcal{S}}^*} \right\|_0 \leq \tilde{s}$ . And by the KKT condition, we have formula (27). By the strictly non-expansive property of the proximal operator, we obtain the formula (30).

---


$$\boldsymbol{\Sigma}_{t+1}^k = \arg\min_{\boldsymbol{\Sigma}} \left\langle \nabla f(\boldsymbol{\Sigma}_t^k), \boldsymbol{\Sigma} - \boldsymbol{\Sigma}_t^k \right\rangle + \frac{1}{2} \left\| \boldsymbol{\Sigma} - \boldsymbol{\Sigma}_t^k \right\|_{\nabla^2 f(\boldsymbol{\Sigma}_t^k)}^2 + \left\| \boldsymbol{\Lambda}^{k-1} \odot \boldsymbol{\Sigma} \right\|_1 \tag{26}$$


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$$\hat{\boldsymbol{\Sigma}}^k = \arg\min_{\boldsymbol{\Sigma}} \left\langle \nabla f(\hat{\boldsymbol{\Sigma}}^k), \boldsymbol{\Sigma} - \hat{\boldsymbol{\Sigma}}^k \right\rangle + \frac{1}{2} \left\| \boldsymbol{\Sigma} - \hat{\boldsymbol{\Sigma}}^k \right\|_{\nabla^2 f(\boldsymbol{\Sigma}_t^k)}^2 + \left\| \boldsymbol{\Lambda}^{k-1} \odot \hat{\boldsymbol{\Sigma}}^k \right\|_1 \tag{27}$$

$$\begin{aligned} \left\| \Sigma_{t+1}^k - \hat{\Sigma}^k \right\|_{\nabla^2 f(\hat{\Sigma}^k)}^2 &\leq \text{vec}^\top \left( \Sigma_{t+1}^k - \hat{\Sigma}^k \right) \left[ \nabla^2 f \left( \Sigma_t^k \right) \text{vec} \left( \Sigma_t^k - \hat{\Sigma}^k \right) + \left( \nabla f \left( \hat{\Sigma}^k \right) - \nabla f \left( \Sigma_t^k \right) \right) \right] \\ &\leq \left\| \Sigma_{t+1}^k - \hat{\Sigma}^k \right\|_F \left\| \nabla^2 f \left( \Sigma_t^k \right) \text{vec} \left( \Sigma_t^k - \hat{\Sigma}^k \right) + \left( \nabla f \left( \hat{\Sigma}^k \right) - \nabla f \left( \Sigma_t^k \right) \right) \right\|_2 \end{aligned} \quad (30)$$

$$\begin{aligned} \left\| \Sigma_{t+1}^k - \hat{\Sigma}^k \right\|_F &\leq \frac{1}{\rho_{2s^*+2\tilde{s}}} \left\| \nabla^2 f \left( \Sigma_t^k \right) \text{vec} \left( \Sigma_t^k - \hat{\Sigma}^k \right) + \left( \nabla f \left( \hat{\Sigma}^k \right) - \nabla f \left( \Sigma_t^k \right) \right) \right\|_2 \\ &= \frac{1}{\rho_{2s^*+2\tilde{s}}} \left\| \int_0^1 \left[ \nabla^2 f \left( \Sigma_t^k + \theta \left( \hat{\Sigma}^k - \Sigma_t^k \right) \right) - \nabla^2 f \left( \Sigma_t^k \right) \right] \cdot \text{vec} \left( \hat{\Sigma}^k - \Sigma_t^k \right) d\theta \right\|_2 \\ &\leq \frac{1}{\rho_{2s^*+2\tilde{s}}} \int_0^1 \left\| \left[ \nabla^2 f \left( \Sigma_t^k + \theta \left( \hat{\Sigma}^k - \Sigma_t^k \right) \right) - \nabla^2 f \left( \Sigma_t^k \right) \right] \cdot \text{vec} \left( \hat{\Sigma}^k - \Sigma_t^k \right) \right\|_2 d\theta \\ &\stackrel{(i)}{\leq} \frac{\tau \kappa^3}{\rho_{2s^*+2\tilde{s}}} \left\| \Sigma_t^k - \hat{\Sigma}^k \right\|_F^2 \end{aligned} \quad (32)$$

Note that both  $\|\Sigma_{t+1}\|_0 \leq \tilde{s}$  and  $\|\hat{\Sigma}^k\|_0 \leq \tilde{s}$ . On the other hand, from the SE properties, we have

$$\left\| \Sigma_{t+1}^k - \hat{\Sigma}^k \right\|_{\nabla^2 f(\hat{\Sigma}^k)}^2 \geq \rho_{2s^*+2\tilde{s}}^- \left\| \Sigma_{t+1}^k - \hat{\Sigma}^k \right\|_F^2. \quad (31)$$

Combining (30) and (31), we have formula (32), where (i) is due to the local restricted Hessian smoothness of  $f$ . Then the proof is finished.  $\square$

In the subsequent analysis, it is essential to utilize the property that the iterates  $\Sigma_t \in \mathcal{B}(\hat{\Sigma}^k, 2r)$  rather than  $\Sigma_t \in \mathcal{B}(\Sigma^*, r)$  for the convergence analysis of the proximal Newton method. This property is valid due to the simultaneous inclusion of  $\Sigma_t \in \mathcal{B}(\Sigma^*, r)$  and  $\hat{\Sigma}^k \in \mathcal{B}(\Sigma^*, r)$ . Consequently, it follows that  $\Sigma_t \in \mathcal{B}(\hat{\Sigma}^k, 2r)$ , where  $2r = \frac{\rho_{2s^*+2\tilde{s}}^-}{2\tau\kappa}$  represents the radius of the quadratic convergence region for the proximal Newton algorithm.

**Lemma 38.** *Suppose that Assumptions 4, 7, 3, and 8 hold. If  $\Sigma_t \in \mathcal{B}(\hat{\Sigma}^k, 2r)$  and  $\|(\Sigma_t)_{\bar{S}^*}\|_0 \leq \tilde{s}$  at each stage  $k \geq 2$  with  $\alpha = 0.3$ , then  $\eta_t = 1$ . Further, we have*

$$F \left( \Sigma_{t+1}^k \right) \leq F \left( \Sigma_t^k \right) + \frac{1}{4} \delta_t.$$

*Proof.* Recall  $\Sigma_{t+\frac{1}{2}}$  defined in (5) and denote  $\Delta \Sigma_t = \Sigma_{t+\frac{1}{2}} - \Sigma_t$ . Then we have

$$\begin{aligned} &\left\| \Delta \Sigma_t^k \right\|_F^2 \\ &\stackrel{(i)}{\leq} \left\| \Sigma_t^k - \hat{\Sigma}^k \right\|_F + \left\| \Sigma_{t+\frac{1}{2}}^k - \hat{\Sigma}^k \right\|_F \\ &\stackrel{(ii)}{\leq} \left\| \Sigma_t^k - \hat{\Sigma}^k \right\|_F + \frac{\tau \kappa^3}{\rho_{2s^*+2\tilde{s}}} \left\| \Sigma_t^k - \hat{\Sigma}^k \right\|_F^2 \\ &\stackrel{(iii)}{\leq} \frac{3}{2} \left\| \Sigma_t^k - \hat{\Sigma}^k \right\|_F, \end{aligned} \quad (33)$$

where (i) is from triangle inequality, (ii) is from Lemma 37, and (iii) is from  $\left\| \Sigma_t^k - \hat{\Sigma}^k \right\|_F \leq R \leq \frac{\rho_{2s^*+2\tilde{s}}^-}{2\tau\kappa^3}$ .

By Lemma 36, we have  $\|(\Sigma_t)_{\bar{S}^*}\|_0 \leq 2\tilde{s}$ . Then by expanding  $F$ , we have formula (34), where (i) is from the restricted Hessian smooth condition, (ii) and (iv) are from Lemma 35, (iii) is from the same argument of (31), and (v) is from (33),  $\delta_t < 0$ , and  $\left\| \Sigma_t^k - \hat{\Sigma}^k \right\|_F \leq r \leq \frac{\rho_{2s^*+2\tilde{s}}^-}{2\tau\kappa^3}$ . The  $\eta_t = 1$  can be demonstrated analogously from the analysis in [87] and [89], thus we omit it.  $\square$

**Lemma 39.** *Suppose that Assumptions 4, 7, 3, 8, and 9 hold. To achieve the approximate KKT condition  $\omega(\Sigma_t) \leq \varepsilon$  for any  $\varepsilon > 0$  at each stage  $k \geq 2$ , the number of iterations for proximal Newton updates is at most*

$$\log \log (3\rho_{2s^*+2\tilde{s}}^+/\varepsilon).$$

*Proof.* Based on the solution  $\Sigma_{t-1}$  obtained from the  $(t-1)$ -th iteration, we consider the optimal solution at  $t$ -th iteration. By the KKT condition, we have

$$\text{mat} \left( \nabla^2 f(\Sigma_{t-1}) \text{vec}(\Sigma_t - \Sigma_{t-1}) + \nabla f(\Sigma_{t-1}) + \Lambda \odot \Xi_t \right) = \mathbf{0},$$

where  $\Xi_t \in \partial \|\Sigma_t\|_1$ . Given any  $\mathbf{V} \in \mathbb{R}^{d \times d}$  satisfying  $\|\mathbf{V}\|_2 \leq \|\mathbf{V}\|_1 = 1$  and  $\|\mathbf{V}\|_0 \leq 2s^* + 2\tilde{s}$ , we derive formula (35), where (i) is from mean value theorem with some  $\check{\Sigma} = (1-\theta)\Sigma_{t-1} + \theta\Sigma_t$  for some  $\theta \in [0, 1]$  and Cauchy-Schwarz inequality, and (ii) is from the SE properties. By taking the supremum of the L.H.S of (35) with respect to  $\mathbf{V}$ , we have

$$\|\nabla f(\Sigma_t) + \Lambda \odot \Xi_t\|_{\max} \leq 2\rho_{2s^*+2\tilde{s}}^+ \|\Sigma_t - \Sigma_{t-1}\|_F. \quad (36)$$

Then from Lemma (37), we have

$$\begin{aligned} &\left\| \Sigma_{t+1}^k - \hat{\Sigma}^k \right\|_F \\ &\leq (\tau \kappa^3 / \rho_{2s^*+2\tilde{s}}^-)^{1+2+4+\dots+2^{t-1}} \left\| \Sigma_0^k - \hat{\Sigma}^k \right\|_F^2 \\ &\leq \left( (\tau \kappa^3 / \rho_{2s^*+2\tilde{s}}^-) \left\| \Sigma_0^k - \hat{\Sigma}^k \right\|_F^2 \right)^{2^t}. \end{aligned}$$

$$\begin{aligned}
& F(\Sigma_t^k + \Delta \Sigma_t^k) - F(\Sigma_t^k) \\
&= f(\Sigma_t^k + \Delta \Sigma_t^k) - f(\Sigma_t^k) + \|\Lambda^{k-1} \odot (\Sigma_t^k + \Delta \Sigma_t^k)\|_1 - \|\Lambda^{k-1} \odot \Sigma_t^k\|_1 \\
&\stackrel{(i)}{\leq} \langle \nabla f(\Sigma_t^k), \Delta \Sigma_t^k \rangle + \frac{1}{2} \|\Delta \Sigma_t^k\|_{\nabla^2 f(\Sigma_t^k)}^2 + \frac{\tau \kappa^3}{3} \|\Delta \Sigma_t^k\|_F^3 + \|\Lambda^{k-1} \odot (\Sigma_t^k + \Delta \Sigma_t^k)\|_1 - \|\Lambda^{k-1} \odot \Sigma_t^k\|_1 \\
&\stackrel{(ii)}{\leq} \delta_t - \frac{1}{2} \delta_t + \frac{\tau \kappa^3}{3} \|\Delta \Sigma_t^k\|_F^3 \stackrel{(iii)}{\leq} \frac{1}{2} \delta_t + \frac{\tau \kappa^3}{3 \rho_{2s^*+2\tilde{s}}} \|\Delta \Sigma_t^k\|_{\nabla^2 f(\Sigma)}^2 \|\Delta \Sigma_t^k\|_F \\
&\stackrel{(iv)}{\leq} \left( \frac{1}{2} - \frac{\tau \kappa^3}{3 \rho_{2s^*+2\tilde{s}}} \|\Delta \Sigma_t^k\|_F \right) \delta_t \stackrel{(v)}{\leq} \frac{1}{4} \delta_t
\end{aligned} \tag{34}$$

$$\begin{aligned}
\langle \nabla f(\Sigma_t) + \Lambda \odot \Xi_t, \mathbf{V} \rangle &= \langle \nabla f(\Sigma_t) - \nabla f(\Sigma_{t-1}) - \text{mat}(\nabla^2 f(\Sigma_{t-1}) \text{vec}(\Sigma_t - \Sigma_{t-1})), \mathbf{V} \rangle \\
&= \langle \nabla f(\Sigma_t) - \nabla f(\Sigma_{t-1}), \mathbf{V} \rangle - (\nabla^2 f(\Sigma_{t-1}) \text{vec}(\Sigma_t - \Sigma_{t-1}))^\top \text{vec}(\mathbf{V}) \\
&\stackrel{(i)}{\leq} \left\| \left( \nabla^2 f(\tilde{\Sigma}) \right)^{\frac{1}{2}} \text{vec}(\Sigma_t - \Sigma_{t-1}) \right\|_2 \cdot \left\| \text{vec}^\top(\mathbf{V}) \left( \nabla^2 f(\tilde{\Sigma}) \right)^{\frac{1}{2}} \right\|_2 \\
&\quad + \left\| \left( \nabla^2 f(\Sigma_{t-1}) \right)^{\frac{1}{2}} \text{vec}(\Sigma_t - \Sigma_{t-1}) \right\|_2 \cdot \left\| \text{vec}^\top(\mathbf{V}) \left( \nabla^2 f(\Sigma_{t-1}) \right)^{\frac{1}{2}} \right\|_2 \\
&\stackrel{(ii)}{\leq} 2 \rho_{2s^*+2\tilde{s}}^+ \|\Sigma_t - \Sigma_{t-1}\|_F,
\end{aligned} \tag{35}$$

By (35) and (33) by taking  $\Delta \Sigma_{t-1} = \Sigma_t - \Sigma_{t-1}$ , we obtain *hold for all  $t \geq T$ :*

$$\begin{aligned}
& \omega_{\Lambda^{k-1}}(\Sigma_t^k) \\
&\leq 2 \rho_{2s^*+2\tilde{s}}^+ \|\Sigma_t^k - \Sigma_{t-1}^k\|_F \\
&\leq 3 \rho_{2s^*+2\tilde{s}}^+ \|\Sigma_{t-1}^k - \hat{\Sigma}^k\|_F \\
&\leq 3 \rho_{2s^*+2\tilde{s}}^+ \left( (\tau \kappa^3 / \rho_{2s^*+2\tilde{s}}^-) \|\Sigma_0^k - \hat{\Sigma}^k\|_F \right)^{2^t}.
\end{aligned}$$

By requiring the R.H.S equal to  $\varepsilon$  we obtain

$$\begin{aligned}
t &= \log \left( \log \left( \frac{3 \rho_{2s^*+2\tilde{s}}^+}{\varepsilon} \right) / \log \left( \frac{\rho_{2s^*+2\tilde{s}}^-}{\tau \kappa^3 \|\Sigma_0^k - \hat{\Sigma}^k\|_F} \right) \right) \\
&= \log \log \left( \frac{3 \rho_{2s^*+2\tilde{s}}^+}{\varepsilon} \right) - \log \log \left( \frac{\rho_{2s^*+2\tilde{s}}^-}{\tau \kappa^3 \|\Sigma_0^k - \hat{\Sigma}^k\|_F} \right) \\
&\stackrel{(i)}{\leq} \log \log \left( \frac{3 \rho_{2s^*+2\tilde{s}}^+}{\varepsilon} \right) - \log \log 4 \\
&\leq \log \log \left( \frac{3 \rho_{2s^*+2\tilde{s}}^+}{\varepsilon} \right),
\end{aligned}$$

where (i) is from the fact that  $\|\Sigma_0^k - \hat{\Sigma}^k\|_F \leq r = \frac{\rho_{2s^*+2\tilde{s}}^-}{4 \tau \kappa^3}$ .  $\square$

#### D. Proof of Theorem 17

**Lemma 40.** Suppose that Assumptions 4, 7, 3, 8, and 9 hold. After sufficiently many iterations  $T < \infty$ , the following results

$$\|(\Sigma_t)_{\overline{\mathcal{S}^*}}\|_0 \leq \tilde{s} \text{ and } F_{\Lambda^{(0)}}(\Sigma_t^1) \leq F_{\Lambda^{(0)}}(\Sigma^*) + \frac{15 \lambda^2 s^*}{4 \rho_{2s^*+2\tilde{s}}^-}.$$

*Proof.* Building upon the framework established in Lemma 35, Lemma 38, it can be concluded that the objective  $F_{\Lambda^{(0)}}$  exhibits a sufficient decrease in each iteration of the proximal Newton step, as also discussed in [90]. Consequently, there exists a constant  $T$  such that for all  $t \geq T$ , we have

$$F_{\Lambda^{(0)}}(\Sigma_t^1) \leq F_{\Lambda^{(0)}}(\Sigma^*) + \frac{\lambda}{4} \|\Sigma_t^1 - \Sigma^*\|_1,$$

where  $\|\Sigma_t^1 - \Sigma^*\|_1 \leq \frac{c_1 \lambda \sqrt{s^*}}{\rho_{2s^*+2\tilde{s}}^-}$  derived from a similar argument presented in the proof of Lemma 25. The subsequent analysis follows a pattern analogous to that of Lemma 36. Let  $S_n = \left\{ (i, j) \mid \left| \nabla f(\tilde{\Sigma})_{ij} \right| = \lambda \right\}$  and consider the set  $\left\{ (i, j) \mid (\tilde{\Sigma})_{ij} \neq 0 \right\} \subseteq S_n$ . It is required to show  $|S_n| \leq s^* + \tilde{s}$ . To demonstrate this, we decompose  $S_n$  into two distinct parts:

- $S_n^1 : \left\{ (i, j) \notin \overline{\mathcal{S}^*} \mid \left| (\nabla f(\tilde{\Sigma}) - \nabla f(\Sigma^*))_{ij} \right| \geq \frac{\lambda}{2} \right\},$
- $S_n^2 : \left\{ (i, j) \notin \overline{\mathcal{S}^*} \mid \left| (\nabla f(\Sigma^*))_{ij} \right| > \frac{\lambda}{2} \right\}.$

We have  $S_n \subseteq \mathcal{S}^* \cup S_n^1 \cup S_n^2$  and establish upper bounds for each.

1 For  $S_n^1$ , consider  $S'$  with maximum size  $s' = |S'| \leq \tilde{s}$  such that  $S' \subseteq S_n^1$ . Then there exists a matrix  $\mathbf{V}$  such that  $\|\mathbf{V}\|_{\max} = 1$  and  $\|\mathbf{V}\|_0 = s'$ , satisfying

$$\frac{\lambda s'}{2} \leq \left\langle \nabla f(\tilde{\Sigma}) - \nabla f(\Sigma^*), \mathbf{V} \right\rangle.$$

By the mean value theorem, there exists some  $\theta \in [0, 1]$  such that

$$\begin{aligned} & \nabla f(\tilde{\Sigma}) - \nabla f(\Sigma^*) \\ &= \text{mat} \left( \nabla^2 f \left( \theta \tilde{\Sigma} + (1 - \theta) \Sigma^* \right) \text{vec}(\tilde{\Sigma} - \Sigma^*) \right). \end{aligned}$$

For simplicity, we define  $\mathbf{H} = \nabla^2 f(\rho \tilde{\Sigma} + (1 - \rho) \Sigma^*)$ . Applying the Cauchy-Schwartz inequality, we have

$$\begin{aligned} \frac{\lambda s'}{2} &\leq \text{vec}^\top(\mathbf{V}) \mathbf{H} \text{vec}(\tilde{\Sigma} - \Sigma^*) \\ &\leq \underbrace{\left\| \text{vec}^\top(\mathbf{V}) (\mathbf{H})^{\frac{1}{2}} \right\|_2}_{\mathbf{I}} \underbrace{\left\| (\mathbf{H})^{\frac{1}{2}} \text{vec}(\tilde{\Sigma} - \Sigma^*) \right\|_2}_{\mathbf{II}}. \end{aligned} \quad (37)$$

We now establish upper bounds for terms **I** and **II** respectively. Let  $\tilde{\Sigma}, \Sigma^* \in \mathcal{B}(\Sigma^*, r)$ , any convex combination of  $\tilde{\Sigma}$  and  $\Sigma^*$  also falls in  $\mathcal{B}(\Sigma^*, r)$ . The localized sparse eigenvalue condition can be applied to  $\mathbf{H}$ .

i) For term **I**, based on Definition (6) and Assumption 7, we have

$$\begin{aligned} \left\| \text{vec}^\top(\mathbf{U}) (\mathbf{H})^{\frac{1}{2}} \right\|_2 &\leq \sqrt{\rho_{2s^*+2\tilde{s}}^+} \|\mathbf{U}\|_{\text{F}}^2 \\ &\leq \sqrt{\rho_{2s^*+2\tilde{s}}^+} (\|\mathbf{U}\|_1 \|\mathbf{U}\|_{\max})^{\frac{1}{2}} \\ &\leq \sqrt{\rho_{2s^*+2\tilde{s}}^+} \sqrt{s'}. \end{aligned}$$

ii) For **II**, using Lemma 32, we have

$$\begin{aligned} & \left\| (\mathbf{H})^{\frac{1}{2}} \text{vec}(\tilde{\Sigma} - \Sigma^*) \right\|_2 \\ &\leq \left\langle \nabla f(\tilde{\Sigma}) - \nabla f(\Sigma^*), \tilde{\Sigma} - \Sigma^* \right\rangle \\ &\leq \frac{c_1 \lambda^2 s^*}{\rho_{2s^*+2\tilde{s}}^-}. \end{aligned}$$

By substituting the bounds for **I** and **II** back into (37), we obtain

$$\frac{\lambda s'}{2} \leq \sqrt{\rho_{2s^*+2\tilde{s}}^+} \sqrt{s'} \times \frac{c_1 \lambda \sqrt{s^*}}{\rho_{2s^*+2\tilde{s}}^-}.$$

Multiplying both sides of this inequality by  $(\frac{\lambda}{2})^{\frac{1}{2}}$  and squaring gives us

$$s' \leq \frac{4\rho_{2s^*+2\tilde{s}}^+ c_2 s^*}{(\rho_{2s^*+2\tilde{s}}^-)^2} < \tilde{s}, \quad (38)$$

where the last inequality follows from our assumption. Since  $s' = |S'|$  achieves the maximum possible value such that  $s' \leq \tilde{s}$  for any subset  $S'$  of  $S_n^1$  and 38 shows that  $s' < \tilde{s}$ , we conclude that  $S' = S_n^1$ , and thus

$$|S_n^1| = s' \leq \left\lfloor \frac{4\rho_{2s^*+2\tilde{s}}^+ c_2 s^*}{(\rho_{2s^*+2\tilde{s}}^-)^2} \right\rfloor < \tilde{s}.$$

(2) For  $S_n^2$ , Given  $\|\nabla f(\Sigma^*)\|_{\max} + \varepsilon \leq \frac{\lambda}{4}$ , it follows that  $S_n^2 = \emptyset$  and thus  $|S_n^2| = 0$ .

Combining all the results, we have  $\left\| \Sigma_{S^*}^{(t)} \right\|_0 \leq \tilde{s}$ .  $\square$

**Lemma 41.** Suppose that Assumptions 7, 8 and 9 hold. If  $\left\| (\Sigma_t^1)_{\tilde{S}^*} \right\|_0 \leq \tilde{s}$ , and  $F_{\Lambda^{(0)}}(\Sigma_t^1) \leq F_{\Lambda^{(0)}}(\Sigma^*) + \frac{15\lambda^2 s^*}{4\rho_{2s^*+2\tilde{s}}^-}$ , we have

$$\left\| \Sigma_t^1 - \Sigma^* \right\|_2 \leq \frac{c_1 \lambda \sqrt{s^*}}{\rho_{2s^*+2\tilde{s}}^-}$$

for some constant  $c_1$  and

$$\left\| \Sigma_{t+1}^1 - \hat{\Sigma}^1 \right\|_2 \leq \frac{\tau \kappa^3}{\rho_{2s^*+2\tilde{s}}^-} \left\| \Sigma_t^1 - \hat{\Sigma}^1 \right\|_2^2.$$

*Proof.* The estimation error is derived analogously from [71] and [89]; thus, we omit it here. The claim of quadratic convergence follows directly from Lemma (37) given sparse solutions.  $\square$

**Lemma 42.** Suppose that Assumptions 7, 8, and 9 hold. If  $\left\| (\Sigma_T^1)_{\tilde{S}^*} \right\|_0 \leq \tilde{s}$ , and  $F_{\Lambda^{(0)}}(\Sigma_T^1) \leq F_{\Lambda^1}(\Sigma^*) + \frac{15\lambda^2 s^*}{4\rho_{2s^*+2\tilde{s}}^-}$  at some iteration  $T$ , we need at most

$$T_1 \leq \log \log \left( \frac{3\rho_{2s^*+2\tilde{s}}^+}{\varepsilon} \right)$$

extra iterations of the proximal Newton updates such that  $\omega_{\Lambda^{(0)}}(\Sigma_{T+T_1}^1) \leq \frac{\lambda}{8}$ .

*Proof.* The maximum number of iterations for the proximal Newton update is determined by integrating Lemma (40) and Lemma (39). Note that

$$T_1 \leq \log \frac{\log \left( \frac{3\rho_{2s^*+2\tilde{s}}^+}{\varepsilon} \right)}{\log \left( \frac{\rho_{2s^*+2\tilde{s}}^-}{\tau \kappa^3 \left\| \Sigma^{(T+1)} - \hat{\Sigma}^{\{1\}} \right\|_{\text{F}}} \right)}.$$

Then we obtain the result from  $\left\| \Sigma_{T+1}^1 - \hat{\Sigma}^1 \right\|_{\text{F}} \leq r = \frac{\rho_{2s^*+2\tilde{s}}^-}{4\tau \kappa}$ .  $\square$

## APPENDIX E SUPPLEMENTARY TABLES

TABLE I  
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE BANDED MATRIX

|                   | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |
|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |
| $d = 80, n = 50$  |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE               | 27.7825<br>(0.2616) | 24.1565<br>(0.1648) | 22.1561<br>(0.1246) | 19.9998<br>(0.1051) | 24.3152<br>(0.2551) | 20.1542<br>(0.1566) | 19.4295<br>(0.1047) | 18.6415<br>(0.1015) |
| FRE               | 1.3876<br>(0.0546)  | 1.2065<br>(0.0591)  | 1.1066<br>(0.0198)  | 0.9989<br>(0.0095)  | 1.2144<br>(0.0357)  | 1.0066<br>(0.0296)  | 0.9704<br>(0.0210)  | 0.9311<br>(0.0090)  |
| FPR               | 0.1615<br>(0.0702)  | 0.1528<br>(0.0765)  | 0.1137<br>(0.0686)  | 0.1053<br>(0.0552)  | 0.1256<br>(0.0699)  | 0.1038<br>(0.0438)  | 0.0681<br>(0.0423)  | 0.0311<br>(0.0217)  |
| TPR               | 0.8072<br>(0.0368)  | 0.8491<br>(0.0382)  | 0.8774<br>(0.0397)  | 0.8955<br>(0.0399)  | 0.8413<br>(0.0397)  | 0.8618<br>(0.0407)  | 0.8957<br>(0.0412)  | 0.9214<br>(0.0420)  |
| $d = 100, n = 50$ |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE               | 28.1276<br>(0.2492) | 26.5312<br>(0.2151) | 23.1562<br>(0.1813) | 20.1532<br>(0.1513) | 24.8612<br>(0.2263) | 21.0017<br>(0.2052) | 20.1562<br>(0.1673) | 18.8964<br>(0.1233) |
| FRE               | 1.1012<br>(0.0435)  | 1.0387<br>(0.0051)  | 0.9066<br>(0.0736)  | 0.7890<br>(0.0435)  | 0.9734<br>(0.0315)  | 0.8223<br>(0.0184)  | 0.7892<br>(0.0163)  | 0.7398<br>(0.0056)  |
| FPR               | 0.1511<br>(0.0851)  | 0.1341<br>(0.0793)  | 0.1056<br>(0.0712)  | 0.0915<br>(0.0663)  | 0.1105<br>(0.0714)  | 0.0557<br>(0.0627)  | 0.0303<br>(0.0487)  | 0.0030<br>(0.0226)  |
| TPR               | 0.8157<br>(0.0337)  | 0.8581<br>(0.0391)  | 0.9167<br>(0.0401)  | 0.9294<br>(0.0413)  | 0.8780<br>(0.0402)  | 0.8860<br>(0.0412)  | 0.9070<br>(0.0422)  | 0.9421<br>(0.0430)  |

|                    | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |
|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |
| $d = 80, n = 100$  |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE                | 26.5923<br>(0.2436) | 23.8546<br>(0.1534) | 21.4165<br>(0.1132) | 19.4620<br>(0.1052) | 23.7515<br>(0.2435) | 19.5612<br>(0.1530) | 18.5164<br>(0.1106) | 17.8962<br>(0.0945) |
| FRE                | 1.3282<br>(0.0973)  | 1.1914<br>(0.0275)  | 1.0697<br>(0.1016)  | 0.9720<br>(0.0099)  | 1.1863<br>(0.0773)  | 0.9770<br>(0.0176)  | 0.9248<br>(0.0012)  | 0.8938<br>(0.0009)  |
| FPR                | 0.1572<br>(0.0425)  | 0.1315<br>(0.0407)  | 0.1152<br>(0.0319)  | 0.0873<br>(0.0287)  | 0.1137<br>(0.0366)  | 0.0852<br>(0.0287)  | 0.0489<br>(0.0209)  | 0.0241<br>(0.0174)  |
| TPR                | 0.8117<br>(0.0371)  | 0.8546<br>(0.0392)  | 0.8810<br>(0.0401)  | 0.9016<br>(0.0408)  | 0.8455<br>(0.0400)  | 0.8821<br>(0.0412)  | 0.9016<br>(0.0418)  | 0.9425<br>(0.0431)  |
| $d = 100, n = 100$ |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE                | 27.8613<br>(0.2161) | 24.2361<br>(0.2111) | 22.2613<br>(0.1562) | 19.9131<br>(0.1162) | 24.2315<br>(0.2023) | 21.1156<br>(0.1762) | 19.8216<br>(0.1230) | 18.1566<br>(0.1055) |
| FRE                | 1.0908<br>(0.0251)  | 0.9489<br>(0.0156)  | 0.8716<br>(0.0013)  | 0.7796<br>(0.0012)  | 0.9487<br>(0.0130)  | 0.8267<br>(0.0104)  | 0.7761<br>(0.0061)  | 0.7109<br>(0.0023)  |
| FPR                | 0.1462<br>(0.0487)  | 0.1227<br>(0.0415)  | 0.1005<br>(0.0326)  | 0.0905<br>(0.0291)  | 0.1126<br>(0.0392)  | 0.0756<br>(0.0311)  | 0.0448<br>(0.0265)  | 0.0131<br>(0.0059)  |
| TPR                | 0.8307<br>(0.0380)  | 0.8706<br>(0.0401)  | 0.8913<br>(0.0409)  | 0.9212<br>(0.0411)  | 0.8605<br>(0.0416)  | 0.8915<br>(0.0423)  | 0.9172<br>(0.0427)  | 0.9512<br>(0.0436)  |

TABLE II  
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE BLOCK MATRIX

|                   | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |
|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |
| $d = 80, n = 50$  |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE               | 14.2562<br>(0.0172) | 12.5602<br>(0.0107) | 11.5627<br>(0.0095) | 10.1561<br>(0.0073) | 13.5615<br>(0.0147) | 11.4833<br>(0.0081) | 10.1566<br>(0.0072) | 9.1562<br>(0.0056)  |
| FRE               | 0.9612<br>(0.0183)  | 0.8468<br>(0.0113)  | 0.7796<br>(0.0099)  | 0.6847<br>(0.0081)  | 0.9143<br>(0.0176)  | 0.7742<br>(0.0106)  | 0.6848<br>(0.0094)  | 0.6173<br>(0.0075)  |
| FPR               | 0.1771<br>(0.0848)  | 0.1662<br>(0.0821)  | 0.1322<br>(0.0792)  | 0.1162<br>(0.0627)  | 0.1607<br>(0.0873)  | 0.1430<br>(0.0791)  | 0.1017<br>(0.0628)  | 0.0924<br>(0.0597)  |
| TPR               | 0.8462<br>(0.0817)  | 0.8716<br>(0.0794)  | 0.9051<br>(0.0762)  | 0.9243<br>(0.0711)  | 0.8513<br>(0.0808)  | 0.8819<br>(0.0791)  | 0.9158<br>(0.0743)  | 0.9356<br>(0.0706)  |
| $d = 100, n = 50$ |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE               | 15.9891<br>(0.0551) | 15.0652<br>(0.0301) | 14.8767<br>(0.0314) | 13.9917<br>(0.0277) | 14.9291<br>(0.0165) | 14.6727<br>(0.0124) | 13.9806<br>(0.0182) | 13.2177<br>(0.0206) |
| FRE               | 0.8869<br>(0.0244)  | 0.8357<br>(0.0196)  | 0.8252<br>(0.0127)  | 0.7761<br>(0.0101)  | 0.8281<br>(0.0186)  | 0.8139<br>(0.0153)  | 0.7755<br>(0.0121)  | 0.7332<br>(0.0098)  |
| FPR               | 0.1672<br>(0.1024)  | 0.1531<br>(0.0868)  | 0.1310<br>(0.0773)  | 0.1011<br>(0.0545)  | 0.1584<br>(0.0997)  | 0.1386<br>(0.0798)  | 0.0996<br>(0.0661)  | 0.0854<br>(0.0602)  |
| TPR               | 0.8554<br>(0.0993)  | 0.8817<br>(0.0813)  | 0.9111<br>(0.0661)  | 0.9306<br>(0.0523)  | 0.8606<br>(0.0985)  | 0.8996<br>(0.0751)  | 0.9224<br>(0.0561)  | 0.9434<br>(0.0413)  |

|                    | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |
|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |
| $d = 80, n = 100$  |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE                | 13.9231<br>(0.0133) | 12.1569<br>(0.0110) | 10.9158<br>(0.0089) | 9.7123<br>(0.0071)  | 12.8873<br>(0.0109) | 11.8153<br>(0.0102) | 10.0862<br>(0.0074) | 9.0561<br>(0.0062)  |
| FRE                | 0.9387<br>(0.0164)  | 0.8196<br>(0.0161)  | 0.7359<br>(0.0088)  | 0.6548<br>(0.0067)  | 0.8689<br>(0.0154)  | 0.7966<br>(0.0106)  | 0.6800<br>(0.0079)  | 0.6106<br>(0.0058)  |
| FPR                | 0.1682<br>(0.0692)  | 0.1531<br>(0.0665)  | 0.1285<br>(0.0594)  | 0.1056<br>(0.0556)  | 0.1502<br>(0.0671)  | 0.1428<br>(0.0598)  | 0.0977<br>(0.0541)  | 0.0887<br>(0.0426)  |
| TPR                | 0.8532<br>(0.0809)  | 0.8762<br>(0.0786)  | 0.9132<br>(0.0758)  | 0.9287<br>(0.0707)  | 0.8762<br>(0.0802)  | 0.8816<br>(0.0775)  | 0.9266<br>(0.0752)  | 0.9532<br>(0.0701)  |
| $d = 100, n = 100$ |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE                | 15.4827<br>(0.0369) | 14.6944<br>(0.0504) | 14.4958<br>(0.0227) | 13.5837<br>(0.0187) | 14.4938<br>(0.0178) | 13.9848<br>(0.0112) | 13.1766<br>(0.0143) | 12.5748<br>(0.0126) |
| FRE                | 0.8588<br>(0.0199)  | 0.8151<br>(0.0185)  | 0.8041<br>(0.0182)  | 0.7535<br>(0.0113)  | 0.8040<br>(0.0178)  | 0.7757<br>(0.0140)  | 0.7309<br>(0.0099)  | 0.6975<br>(0.0067)  |
| FPR                | 0.1593<br>(0.0924)  | 0.1488<br>(0.0803)  | 0.1247<br>(0.0722)  | 0.1006<br>(0.0521)  | 0.1477<br>(0.0911)  | 0.1323<br>(0.0754)  | 0.1109<br>(0.0612)  | 0.0909<br>(0.0509)  |
| TPR                | 0.8595<br>(0.0862)  | 0.8906<br>(0.0615)  | 0.9323<br>(0.0515)  | 0.9344<br>(0.0356)  | 0.8874<br>(0.0731)  | 0.9001<br>(0.0515)  | 0.9389<br>(0.0417)  | 0.9562<br>(0.0258)  |

TABLE III  
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE TOEPLITZ MATRIX

|                   | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |
|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |
| $d = 80, n = 50$  |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE               | 20.3550<br>(0.1918) | 16.7333<br>(0.1217) | 15.3850<br>(0.0933) | 11.6680<br>(0.0421) | 17.7370<br>(0.1065) | 15.2278<br>(0.0803) | 14.1555<br>(0.0622) | 11.1678<br>(0.0531) |
| FRE               | 1.2168<br>(0.0218)  | 1.0003<br>(0.0051)  | 0.9197<br>(0.0062)  | 0.6975<br>(0.0007)  | 1.0603<br>(0.0045)  | 0.9103<br>(0.0003)  | 0.8462<br>(0.0011)  | 0.6676<br>(0.0002)  |
| $d = 100, n = 50$ |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE               | 24.1613<br>(1.0406) | 19.2561<br>(0.7621) | 17.6243<br>(0.0773) | 14.9918<br>(0.0712) | 20.3272<br>(0.7348) | 17.3691<br>(0.0804) | 15.6213<br>(0.0674) | 12.1561<br>(0.0526) |
| FRE               | 1.2891<br>(0.0982)  | 1.0274<br>(0.0461)  | 0.9404<br>(0.0100)  | 0.7999<br>(0.0071)  | 1.0846<br>(0.0021)  | 0.9267<br>(0.0002)  | 0.8335<br>(0.0011)  | 0.6486<br>(0.0003)  |

|                    | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |
|--------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                    | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |
| $d = 80, n = 100$  |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE                | 18.7959<br>(0.1241) | 15.8283<br>(0.0887) | 15.0287<br>(0.0450) | 11.3017<br>(0.0326) | 16.5025<br>(0.0915) | 14.5135<br>(0.0754) | 12.9327<br>(0.0301) | 9.8112<br>(0.0222)  |
| FRE                | 1.1236<br>(0.0062)  | 0.9462<br>(0.0047)  | 0.8984<br>(0.0010)  | 0.6756<br>(0.0005)  | 0.9865<br>(0.0071)  | 0.8676<br>(0.0033)  | 0.7731<br>(0.0009)  | 0.5865<br>(0.0003)  |
| $d = 100, n = 100$ |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE                | 20.3576<br>(1.0006) | 18.0317<br>(0.6862) | 16.7666<br>(0.2165) | 13.5243<br>(0.0465) | 18.0767<br>(0.8611) | 16.6280<br>(0.5626) | 14.1802<br>(0.0560) | 10.6530<br>(0.0447) |
| FRE                | 1.0862<br>(0.0861)  | 0.9621<br>(0.0355)  | 0.8946<br>(0.0086)  | 0.7216<br>(0.0045)  | 0.9645<br>(0.0633)  | 0.8872<br>(0.0072)  | 0.7566<br>(0.0041)  | 0.5684<br>(0.0009)  |

FPR and TPR are not reported due to the fact that it is not sparse.

TABLE IV  
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE PROBABILITY MATRIX WITH  $\mathbf{P} = 0.97$

|                   | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |  | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |
|-------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                   | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |  | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |                     |                     |                     |                     |
| $d = 80, n = 50$  |                     |                     |                     |                     |                     |                     |                     |                     |  |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE               | 38.9147<br>(0.2249) | 26.4884<br>(0.1971) | 14.8860<br>(0.1666) | 14.2507<br>(0.1215) | 33.0702<br>(0.2355) | 18.7424<br>(0.1998) | 13.3777<br>(0.1572) | 9.6156<br>(0.1147)  |  | 36.6029<br>(0.2201) | 24.7278<br>(0.1615) | 14.5060<br>(0.1322) | 14.0390<br>(0.1162) | 29.3833<br>(0.1981) | 17.0730<br>(0.1455) | 12.2319<br>(0.1106) | 8.3190<br>(0.0893)  |
| FRE               | 1.3416<br>(0.0107)  | 0.9132<br>(0.0095)  | 0.5132<br>(0.0074)  | 0.4913<br>(0.0052)  | 1.1401<br>(0.0100)  | 0.6462<br>(0.0083)  | 0.4612<br>(0.0065)  | 0.3315<br>(0.0048)  |  | 1.2619<br>(0.0112)  | 0.8525<br>(0.0089)  | 0.5001<br>(0.0083)  | 0.4840<br>(0.0062)  | 1.0130<br>(0.0099)  | 0.5886<br>(0.0075)  | 0.4217<br>(0.0030)  | 0.2868<br>(0.0024)  |
| FPR               | 0.1216<br>(0.1106)  | 0.1027<br>(0.0917)  | 0.0892<br>(0.0432)  | 0.0774<br>(0.0327)  | 0.1163<br>(0.1023)  | 0.0994<br>(0.0755)  | 0.0652<br>(0.0523)  | 0.0488<br>(0.0348)  |  | 0.1186<br>(0.0812)  | 0.1003<br>(0.0856)  | 0.0844<br>(0.0561)  | 0.0698<br>(0.0222)  | 0.1026<br>(0.0856)  | 0.0952<br>(0.0413)  | 0.0774<br>(0.0232)  | 0.0588<br>(0.0103)  |
| TPR               | 0.8314<br>(0.0562)  | 0.8612<br>(0.0231)  | 0.8913<br>(0.0213)  | 0.9128<br>(0.0127)  | 0.8510<br>(0.0856)  | 0.8864<br>(0.0335)  | 0.9176<br>(0.0245)  | 0.9355<br>(0.0156)  |  | 0.8531<br>(0.0756)  | 0.8751<br>(0.0565)  | 0.9023<br>(0.0216)  | 0.9242<br>(0.0163)  | 0.8644<br>(0.0565)  | 0.8923<br>(0.0326)  | 0.9214<br>(0.0261)  | 0.9462<br>(0.0152)  |
| $d = 100, n = 50$ |                     |                     |                     |                     |                     |                     |                     |                     |  |                     |                     |                     |                     |                     |                     |                     |                     |
| FAE               | 55.2209<br>(0.2115) | 36.1787<br>(0.1956) | 28.7683<br>(0.1561) | 20.2725<br>(0.1193) | 48.5060<br>(0.1915) | 25.6995<br>(0.1566) | 17.0870<br>(0.1325) | 12.3318<br>(0.1033) |  | 47.8352<br>(0.1986) | 31.5048<br>(0.1568) | 27.4390<br>(0.1262) | 17.3699<br>(0.1091) | 37.5328<br>(0.1535) | 21.2660<br>(0.1217) | 15.8426<br>(0.1066) | 10.4085<br>(0.0997) |
| FRE               | 1.5619<br>(0.0096)  | 1.0233<br>(0.0081)  | 0.8137<br>(0.0063)  | 0.5734<br>(0.0047)  | 1.3720<br>(0.0089)  | 0.7269<br>(0.0066)  | 0.4833<br>(0.0019)  | 0.3488<br>(0.0010)  |  | 1.3530<br>(0.0101)  | 0.8911<br>(0.0076)  | 0.7761<br>(0.0058)  | 0.4913<br>(0.0036)  | 1.0616<br>(0.0084)  | 0.6015<br>(0.0061)  | 0.4481<br>(0.0029)  | 0.2944<br>(0.0011)  |
| FPR               | 0.1162<br>(0.1097)  | 0.0986<br>(0.0896)  | 0.0848<br>(0.0653)  | 0.0725<br>(0.0313)  | 0.1056<br>(0.0995)  | 0.0955<br>(0.0766)  | 0.0769<br>(0.0575)  | 0.0643<br>(0.0243)  |  | 0.1098<br>(0.0995)  | 0.0926<br>(0.0766)  | 0.0813<br>(0.0589)  | 0.0699<br>(0.0304)  | 0.0999<br>(0.0865)  | 0.0896<br>(0.0561)  | 0.0753<br>(0.0237)  | 0.0619<br>(0.0111)  |
| TPR               | 0.8426<br>(0.0155)  | 0.8847<br>(0.0274)  | 0.9017<br>(0.0136)  | 0.9273<br>(0.0115)  | 0.8553<br>(0.0117)  | 0.8962<br>(0.0206)  | 0.9172<br>(0.0184)  | 0.9326<br>(0.0095)  |  | 0.8543<br>(0.0143)  | 0.8914<br>(0.0103)  | 0.9116<br>(0.0085)  | 0.9306<br>(0.0062)  | 0.8668<br>(0.0097)  | 0.9006<br>(0.0016)  | 0.9233<br>(0.0008)  | 0.9447<br>(0.0004)  |

TABLE V  
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE PROBABILITY MATRIX WITH  $\mathbf{P} = 0.85$

|     | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |  | $\ell_1$            |                     |                     |                     | MCP                 |                     |                     |                     |  |
|-----|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|--|
|     | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |  | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          | $m = 100$           | $m = 500$           | $m = 1500$          | $m = 3000$          |  |
|     | $d = 80, n = 50$    |                     |                     |                     |                     |                     |                     |                     |  | $d = 80, n = 100$   |                     |                     |                     |                     |                     |                     |                     |  |
| FAE | 74.5959<br>(0.2752) | 49.0515<br>(0.1538) | 41.2813<br>(0.1210) | 24.7726<br>(0.1006) | 57.8763<br>(0.2647) | 41.2046<br>(0.1433) | 29.7818<br>(0.1157) | 19.9504<br>(0.0948) |  | 71.2069<br>(0.2556) | 46.3288<br>(0.1315) | 37.8156<br>(0.1057) | 23.3010<br>(0.0839) | 52.7664<br>(0.2123) | 36.9864<br>(0.1625) | 25.7169<br>(0.1100) | 16.5902<br>(0.0872) |  |
| FRE | 1.5562<br>(0.0261)  | 1.0233<br>(0.0175)  | 0.8612<br>(0.0099)  | 0.5168<br>(0.0043)  | 1.2074<br>(0.0197)  | 0.8596<br>(0.0088)  | 0.6213<br>(0.0074)  | 0.4162<br>(0.0016)  |  | 1.4855<br>(0.0085)  | 0.9665<br>(0.0078)  | 0.7889<br>(0.0064)  | 0.4861<br>(0.0051)  | 1.1008<br>(0.0080)  | 0.7716<br>(0.0063)  | 0.5365<br>(0.0035)  | 0.3461<br>(0.0017)  |  |
| FPR | 0.1398<br>(0.0258)  | 0.1245<br>(0.0284)  | 0.1135<br>(0.0443)  | 0.1027<br>(0.0209)  | 0.1316<br>(0.0411)  | 0.1160<br>(0.0169)  | 0.0983<br>(0.0237)  | 0.0937<br>(0.0652)  |  | 0.1264<br>(0.0436)  | 0.1152<br>(0.0674)  | 0.1008<br>(0.0156)  | 0.0983<br>(0.0175)  | 0.1163<br>(0.0178)  | 0.1134<br>(0.0454)  | 0.0934<br>(0.0137)  | 0.0863<br>(0.0112)  |  |
| TPR | 0.8453<br>(0.0381)  | 0.8642<br>(0.0183)  | 0.8894<br>(0.0241)  | 0.9204<br>(0.0239)  | 0.8637<br>(0.0216)  | 0.9092<br>(0.0128)  | 0.9464<br>(0.0337)  | 0.9513<br>(0.0278)  |  | 0.8755<br>(0.0297)  | 0.8972<br>(0.0240)  | 0.8981<br>(0.0281)  | 0.9257<br>(0.0226)  | 0.9111<br>(0.0331)  | 0.9268<br>(0.0205)  | 0.9415<br>(0.0184)  | 0.9574<br>(0.0121)  |  |
|     | $d = 100, n = 50$   |                     |                     |                     |                     |                     |                     |                     |  | $d = 100, n = 100$  |                     |                     |                     |                     |                     |                     |                     |  |
| FAE | 91.4426<br>(0.2482) | 61.4585<br>(0.2688) | 47.5946<br>(0.1369) | 35.9809<br>(0.1346) | 79.8289<br>(0.1405) | 48.5239<br>(0.1269) | 38.6520<br>(0.1343) | 24.7354<br>(0.1086) |  | 85.0191<br>(0.1232) | 57.6535<br>(0.1666) | 43.5675<br>(0.1349) | 33.6955<br>(0.1282) | 73.4463<br>(0.1246) | 41.8549<br>(0.0958) | 34.2157<br>(0.1174) | 18.4405<br>(0.1039) |  |
| FRE | 1.5645<br>(0.0234)  | 1.0515<br>(0.0209)  | 0.8143<br>(0.0301)  | 0.6156<br>(0.0146)  | 1.3658<br>(0.0105)  | 0.8302<br>(0.0149)  | 0.6613<br>(0.0134)  | 0.4232<br>(0.0129)  |  | 1.4546<br>(0.0197)  | 0.9864<br>(0.0166)  | 0.7454<br>(0.0108)  | 0.5765<br>(0.0114)  | 1.2566<br>(0.0121)  | 0.7161<br>(0.0097)  | 0.5854<br>(0.0132)  | 0.3155<br>(0.0086)  |  |
| FPR | 0.1263<br>(0.0342)  | 0.1122<br>(0.0228)  | 0.1061<br>(0.0139)  | 0.1009<br>(0.0114)  | 0.1038<br>(0.0143)  | 0.0988<br>(0.0231)  | 0.0852<br>(0.0127)  | 0.0787<br>(0.0113)  |  | 0.1188<br>(0.0175)  | 0.1059<br>(0.0202)  | 0.0905<br>(0.0163)  | 0.0878<br>(0.0145)  | 0.1037<br>(0.0128)  | 0.0963<br>(0.0106)  | 0.0843<br>(0.0086)  | 0.0639<br>(0.0079)  |  |
| TPR | 0.8163<br>(0.0349)  | 0.8207<br>(0.0224)  | 0.8427<br>(0.0131)  | 0.8452<br>(0.0101)  | 0.8309<br>(0.0133)  | 0.8596<br>(0.0116)  | 0.8978<br>(0.0147)  | 0.9071<br>(0.0122)  |  | 0.8288<br>(0.0113)  | 0.8364<br>(0.0157)  | 0.8553<br>(0.0118)  | 0.8774<br>(0.0107)  | 0.8538<br>(0.0184)  | 0.8739<br>(0.0197)  | 0.9126<br>(0.0108)  | 0.9232<br>(0.0134)  |  |

TABLE VI  
QUANTITATIVE COMPARISON AMONG DIFFERENT PENALTIES FOR THE SPRANDSYM MATRIX

|     | $\ell_1$           |                    |                    |                    | MCP                |                    |                    |                    |                    | $\ell_1$           |                    |                    |                    | MCP                |                    |                     |            |  |
|-----|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|---------------------|------------|--|
|     | $m = 100$          | $m = 500$          | $m = 1500$         | $m = 3000$         | $m = 100$          | $m = 500$          | $m = 1500$         | $m = 3000$         |                    | $m = 100$          | $m = 500$          | $m = 1500$         | $m = 3000$         | $m = 100$          | $m = 500$          | $m = 1500$          | $m = 3000$ |  |
|     | $d = 80, n = 50$   |                    |                    |                    |                    |                    |                    |                    |                    | $d = 80, n = 100$  |                    |                    |                    |                    |                    |                     |            |  |
| FAE | 5.6633<br>(0.2219) | 4.0051<br>(0.1039) | 2.5691<br>(0.0751) | 1.5326<br>(0.0222) | 4.7250<br>(0.1949) | 2.5927<br>(0.0416) | 1.3261<br>(0.0285) | 0.9231<br>(0.0143) | 5.1681<br>(0.2234) | 3.7130<br>(0.1037) | 2.3787<br>(0.0453) | 1.3010<br>(0.0222) | 4.7534<br>(0.1669) | 2.3718<br>(0.0949) | 1.1866<br>(0.0307) | 0.7533<br>(0.0144)  |            |  |
| FRE | 1.1540<br>(0.0077) | 0.8161<br>(0.0058) | 0.5233<br>(0.0049) | 0.3123<br>(0.0014) | 0.9628<br>(0.0081) | 0.5283<br>(0.0073) | 0.2702<br>(0.0012) | 0.1881<br>(0.0007) | 1.0531<br>(0.0072) | 0.7566<br>(0.0035) | 0.4847<br>(0.0027) | 0.2651<br>(0.0033) | 0.9686<br>(0.0061) | 0.4833<br>(0.0028) | 0.2418<br>(0.0014) | 0.1535<br>(0.0009)  |            |  |
| FPR | 0.1368<br>(0.1124) | 0.1346<br>(0.0965) | 0.0805<br>(0.0223) | 0.0649<br>(0.0118) | 0.1271<br>(0.1009) | 0.1131<br>(0.0897) | 0.0761<br>(0.0134) | 0.0546<br>(0.0106) | 0.1322<br>(0.1107) | 0.1295<br>(0.0901) | 0.0769<br>(0.0139) | 0.0608<br>(0.0113) | 0.1238<br>(0.0983) | 0.1117<br>(0.0741) | 0.0643<br>(0.0139) | 0.0445<br>(0.0085)  |            |  |
| TPR | 0.8597<br>(0.0224) | 0.8887<br>(0.0131) | 0.9068<br>(0.0101) | 0.9107<br>(0.0173) | 0.8641<br>(0.0158) | 0.8946<br>(0.0118) | 0.9176<br>(0.0103) | 0.9282<br>(0.0131) | 0.8611<br>(0.0147) | 0.8913<br>(0.0107) | 0.9139<br>(0.0124) | 0.9276<br>(0.0105) | 0.8748<br>(0.0110) | 0.9055<br>(0.0098) | 0.9226<br>(0.0091) | 0.9374<br>(0.00101) |            |  |
|     | $d = 100, n = 50$  |                    |                    |                    |                    |                    |                    |                    | $d = 100, n = 100$ |                    |                    |                    |                    |                    |                    |                     |            |  |
| FAE | 6.3763<br>(0.2741) | 4.6956<br>(0.1452) | 3.0768<br>(0.1088) | 1.7689<br>(0.0820) | 5.8840<br>(0.2610) | 3.4189<br>(0.0620) | 1.7551<br>(0.0454) | 1.0624<br>(0.0218) | 5.7895<br>(0.2110) | 4.1438<br>(0.1351) | 2.8588<br>(0.0921) | 1.6846<br>(0.0742) | 5.4194<br>(0.1374) | 3.0847<br>(0.1143) | 1.5530<br>(0.0971) | 0.7971<br>(0.0669)  |            |  |
| FRE | 1.1632<br>(0.0057) | 0.8566<br>(0.0047) | 0.5613<br>(0.0031) | 0.3227<br>(0.0015) | 1.0734<br>(0.0087) | 0.6237<br>(0.0021) | 0.3202<br>(0.0016) | 0.1938<br>(0.0006) | 1.0561<br>(0.0033) | 0.7559<br>(0.0027) | 0.5215<br>(0.0019) | 0.3073<br>(0.0011) | 0.9886<br>(0.0032) | 0.5627<br>(0.0018) | 0.2833<br>(0.0009) | 0.1454<br>(0.0006)  |            |  |
| FPR | 0.1226<br>(0.0119) | 0.1245<br>(0.0105) | 0.0837<br>(0.0096) | 0.0513<br>(0.0083) | 0.1148<br>(0.0105) | 0.1045<br>(0.0104) | 0.0616<br>(0.0091) | 0.0506<br>(0.0071) | 0.1133<br>(0.0095) | 0.1076<br>(0.0076) | 0.0711<br>(0.0063) | 0.0439<br>(0.0043) | 0.0981<br>(0.0085) | 0.0861<br>(0.0061) | 0.0646<br>(0.0054) | 0.0337<br>(0.0038)  |            |  |
| TPR | 0.8645<br>(0.0317) | 0.8971<br>(0.0139) | 0.9147<br>(0.0116) | 0.9225<br>(0.0108) | 0.8861<br>(0.0132) | 0.9091<br>(0.0130) | 0.9159<br>(0.0104) | 0.9286<br>(0.0092) | 0.8748<br>(0.0186) | 0.8996<br>(0.0126) | 0.9195<br>(0.0109) | 0.9278<br>(0.0099) | 0.8849<br>(0.0176) | 0.9108<br>(0.0112) | 0.9218<br>(0.0081) | 0.9372<br>(0.0072)  |            |  |

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